01_张弛老师第一作者语言样本

用途

本包按作者角色和证据用途分组。分隔页说明来源、张老师作者角色和证据等级；随后保留每篇论文的完整原始页面。

论文正文属于已发表作品，仅用于本项目中的学术研究和写作分析。请引用原论文。

内容

SRC-001: Monitoring microbial sulfate reduction in porous media using multipurpose electrodes

SRC-002: Spectral Induced Polarization Signatures of Hydroxide Adsorption and Mineral Precipitation in Porous Media

SRC-003: Complex Dielectric Properties of Sulfate-Reducing Bacteria Suspensions

SRC-004: Quadrature conductivity: A quantitative indicator of bacterial abundance in porous media

SRC-015: A critical mini-review on the low-field nuclear magnetic resonance investigation of pore coupling effects in near-surface environments

SRC-017: Recent research advances in enhanced CO2 mineralization and geologic CO2 storage

01_张弛老师第一作者语言样本.pdf

文献 1/6

Monitoring microbial sulfate reduction in porous media using multipurpose electrodes

Chi Zhang; Dimitrios Ntarlagiannis; Lee Slater; Rory Doherty

Journal of Geophysical Research: Biogeosciences (2010); DOI: 10.1029/2009JG001157

证据用途：第一作者直接语言样本；张弛老师角色：第一作者；证据等级：A

JOURNAL OF GEOPHYSICAL RESEARCH, VOL. 115, G00G09, doi:10.1029/2009JG001157, 2010

Click Here for Full Article

Monitoring microbial sulfate reduction in porous media using multipurpose electrodes

Chi Zhang,[1] Dimitrios Ntarlagiannis,[1] Lee Slater,[1] and Rory Doherty[2]

Received 16 September 2009; revised 10 March 2010; accepted 25 March 2010; published 21 July 2010.

[1] There is growing interest in the application of electrode-based measurements for monitoring microbial processes in the Earth using biogeophysical methods. In this study, reactive electrode measurements were combined to electrical geophysical measurements during microbial sulfate reduction occurring in a column of silica beads saturated with natural river water. Electrodic potential (EP), self potential (SP) and complex conductivity signals were recorded using a dual electrode design (Ag/AgCl metal as sensing/EP electrode, Ag/AgCl metal in KCl gel as reference/SP electrode). Open-circuit potentials, representing the tendency for electrochemical reactions to occur on the electrode surfaces, were recorded between sensing/EP electrode and reference/SP electrode and showed significant spatiotemporal variability associated with microbial activity. The dual electrode design isolates the microbial driven sulfide reactions to the sensing electrode and permits removal of any SP signal from the EP measurement. Based on the known sensitivity of a Ag electrode to dissolved sulfide, we interpret EP signals exceeding 550 mV recorded in this experiment in terms of bisulfide (HS^-) concentration near multiple sensing electrodes. Complex conductivity measurements capture an imaginary conductivity (σ'') signal interpreted as the response of microbial growth and biomass formation in the column. Our results suggest that the implementation of multipurpose electrodes, combining reactive measurements with electrical geophysical measurements, could improve efforts to monitor microbial processes in the Earth using electrodes.

Citation: Zhang, C., D. Ntarlagiannis, L. Slater, and R. Doherty (2010), Monitoring microbial sulfate reduction in porous media using multipurpose electrodes, *J. Geophys. Res.*, *115*, G00G09, doi:10.1029/2009JG001157.

1. Introduction

[2] Biogeophysics is a rapidly evolving Earth science discipline concerned with the links between dynamic subsurface microbial processes, microbial-induced alterations to geologic materials, and geophysical signatures [*Atekwana and Slater*, 2009]. There is growing interest in applying electrical geophysical techniques (resistivity, complex conductivity, and self potential (SP)) for biogeophysical projects as they have repeatedly been shown to be sensitive to bacterial cells, microbial growth and microbe-mineral alterations [*Atekwana et al.*, 2004a, 2004b; *Naudet et al.*, 2004; *Personna et al.*, 2008; *Slater et al.*, 2008]. Previous studies have confirmed the potential of electrical geophysics for characterizing microbial production of metabolic byproducts and resulting alteration of mineral surfaces [*Werkema et al.*, 2003; *Atekwana et al.*, 2004a, 2004b; *Allen et al.*, 2007; *Che-Alota et al.*, 2009], evaluating hydrological and biotransformations due to microbial reduction of heavy metals [*Hubbard et al.*, 2008; *Williams et al.*, 2009], improving understanding of metal and nutrient cycling driven by microbe-mineral transformations [*Ntarlagiannis et al.*, 2005b; *Williams et al.*, 2005; *Personna et al.*, 2008], and detection of microbes, microbial growth and biofilm formation [*Abdel Aal et al.*, 2004; *Prodan et al.*, 2004; *Ntarlagiannis et al.*, 2005a; *Davis et al.*, 2006; *Abdel Aal et al.*, 2009; *Ntarlagiannis and Ferguson*, 2009; *Slater et al.*, 2009].

[3] Biogeophysical signals are inherently non-unique, as complex and coupled biogeochemical processes alter subsurface physical properties in many ways and can drive multiple geophysical signatures over a wide range of spatial and temporal scales [*Atekwana and Slater*, 2009]. Additional biological, chemical, and physical information is typically required to reliably constrain interpretation of biogeophysical signals. One approach to constrain the interpretation is to simultaneously collect and analyze multiple electrical geophysical measurements (e.g., joint acquisition of complex conductivity and SP), along with any available aqueous geochemistry data (e.g., pH, Eh, and ion concentration). However, direct sampling of aqueous geochemistry is invasive, time consuming and expensive. Real-time monitoring of aqueous geochemistry using novel sensors offers promise in capturing temporal evolution of biogeochemical processes [*Taillefert*

[1]Department of Earth and Environmental Sciences, Rutgers, State University of New Jersey, Newark, New Jersey, USA.
[2]Environmental Engineering Research Centre, School of Planning, Architecture and Civil Engineering, Queen's University of Belfast, Belfast, UK.

Copyright 2010 by the American Geophysical Union.
0148-0227/10/2009JG001157

et al., 2000; *Viollier et al.*, 2003], but is usually impractical and cost-prohibitive to perform at high sampling density over large spatial domains.

[4] Recent studies have shown that electrodic potential (EP) measurements, using simple in construction and inexpensive electrodes, can be utilized to capture temporal and spatial variability in sulfide chemistry associated with microbial sulfate reduction under anaerobic conditions [*Williams et al.*, 2007; *Personna et al.*, 2008; *Slater et al.*, 2008]. The EP measurements record the tendency for spontaneous redox reactions to occur on the surfaces of metal and/or metal-metal salt electrodes. *Williams et al.* [2007], *Personna et al.* [2008] and *Slater et al.* [2008] showed that, in the presence of a bisulfide (HS^-) concentration gradient across two Ag-AgCl metal electrodes, a galvanic cell (GC) potential is recorded, being the potential of an electrochemical cell with a known reactivity between the target compound and the electrodes used. We stress here that this EP method may provide only an approximation of HS^- concentration, as the accurate interpretation of HS^- may be limited due to possible complex electrochemical reactions in the system. More precise and diagnostic reactive electrode-based techniques, such as voltammetry, have been used with great success to determine chemical speciation associated with sulfide redox chemistry in the deep-sea [*Luther et al.*, 2001]. However, the anode and cathode in EP measurements are physically separated, and rely on very simple in construction and inexpensive electrodes that could potentially be deployed over large distances to provide spatially rich data sets collected in tandem with geophysical data sets.

[5] In this paper, we investigate the application of simultaneous EP/electrical geophysical measurements for monitoring microbial sulfate reduction. Unlike earlier studies [*Williams et al.*, 2007; *Personna et al.*, 2008; *Slater et al.*, 2008], we employ a dual electrode (Ag-AgCl metal as sensing/EP electrode; Ag-AgCl metal in KCl gel as reference/SP electrode) technique that constrains EP interpretation of HS^- concentration by using a non polarizing (SP) electrode as the reference cathode. This dual electrode strategy results in EP measurements with a constant stable reference potential, facilitating HS^- concentration predictions at the active electrode locations. In addition, this approach overcomes the overlooked limitations of earlier studies in that EP signals would be in error in the presence of significant SP signals, as have been repeatedly postulated to result from microbial activity [*Naudet et al.*, 2003, 2004; *Minsley et al.*, 2007; *Ntarlagiannis et al.*, 2007]. Finally, we demonstrate how complex conductivity, SP, and EP measurements can be jointly recorded in order to improve the understanding of microbial processes relative to electrical geophysical measurements alone.

[6] We describe a column experiment where EP, SP, complex conductivity, and resistivity measurements were made during microbial sulfate reduction in a column of silica beads saturated with natural river water amended with lactate. We show how EP measurements using geophysical instrumentation capture spatiotemporal evolution of sulfide production driven by the microbial activity. We also record a complex conductivity response consistent with previous studies [*Davis et al.*, 2006], further supporting the concept that complex conductivity can be used as a non-invasive indicator of microbial growth and biomass formation in porous media, even in silica beads and in the absence of biodegradation. We discuss how the simultaneous collection of EP and complex conductivity data helps to better understand the processes in our microbial active system.

2. Experimental Methods

2.1. Electrodic Potential Measurements

[7] Point electrodes are commonly used for electrochemical applications to measure geochemical parameters, such as Eh and pH, as well as specific chemical concentrations (e.g., Br^- using a bromide specific electrode). An open-circuit potential difference (high input impedance of the recording device impedes significant current flow) is recorded between a metal electrode in contact with the pore-filling electrolyte and a reference electrode in close proximity and connected to the metal electrode by a salt bridge. These open circuit potentials represent the tendency for an electrochemical (galvanic cell) reaction, associated with reactive compounds at the metal electrode, to proceed. A common target of such point electrodes is sulfide concentration, which can be estimated using Ag-Ag_2S electrodes [*Berner*, 1963; *Mirna*, 1971; *Whitfield*, 1971; *Revsbech et al.*, 1983]. *Williams et al.* [2007], *Personna et al.* [2008], and *Slater et al.* [2008] showed how a pair of physically separated Ag-AgCl metal electrodes, as used in biogeophysical measurements, similarly record an open circuit potential between two points in a porous medium when the electrodes straddle a microbe-induced gradient in HS^- concentration. Being a simple approach, the EP measurements have inherent uncertainties discussed later, and are unlikely to give precise estimates of HS^- concentration. However, *Williams et al.* [2007] did show a positive correlation between sulfide concentration and measured EP response, supporting the concept of using these simple in construction and inexpensive electrodes, physically spaced over large distances, to capture spatiotemporal variability in HS^- concentration.

[8] Here we modify the EP method to improve (relative to prior EP studies) monitoring of spatiotemporal variations in microbial-driven aqueous sulfide chemistry. Similar to previous work, metal Ag-AgCl electrodes serve as sensing electrodes (anodic reaction). However, we use a non-polarizing (Ag-AgCl metal in KCl gel) electrode as reference electrode, such that the cathodic reaction occurs under known chemical conditions. The reference electrode therefore maintains a constant potential that is independent of changes in fluid properties in the column. The approach is therefore more akin to the measurement obtained with a point electrode (Figure 1b), except that the active electrode is distant from the reference electrode, and multiple active electrodes can be referenced to a single reference (Figure 1a). The assumed cathodic and anodic reactions, as well as overall galvanic cell reactions are summarized in Table 1, along with the individual standard potentials calculated from standard free energies of formation [*Stumm and Morgran*, 1996].

2.2. Complex Conductivity

[9] Complex conductivity measures the frequency (ω) dependent electrical behavior of a sample (e.g., a porous

21562202g, 2010, G3, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2009JG001157 by CochraneAustria, Wiley Online Library on [19/07/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

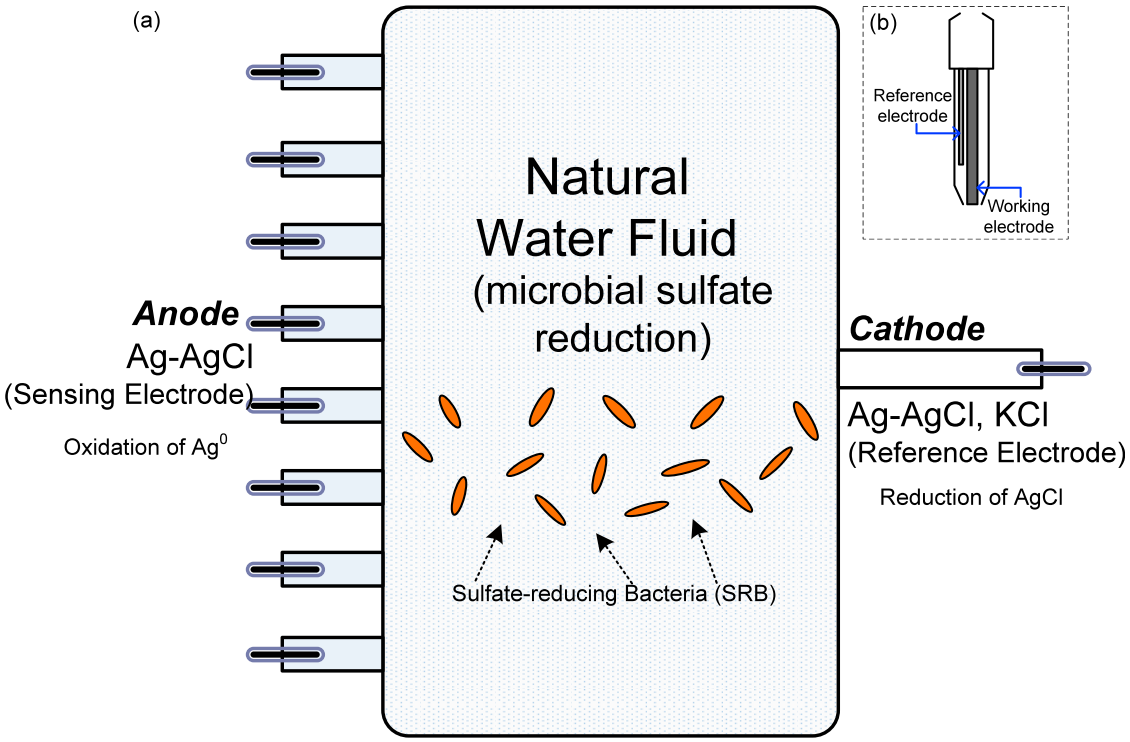


Figure 1. (a) In the column, sulfate-reducing bacteria (SRB) utilize sulfate as electron acceptor, reducing it to HS^-, while organic molecules (such as lactate) are used as the carbon source and oxidized to acetate. The Ag-AgCl in KCl gel (SP) electrode serves as a reference electrode whereby a cathodic reaction involves the reduction of AgCl coating. The anodic reaction at the sensing Ag-AgCl (EP) electrode is an oxidation of Ag^0 as a result of a reaction with sulfide. The EP/SP electrode pair serves as a redox probe sensitive to HS^- concentration. (b) The dashed frame area represents a simple schematic of point electrode used in electrochemical-based techniques for determination of aqueous ion concentration for comparison.

medium or a suspension of cells). The measured complex conductivity $\sigma^*(\omega)$ of a sample can be expressed as

$$\sigma^*(\omega) = \sigma'(\omega) + i\sigma''(\omega) \quad (1)$$

where σ' is the measured real part of $\sigma^*(\omega)$, being the conduction (energy loss) term, σ'' is the measured imaginary part of $\sigma^*(\omega)$, being the polarization (energy storage) term, and $i = \sqrt{-1}$. At low frequencies (<1000 Hz), $\sigma^*(\omega)$ of a saturated porous material can be modeled as the sum of an electrolytic conductivity σ_{el} resulting from ohmic conduction within the interconnected pore space, combined with complex mechanisms (σ^*_{surf}) associated with grain-fluid surfaces. Two electrochemical interfacial polarization mechanisms assumed to occur in porous media are (1) polarization of ions in the electrical double layer at the mineral-fluid interface, and (2) polarization resulting from differences

Table 1. Half-Cell Electrochemical Reactions and Standard Potentials on Cathode and Anode, and Overall Reactions and Standard Potential of the Galvanic Cell[a]

	Electrodes	Reaction	Standard Potential (E^0)
Cathode	Ag/AgCl, KCl (1M) (reference electrode)	$AgCl_{(s)} + e^- \leftrightarrow Ag_{(s)} + Cl^-$ (Reduction of AgCl)	0.222 V
Anode	Ag/AgCl (sensing electrode)	$2Ag_{(s)} + HS^- \leftrightarrow Ag_2S_{(s)} + H^+ + 2e^-$ (Oxidation of Ag^0); $2Ag_{(s)} + H_2S_{(aq)} \leftrightarrow Ag_2S_{(s)} + 2H^+ + 2e^-$ ($E^0 = 0.036$V);[b] $2Ag_{(s)} + S^{2-} \leftrightarrow Ag_2S_{(s)} + 2e^-$ ($E^0 = 0.69$V)[b]	0.273 V
Overall galvanic cell		$2AgCl_{(s)} + HS^- \leftrightarrow Ag_2S_{(s)} + H^+ + 2Cl^-$	0.495 V

[a]Reactions are thermodynamically favorable ($E^0 > 0$), and standard potentials (versus standard hydrogen electrode) are calculated from standard free energies of formation at 25°C [*Stumm and Morgran*, 1996].

[b]Anodic reactions with other major possible sulfate reduction products (H_2S and S^{2-}) and standard potentials for half-cell reactions are also provided, which are not the predicted reactions in our study.

21562202g, 2010, G3, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2009GC001157 by CochraneAustria, Wiley Online Library on [19/07/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

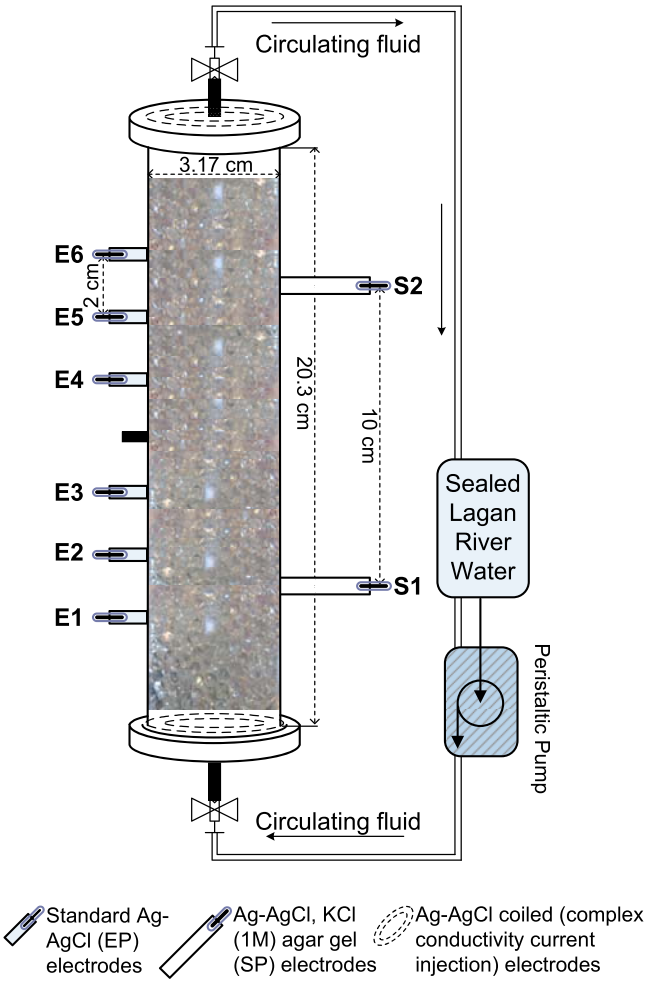


Figure 2. Schematic diagram of column setup showing flow through configuration, location of Ag-AgCl (EP) electrodes, Ag-AgCl in KCl gel (reference/SP) electrodes, and Ag-AgCl coil (complex conductivity current injection) electrodes. Lagan River water was pumped through the column via a peristaltic pump; pH and fluid conductivity (σ_f) were monitored at the outflow.

in mobility of ions in the electrolyte driven by variable pore throat diameters.

[10] The sensitivity of complex conductivity measurements to subtle changes in the surface physical properties of geologic media makes it a suitable technique for investigating microbial growth and biofilm formation [*Abdel Aal et al.*, 2004; *Ntarlagiannis et al.*, 2005a; *Abdel Aal et al.*, 2009; *Slater et al.*, 2009], and detecting microbial growth and biomineralization transformations in porous media [*Ntarlagiannis et al.*, 2005b; *Williams et al.*, 2005; *Personna et al.*, 2008].

2.3. Self Potential Method

[11] Self potential signals are voltages associated with a gradient in an electric field generated by internal current sources in the Earth. Self potentials are recorded using nonpolarizing electrodes where both metallic electrodes are removed (by embedding in a gel or aqueous solution with identical chemistry at both metallic electrodes) from contact with the pore fluids. This ensures that any potential recorded on the SP electrodes must result from internal current sources and not from galvanic potentials associated with reactions between compounds in the pore fluid and the metal electrodes.

[12] Self potential signals arise from multiple mechanisms. Streaming potentials are recorded in the presence of a streaming current source term resulting from the transport of excess charge in the electrical double layer at the solid-fluid interface of a porous medium in response to the viscous drag exerted by fluid flow through the pores. Streaming potentials have been employed with considerable success to monitor hydrological processes [*Revil et al.*, 2002; *Rizzo et al.*, 2004; *Linde et al.*, 2007]. Another SP source mechanism is electrodiffusion, arising due to gradients in the chemical potentials of charge carriers. More important in biogeophysics research is the geobattery mechanism [*Sato and Mooney*, 1960]. Geo-batteries can develop when an electron conductor bridges electron donors and electron acceptors, resulting in current flow in response to the redox gradient (the thermodynamic driving force) [*Sato and Mooney*, 1960; *Bigalke and Grabner*, 1997]. Previous studies have suggested the potential for indirect detection of microbial activity from redox gradients produced using SP methods, with strong SP signals (+200 mV) observed at sites where microbial degradation of hydrocarbons is occurring [*Nyquist and Corry*, 2002; *Naudet et al.*, 2003, 2004; *Minsley et al.*, 2007].

3. Experimental Procedures

3.1. Sample Preparation and Column Setup

[13] The objective of our design was to capture EP and electrical geophysical signatures in response to sulfate reduction occurring in a porous medium composed of glass beads. We utilized two identical columns for this experiment; one was used as the experimental (biological active) column while the second served as a control. Lexan columns were constructed with inner diameter of 3.17 cm and a length of 20.3 cm. Both columns were dry packed with 3 mm glass beads (SiLibeads-Type M) with a measured porosity of 0.38 ± 0.02 and density of 2.5 kg/m^3. Silica beads were chosen, in favor of quartzitic sand, since they offer high chemical resistance, excellent roundness, constant size and maintain the pore geometry before and after saturation by dry packing [*Ntarlagiannis and Ferguson*, 2009]. The experimental design is schematically illustrated in Figure 2.

[14] We used water from River Lagan (Belfast, Northern Ireland, UK) as the saturating fluid. The River Lagan, which feeds into Belfast Lough, is the recipient of numerous sewage and industrial discharges. The water sample contained suspended sediments and particles, known to contain sulfate-reducing bacteria (SRB) (K. P. Singh, Queen's University-Belfast, unpublished data, 2008). Chemical analysis of the collected river water involved ion chromatography (IC) and inductively coupled plasma mass spectrometry (ICP-MS) for the determination of major ions and metals, respectively. This analysis showed the water contains 29 mg/L Cl^-, 17 mg/L of SO_4^{2-}, 9.3 mg/L of NO_3^-, 0.60 mg/L of PO_4^{3-}, 0.40 mg/L of NO_2^-, 0.06 mg/L of F^-, 8.4 μg/L of aqueous Cu, 7 μg/L of aqueous Ni, Zn, and Cr, 5.8 μg/L of aqueous Se, 0.7 μg/L of aqueous As, and trace amounts of other metals. Two days prior to the experiment, the river water sample was spiked

21562202g, 2010, G3, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2009JG001157 by CochraneAustria, Wiley Online Library on [19/07/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

with 2.24 g/L sodium lactate solution, and then placed in a dark airproof container to decrease oxygen penetration [*Ramsing et al.*, 1993] and promote SRB growth.

[15] The river water, amended with lactate to stimulate microbial growth, was used to saturate the experimental column. The control column was saturated with the same treated river water, but autoclaved to prevent biological activity. Both columns were positioned vertically and attached to separate closed circulation systems with 1L treated river water (experimental column) and 1L autoclaved treated river water (control column) respectively as the flow-through medium. A multichannel peristaltic pump was used to maintain a steady flow rate of ~2 pore volumes/day (~64 ml/day) for both columns. All tubing and column components were sterilized before the experiment by autoclaving or rinsing with 70% ethanol to minimize contamination.

3.2. Electrodic Potential and Self-Potential Measurements

[16] The experiment was performed at room temperature for a period of ~22 days, although measurements for Days 10–18 were lost as a result of hardware problems. Measurements on the control column started with a two-day delay relative to the active column.

[17] The EP measurements were recorded on six metal Ag-AgCl electrodes installed in electrolyte filled chambers, in electrolytic contact with the column, placed 2 cm apart along one column side (Figure 2). These Ag-AgCl electrodes were made from high grade Ag wire (99.999%) that was immersed in a strong chlorine solution (6.5% NaOCl) until a continuous solid gray AgCl coating formed on the Ag surface. The SP signals were recorded using electrodes placed on the other side of each column, one of which also served as the reference electrode for the EP measurements (Figure 2). These electrodes were constructed by immersing a single Ag-AgCl metal electrode into 1M KCl agar gel and housed in a 15 ml pipette tip. The agar gel was prepared by dissolving 14 g/L agar powder and 74.5 g/L KCl into heated DI water. The solution was then stirred until it became clear, when 0.27 g/L $HgCl_2$ of biocide was added to inhibit the erosion of agar gel by microbes. The final solution was left to cool in the pipette tips. The fine open end of the pipette tip ensured electrolytic contact between the electrode and the electrolyte in the column.

[18] The EP and SP measurements were obtained with a high-impedance (>10 Mohm ± 1%) Keithley 2701 digital multimeter (DMM) data logging system. We collected measurement every 15 min during the experimental period, with the exception of 30-min intervals when complex conductivity measurements were performed. For all measurements we use the same reference electrode, S1 (Figure 2). The EP measurements were recorded between S1 and E1, E3, E4, and E6, whereas SP measurements were recorded between S1 and S2. In both EP and SP measurements the reference electrode was connected to the negative terminal of the digital multimeter by convention.

3.3. Complex Conductivity Measurements

[19] Complex conductivity measurements were obtained daily (from Day 0 to Day 10) with a two-channel dynamic signal analyzer (DSA, National Instrument 4551) for frequencies between 0.1 and 1000 Hz at 40 logarithmic intervals [*Slater and Lesmes*, 2002]. The commonly used four-electrode configuration technique was utilized; two coiled Ag-AgCl electrodes located at each end of the column were used for current injection (Figure 2). The resulting potential was recorded between SP electrode pair S1 and S2. The magnitude ($|\sigma|$) and the phase shift (φ) were measured relative to a known high precision resistor on channel 1. The real ($\sigma' = |\sigma|\cos\varphi$) and imaginary ($\sigma'' = |\sigma|\sin\varphi$) components of the sample complex conductivity were then calculated. The instrumentation and setup applied in this study is similar to previous studies [*Ntarlagiannis et al.*, 2005a], with phase accuracy of ~0.06 mrad.

3.4. Sampling and Geochemical Analysis

[20] Aqueous geochemical measurements of fluid conductivity (σ_f), Eh, and pH of circulating fluid were taken daily for both experimental (from Day 1) and control columns (from Day 4). A 1.5 ml fluid sample was extracted and anaerobically sealed using a sterile syringe inserted into the tubing, and Eh/pH point probes was immediately immersed in fluid to minimize the contact of sample and ambient air. The σ_f was subsequently measured using a conductivity probe. We chose not to determine sulfide concentrations (e.g., colorimetrically) in this experiment because of potential sampling errors from a closed anaerobic system of limited volume. Sulfide is unstable and easily chemically oxidized under aerobic conditions, leading to sulfate production. In addition, previous studies have already proven the positive correlation between measured EP response and sulfide concentration [*Williams et al.*, 2007] as we discussed in Section 2.1, and this was not our objective here.

3.5. Bisulfide Concentration Estimation From EP

[21] The SRB can utilize H_2 and/or low molecular weight organic matter (e.g., lactate and pyruvate) as an electron donor and sulfate as an electron acceptor under anaerobic conditions while producing sulfide [*Maier et al.*, 2000]. Taking lactate for instance, the reaction could be summarized in equation (2), whereby dissimilatory sulfate reduction by bacteria is the most kinetically favorable mechanism by which sulfide is produced [*Ledin and Pedersen*, 1996],

$$2CH_3CHOHCOO^- + SO_4^{2-} \xrightarrow{SRB} 2CH_3COO^- + HS^- + 2HCO_3^- + H^+ \quad (2)$$

At circumneutral pH, HS^- is the predominant sulfide species in aqueous system, although other major sulfide species are H_2S and S^{2-}. In our experiment with the presence of HS^-, oxidation of silver occurs at the anode (metal Ag-AgCl electrode) along with the concurrent conversion of AgCl to Ag_2S due to the much smaller stability constant (K_{sp}) of Ag_2S. Reduction of the AgCl electrode coating occurs at the non-polarizing reference electrode. The electrochemical reactions on the electrodes and the associated half-cell potentials are summarized in Table 1.

21562202g, 2010, G3, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2009JG001157 by CochraneAustria, Wiley Online Library on [19/07/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

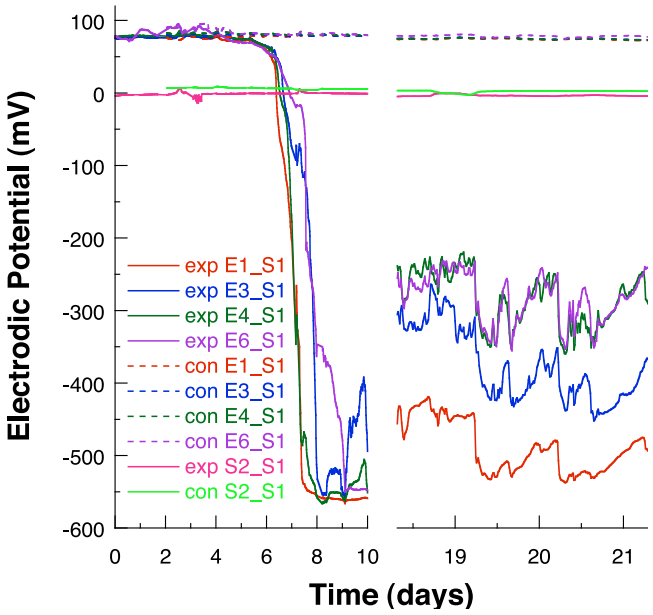


Figure 3. Electrodic potential (EP) and self-potential (SP) measurements over the experiment duration. Solid color lines denote EP signals at different locations in experimental column referenced to S1, and color dashed lines indicate EP signals at the same locations in control column. Magenta and bright green lines show the SP signals at pair S2_S1 in experimental and control column, respectively.

[22] Utilizing the Nernst equation, the HS^- concentration ($[HS^-]$) can be estimated given EP and pH measurements from

$$\Delta E_{cell} = E^0_{cell} - \frac{RT}{nF}\log\frac{[H^+][Cl^-]^2}{[HS^-]} \quad (3)$$

where E^0_{cell} is the standard potential of the overall galvanic cell reactions determined from two half-cell potentials (Table 1), and ΔE_{cell} is the EP of our system. However, the potential recorded between the Ag-AgCl metal electrode and the reference electrode is actually the sum of the EP and any SP signal in the system (a fact that was overlooked in previous studies using EP [*Williams et al.*, 2007; *Personna et al.*, 2008; *Slater et al.*, 2008]). In our setup, the ΔE_{cell} can be determined by subtracting any SP signal (as recorded between the non-polarizing electrodes S1 and S2 in Figure 2) from the total potential recorded between any EP electrode (E1-E6) and S1. The hydrogen concentration ($[H^+]$) is known from pH, which was measured daily and linearly interpolated to predicted values at 15 min intervals (recording interval for EP measurements). As the chloride concentration ($[Cl^-]$) is known (1M KCl in the cathode), $[HS^-]$ is the only unknown in equation (3). It is important to emphasize that the $[HS^-]$ calculated here is the localized concentration immediately adjacent to the sensing EP electrode, and it does not necessarily represent $[HS^-]$ of the bulk fluid. Furthermore, the approach is based on assumptions that we discuss in detail later.

4. Results

4.1. Visual Observation

[23] A darkening of the fluid in the column and tubing was visually observed, being first noted at the bottom of the column (inflow) (Figure 2) at Day 6, and covered the entire active column after Day 8. At the beginning of experiment, both aerobic and anaerobic microorganisms (including SRB) likely co-existed. The observed darkening is indicative of microbial sulfate reduction, expected as the river water contained a significant amount of sulfate (17 mg/L) and SRB communities. In the presence of available metal cations (e.g., Fe^{2+}, Cu^{2+}, and Zn^{2+}), S^{2-}, the main product of sulfate reduction, binds to form insoluble metal sulfide precipitates (MeS). In our experiment the observed darkening was attributed to possible precipitation of such metal-sulfide compounds. The gradual transition from bottom to top may represent the higher availability of nutrients for SRB closer to the inflow. Alternatively, this darkening could also conceivably arise simply from the settling and/or filtration of MeS within the column. This visual evidence of SRB activity was backed up by the characteristic sulfurous smell noted during syringe extraction of fluid from the experimental column. In the control column no change of color or sulfurous smell was observed over the duration of the experiment, indicating no significant SRB activity as expected.

4.2. EP and SP Measurements

[24] Electrodic potentials recorded at E1, E3, E4, and E6 on the experimental column relative to the reference electrode are shown in solid color lines in Figure 3. The measurements taken at the same pairs on the control column are also shown in Figure 3 in color dashed lines. At the beginning of experiment, steady (~75 mV) EP measurements were observed for both the experimental and control column until Day 6 when strong negative potentials in the experimental column developed. The EP readings in the control column (color dashed lines) remained the same thereafter, suggesting that no changes in redox state occurred in the natural waters in the absence of biological mediated redox reactions. In contrast, peak negative EP values (−550 ~ −560 mV) were measured in the experimental column after Day 7 and were maintained until ~Day 10. Between Day 19 (446 h) and the end of the experiment, the EP signals of all pairs in the experimental column slowly decreased in magnitude. During this time, the bottom electrode E1 (pair E1_S1) showed the least change, followed by pair E3, with E4 and E6 behaving similarly and exhibiting the greatest decrease.

[25] Figure 3 also shows SP data at pair S2_S1 for both experimental (magenta solid line) and control (bright green solid line) columns. The measurements were relatively steady (+3 ± 4 mV) for the duration of the experiment. The SP in the control column was slightly (~5 mV) higher than the SP in the experimental column.

21562202g, 2010, G3, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2009JG001157 by CochraneAustria, Wiley Online Library on [19/07/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

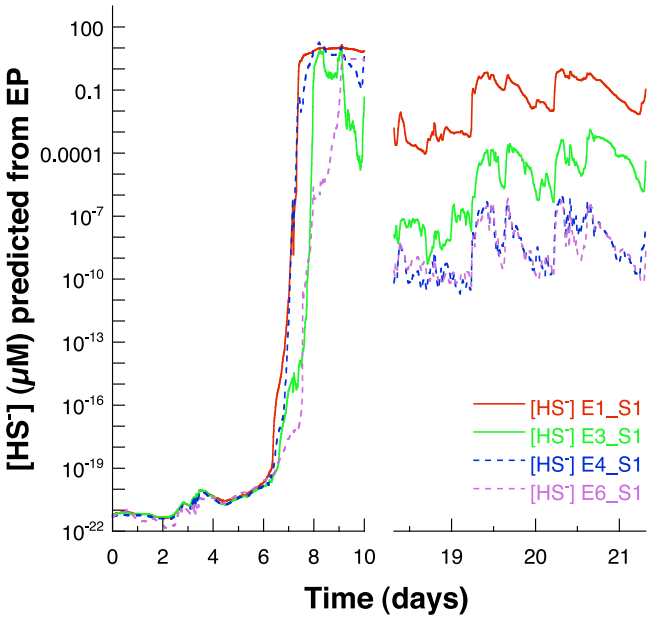


Figure 4. Estimated hydrogen sulfide concentration ($[HS^-]$) at different locations in the experimental column over the experiment duration. The $[HS^-]$ was determined by the Nernst equation as a function of theoretical cell potential (E^0_{cell}), cell potential (ΔE_{cell}, equivalent to measured EP corrected by eliminating SP), and pH, while holding $[Cl^-]$ (1 M) and temperature (25°C) constant. The concentrations shown here are in μM.

4.3. HS^- Concentration

[26] The calculated $[HS^-]$ (in μM) in the experimental column is showed in Figure 4 for pairs E1_S1, E3_S1, E4_S1, and E6_S1. Reflecting the relative small increase in EP early in the experiment, $[HS^-]$ slightly increased from Day 0 to Day 5, the response being nearly the same for each pair. Starting from Day 6, $[HS^-]$ dramatically increased for all pairs. Pair E1_S1 was the first to develop peak values (14.279 μM), followed by pair E4_S1 (peak value 18.15 μM), E3_S1 (peak value 9.96 μM), and E6_S1 (4.13 μM). These peak values maintained a plateau from Day 7 to ~Day 10. During Day ~19 to Day 22, $[HS^-]$ decreased, showing a gradual decline from pair E1_S1 to E6_S1. At the end of this experiment, pair E1_S1 had the highest estimated $[HS^-]$, followed by pair E3_S1, while pairs E4_S1 and E6_S1 had the lowest estimated concentration.

4.4. Complex Conductivity and Fluid Conductivity

[27] The complex conductivity measured between electrode pair S1_S2 from both columns was used to calculate σ' (Figure 5a) and σ'' (Figure 5b) components, respectively (34 Hz values shown). The σ' (Figure 5a) increased from 0.085 S/m to 0.27 S/m from Day 0 to Day 4 and remained steady in the experimental column, while σ' remained at 0.07 S/m in the control column and showed no significant changes throughout the experiment. The σ'' (Figure 5b) in the experimental column increased from 1.16×10^{-5} S/m (Day 0) to 4.29×10^{-4} S/m (Day 5), and then decreased to 4.76×10^{-5} S/m on Day 10. The magnitude of the σ'' response in the control column was constant at 1.1×10^{-5} S/m over the duration of the experiment.

[28] Aqueous geochemical analysis of fluid samples showed σ_f in the experimental column was the same as in the control column at the beginning of experiment, but later increased to ~5 times higher than it in control column. The σ_f value remained steady at 0.23 S/m in the control column, whereas σ_f in the experimental column increased with slight fluctuations. Normalizing the real conductivity by fluid conductivity (σ'/σ_f), resulted in a flat response, indicating that increases in σ' in the experimental column result from associated increases in σ_f.

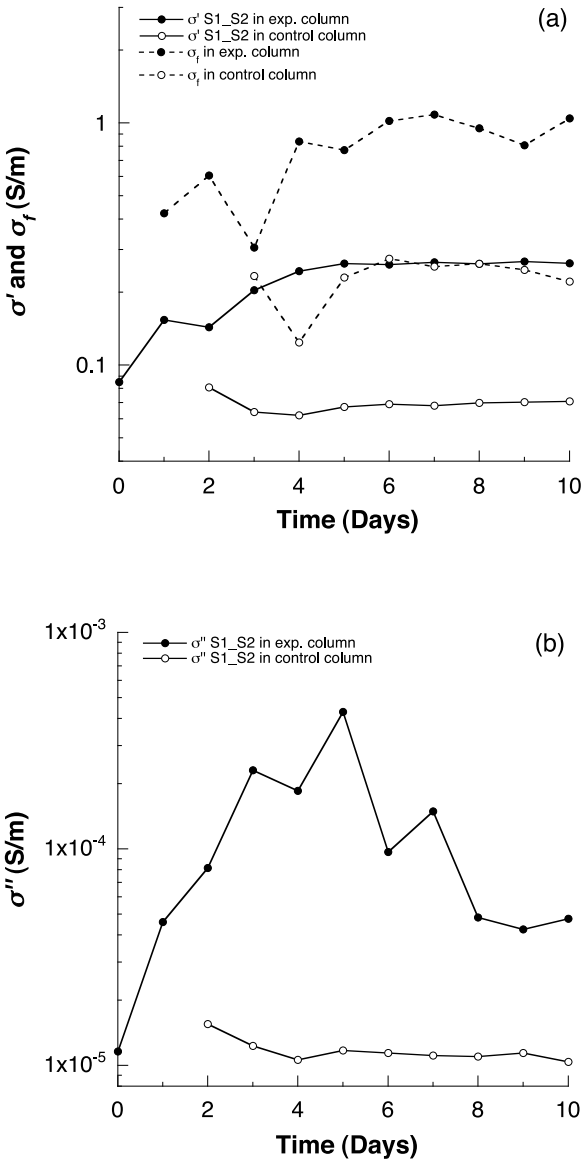


Figure 5. (a) Fluid conductivity σ_f (dashed lines) and results of measured real conductivity σ' (solid lines) for pair S1_S2 in both experimental (closed black circles) and control (open circles) columns. (b) Imaginary conductivity (σ'') for pair S1_S2 in both experimental (closed black circles) and control (open circles) columns. Both σ' and σ'' shown here are at 34 Hz from Day 0 to Day 10.

21562202g, 2010, G3, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2009GC001157 by CochraneAustria, Wiley Online Library on [19/07/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

[29] The pH in the experimental column decreased from 7.81 to 7.29, and it dropped from 8.98 to 7.42 in control column throughout the experimental period. This drop was likely due to the existence of CO_2 and organic acids, generated from the degradation of complex organic material and biological respiration in the column, dissociated to form protons. The Eh (data not shown) recorded at the fluid sample varied considerably, probably reflecting problems with sampling and exposure to ambient air prior to Eh probe measurements.

5. Discussion

5.1. Sulfide Chemistry and Reactive Electrodes

[30] Our results show how microbial-driven sulfide chemistry can be monitored using EP measurements made on simple in construction and inexpensive metal electrodes, that can also be used to make electrical geophysical measurements. We note that the EP response curves recorded in this experiment are quite similar in shape and magnitude when compared to those previously recorded on other experiments using mixed SRB cultures [*Slater et al.*, 2008], as well as those recorded for a pure *D. vulgaris* culture [*Personna et al.*, 2008]. However, we do not expect identical EP responses across these systems as the mineral matrix, growth media and cultures differ significantly. These differences could result in different onsets of SRB activity (e.g., the EP response curves in the work by *Slater et al.* [2008] develop later than in this study) and differences in total sulfide produced. Although laboratory experiments using pure cultures are more easily quantifiable, they do not represent the real world. Our laboratory experiment suggests that we can bridge the gap between laboratory and field scale applications and this technology is transferable to the field scale.

[31] We recorded an open-circuit potential, as in previous work [*Personna et al.*, 2008; *Slater et al.*, 2008], which increases with the difference in HS^- concentration between the cathode and anode. However, in our study we utilize two types of electrodes, allowing for the correction of any SP signals during the EP measurement. We note that the presence of conductive metals between S1 and S2 could facilitate a geobattery SP mechanism [*Sato and Mooney*, 1960; *Bigalke and Grabner*, 1997] within the column given a sufficient redox gradient between these two locations. Even though, we only observe very small SP signals (few mV in both columns), suggesting there is no apparent mechanism associated with this microbial sulfate reduction process that could generate current sources. This result is expected given that (1) the redox gradient between S1 and S2 is small in this closed circulation experiment, and (2) it is unlikely that the MeS precipitates were extensive enough to form continuous conduction paths.

[32] Instead, the SP signal probably represents the streaming potential due to the flow of the saturating fluid through the glass beads. In our case then, the SP correction needed to determine EP from the potentials recorded on the metal electrodes was minor. However, recent experiments suggest that this may not be the case in field-scale studies. Previous studies have identified large (+200 mV) SP readings, postulated to be due to biogeobatteries [*Naudet et al.*, 2003, 2004; *Minsley et al.*, 2007], in the presence of strong microbial-driven redox potential gradients. In such field studies, it would be critical to remove the SP signals from the EP responses prior to estimation of sulfide concentration. The previous EP studies [*Williams et al.*, 2007; *Slater et al.*, 2008] employed only reactive metallic electrodes (Ag-AgCl electrodes), therefore making it impossible to determine the contribution of any SP signal to the potential recorded on the metallic electrodes. The fact that SP was not quantified in these earlier studies has implications for the accuracy of the estimated sulfide concentration and adopted interpretation of the mechanism driving the measured potentials in general.

[33] The use of a constant potential SP electrode in the EP measurement isolates the sulfide chemistry information to a single reactive electrode. In contrast, previous studies [*Williams et al.*, 2007; *Personna et al.*, 2008; *Slater et al.*, 2008] using two reactive electrodes being located in a stable/quiescent, non-reactive environment in order to be isolated from confounding effects of reactive fluid chemistry (e.g., on the column influent for flow-through experiments). The use of a constant potential SP electrode for the reference enhances our confidence in the possible quantitative estimation of HS^- concentration at the anode, as the concentration at the cathode is known to be constant. Our approach requires a reference SP electrode (Ag-AgCl immersed in a KCl gel), where the electrodic reaction is fixed and known. When two reference electrodes are available, SP measurements can also be performed. These permit correction of the measurements made on metallic electrodes for SP signals and estimation of EP.

[34] Our approach to biogeophysics monitoring, couples electrical geophysical measurements with a simple measurement sensitive to pore fluid redox chemistry at the electrodes. The interpretation of biogeophysical signals is inherently uncertain due to the multiple possible sources for the recorded electrical signals; additional measurements, such as aqueous geochemistry measurements, are clearly advantageous for constraining the interpretation. In our experiment we demonstrate a quick and inexpensive means to capture spatial and temporal variability in microbial-driven sulfide chemistry using the same electrodes applied for the electrical geophysical measurements. Geophysical monitoring of microbial processes using large numbers of electrodes is increasingly being conducted [*Daily and Ramirez*, 1995; *Werkema et al.*, 2003; *Atekwana et al.*, 2004b; *Allen et al.*, 2007; *Minsley et al.*, 2007], and our approach could be adopted to determine variations in sulfide concentration at hundreds of locations as defined by the placement of electrodes. Such a sampling density would be prohibitively costly and invasive if on site extraction and sampling was conducted. The suggested EP electrode design is sensitive to other electrochemical reactive species and could be used as a qualitative Eh indicator in environments where no additional information is available; in certain cases where the chemistry is known and the dominant electrochemical species are identified or assumed (e.g., sulfide due to SRB activity in our experiment), EP measurements can be used to quantitatively estimate the concentration of the species of interest.

[35] The quantitative interpretation of the EP signals in terms of sulfide concentration is, however, based on a number of assumptions. First, we assume that the reactions

21562202g, 2010, G3, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2009JG001157 by CochraneAustria, Wiley Online Library on [19/07/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

on the anode effectively alter the anode from Ag-AgCl to Ag-Ag_2S, which is a thermodynamically favorable but unverified reaction here. Second, we treat this system as a single-potential system (although it is probably a mixed potential in reality), and we attribute the EP signals only to HS^-, without accounting for any other possible sulfate reduction products in the sulfide family (e.g., SO_3^{2-}, S^{2-} ions, dissolved H_2S, and etc.) or any transformation between different sulfide species. Third, we assume the reactions describe an ideal Nernstian response over the whole range of HS^- concentrations. In fact, a linear Nernstian slope is only obtained for concentrations from 10^{-6} to ~10^{-4} mol/L of total sulfur, the range depending on pH and ion concentration of the aqueous solution [*Revsbech et al.*, 1983; *Jeroschewski et al.*, 1993; *Lawrence et al.*, 2000]. These problems are not unique to our system, but are recognized difficulties in using point Ag/Ag_2S electrodes to determine sulfide. In addition to deviation from ideal Nernstian behavior, other problems include long response times and poisoning of the reference electrode [*Jeroschewski et al.*, 1996; *Lawrence et al.*, 2000].

[36] Despite these limitations, the Ag/Ag_2S electrode has for a long time offered the only reasonable solution to estimate sulfide at high spatial resolution at the aerobic and anaerobic interface in aquatic sediments and biofilms, where distinct gradients of chemical and physical parameters exist [*Revsbech et al.*, 1986; *Kühl and Jørgensen*, 1992]. Electrode-based electrochemical determination of sulfide in aqueous system has a lower determination limit of a few ppb and an upper limit of hundreds of ppm, depending on the methods and electrodes [*Lawrence et al.*, 2000]. Sulfide concentration in environmental systems varies from a few ppb (e.g., in oxic seawater) to thousands of ppm (e.g., in pore waters from salt marshes). In the latter case, the use of simple in construction and inexpensive multipurpose electrodes, as proposed here, could conceivably capture spatiotemporal changes calibrated in terms of actual sulfide concentration. However, in many environments it will likely only be possible to use EP measurements to infer patterns of sulfide variation, but not quantify absolute concentrations.

[37] Even if the determination of HS^- concentration is not entirely accurate in our system, the EP signals still capture the onset of SRB activity, along with spatiotemporal variability in the processes. Such information is very valuable from a monitoring standpoint, and can also improve understanding of biogeophysical signals associated with microbial processes. In fact, validation of field studies often requires accredited chemical and biological analysis techniques, and the application of EP potentially provides a cost effective real time monitoring method that could supplement expensive chemical and biological analysis. In future studies, the estimation of HS^- using our approach might be improved by recording the real time temperature, pH, and ion concentration of the fluid, or constructing electrodes with a constant reactive area [*Barrett et al.*, 1988]. It may also be possible to calibrate these electrodes for variations in pH and ion concentration.

5.2. Electrical Geophysical Signals

[38] Our complex conductivity measurements revealed temporal increases in σ' in the experimental column resulting from increases in fluid conductivity (as supported by normalizing the σ' data by σ_f). Fluid conductivity is largely controlled by the total dissolved solids representing the number of ions present in the solution. Organic acids (acetic acid) produced as metabolic by-products of degradation of larger molecular organic matter (e.g., lactate) have been shown to cause mineral dissolution resulting in the release of ions into the pore space, thus increasing the conductivity of the pore fluid [*Abdel Aal et al.*, 2004; *Allen et al.*, 2007]. In addition, biomass and decay of bacterial cells may contribute to higher electrical conductivity [e.g., *Slater et al.*, 2009].

[39] More significantly, our complex conductivity measurements also reveal a σ'' signal that is similar to that observed in work of *Davis et al.* [2006] and attributed to microbial growth/attachment followed by subsequent death/detachment. Given that no signals were observed in the control column, we are confident that the source is microbial in origin. The possible presence of metal-sulfide precipitates in the experimental column, which might be the reason for the observed darkening of fluid, cannot explain the σ'' response, as σ'' declined after reaching the peak value at Day 5. Although previous work has shown similar σ'' curves that were attributed to changes in mineral aggregation and variation in the spatial position of sulfide-encrusted cells [*Ntarlagiannis et al.*, 2005b; *Williams et al.*, 2005], these studies were conducted on columns with continuous supply of metals. In contrast, any σ'' response associated with metal-sulfide precipitates in our metal-limited system might be expected to show just the early time response observed in these previous studies, whereby σ'' increases in response to dispersed metal precipitates. This is consistent with elevated σ'' values throughout the experiment and dark fluid being observed until the end of experiment.

[40] We suggest that our σ'' response in the experimental column reflects changes of microbial population in our system. *Davis et al.* [2006] showed a correlation between σ'' and microbial growth in their diesel fuel and nutrient broth saturated sand column experiment. The model of *Davis et al.* [2006] seems plausible in our case, although our study was conducted under much simpler experimental conditions. More recently, *Slater et al.* [2009] demonstrated an electrical conductivity response exhibiting the characteristics of microbial growth models. Our results suggest that a signal associated with microbial growth/attachment to death/detachment cycle can be captured during microbial sulfate reduction in the absence of hydrocarbon biodegradation. In our experiment, σ'' reached peak values at Day 5, although the largest EP signals developed on Day 6. This delay in the EP signals relative to the σ'' peak may suggest that 1) sulfate reduction by SRB was not initiated until oxygen is depleted, and/or 2) significant time is needed for electrochemical conditions at the electrode to develop for EP signal generation. The microbial activity between Days 2–8 captured with σ'' is likely the response of both aerobic and anaerobic microorganisms (including SRB) existing in the natural fluid. The peak in SRB activity occurred later as captured in the EP response. In this case, σ'' detects the growth of all microbial cells in the column whereas the EP signals are associated only with SRB activity. Our work thus lends strength to the notion that the signal recorded in the work of *Davis et al.* [2006] is an important biogeophysical signal

21562202g, 2010, G3, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2009GC001157 by CochraneAustria, Wiley Online Library on [19/07/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

that is not specific to the conditions of the *Davis et al.* [2006] experiment.

6. Conclusion

[41] Here we have shown how EP measurements, in conjunction with SP and complex conductivity measurements, can be used to monitor spatial and temporal variability in microbial sulfate reduction within a porous medium saturated with natural river water using dual electrode sensors. The dual electrode approach permits an SP correction to the EP measurement, and isolates the microbial driven sulfide chemistry to a sensing/EP electrode while holding the reference/SP electrode constant. We assume these open-circuit EP signals are associated with electrochemical reactions that occur on the Ag-AgCl (sensing/EP) electrode in the presence of dissolved sulfide, although it is possible that other microbial driven redox reactions may also have contributed to the observed signal. The HS^- concentration near the EP electrode surface could be estimated from EP signals based on certain assumptions. Our complex conductivity results are consistent with previous biogeophysics research, showing sensitivity to microbial activity in porous media. Thus EP measurements using Ag-AgCl electrodes offer a simple approach to capture spatial variability in microbial driven sulfide chemistry that can be collected using the same instrumentation and hardware as used for electrical geophysics measurements. The joint use of EP and complex conductivity measurements could facilitate simultaneous monitoring of individual microbial communities (e.g., sulfate reducers), while also monitoring net microbial growth. Our work may lead to the application of multipurpose electrodes to improve the geophysical and geochemical monitoring in biogeophysics research.

[42] **Acknowledgments.** This work was performed at Queen's University Belfast, Belfast (UK) in the Environmental Engineering Research Center. This material is based upon work supported by the National Science Foundation under grant EAR-0433729. We thank Kushal P. Singh (Queen's University Belfast, Belfast, UK) for his assistance in laboratory work. Review comments provided by two reviewers and the Associate Editor significantly improved this manuscript.

References

Abdel Aal, G. Z., E. A. Atekwana, L. D. Slater, and E. A. Atekwana (2004), Effects of microbial processes on electrolytic and interfacial electrical properties of unconsolidated sediments, *Geophys. Res. Lett.*, *31*, L12505, doi:10.1029/2004GL020030.

Abdel Aal, G., E. Atekwana, S. Radzikowski, and S. Rossbach (2009), Effect of bacterial adsorption on low frequency electrical properties of clean quartz sands and iron-oxide coated sands, *Geophys. Res. Lett.*, *36*, L04403, doi:10.1029/2008GL036196.

Allen, J. P., E. A. Atekwana, E. A. Atekwana, J. W. Duris, D. D. Werkema, and S. Rossbach (2007), The microbial community structure in petroleum-contaminated sediments corresponds to geophysical signatures, *Appl. Environ. Microbiol.*, *73*(9), 2860–2870, doi:10.1128/AEM.01752-06.

Atekwana, E. A., and L. D. Slater (2009), Biogeophysics: A new frontier in Earth science research, *Rev. Geophys.*, *47*, RG4004, doi:10.1029/2009RG000285.

Atekwana, E. A., E. A. Atekwana, D. D. Werkema, J. P. Allen, L. A. Smart, J. W. Duris, D. P. Cassidy, W. A. Sauck, and S. Rossbach (2004a), Evidence for microbial enhanced electrical conductivity in hydrocarbon-contaminated sediments, *Geophys. Res. Lett.*, *31*, L23501, doi:10.1029/2004GL021359.

Atekwana, E. A., D. D. Werkema Jr., J. W. Duris, S. Rossbach, E. A. Atekwana, W. A. Sauck, D. P. Cassidy, J. Means, and F. D. Legall (2004b), In-situ apparent conductivity measurements and microbial population distribution at a hydrocarbon-contaminated site, *Geophysics*, *69*(1), 56–63, doi:10.1190/1.1649375.

Barrett, T. J., G. M. Anderson, and J. Lugowski (1988), The solubility of hydrogen sulphide in 0–5 m NaCl solutions at 25°–95°C and one atmosphere, *Geochim. Cosmochim. Acta*, *52*(4), 807–811, doi:10.1016/0016-7037(88)90352-3.

Berner, R. A. (1963), Electrode studies of hydrogen sulfide in marine sediments, *Geochim. Cosmochim. Acta*, *27*(6), 563–575, doi:10.1016/0016-7037(63)90013-9.

Bigalke, J., and E. W. Grabner (1997), The Geobattery model: A contribution to large scale electrochemistry, *Electrochim. Acta*, *42*(23–24), 3443–3452, doi:10.1016/S0013-4686(97)00053-4.

Che-Alota, V., E. A. Atekwana, E. A. Atekwana, W. A. Sauck, and D. D. Werkema Jr. (2009), Temporal geophysical signatures from contaminant-mass remediation, *Geophysics*, *74*(4), B113–B123, doi:10.1190/1.3139769.

Daily, W., and A. Ramirez (1995), Electrical resistance tomography during in-situ trichloroethylene remediation at the Savannah River Site, *J. Appl. Geophys.*, *33*(4), 239–249.

Davis, C. A., E. Atekwana, E. Atekwana, L. D. Slater, S. Rossbach, and M. R. Mormile (2006), Microbial growth and biofilm formation in geologic media is detected with complex conductivity measurements, *Geophys. Res. Lett.*, *33*, L18403, doi:10.1029/2006GL027312.

Hubbard, S. S., K. Williams, M. E. Conrad, B. Faybishenko, J. Peterson, J. Chen, P. Long, and T. Hazen (2008), Geophysical monitoring of hydrological and biogeochemical transformations associated with Cr (VI) bioremediation, *Environ. Sci. Technol.*, *42*(10), 3757–3765, doi:10.1021/es071702s.

Jeroschewski, P., K. Haase, A. Trommer, and P. Gründler (1993), Galvanic sensor for the determination of hydrogen sulphide/sulphide in aqueous media, *Fresenius J. Anal. Chem.*, *346*(10–11), 930–933, doi:10.1007/BF00322753.

Jeroschewski, P., C. Steuckart, and M. Kühl (1996), An amperometric microsensor for the determination of H_2S in aquatic environments, *Anal. Chem.*, *68*(24), 4351–4357, doi:10.1021/ac960091b.

Kühl, M., and B. B. Jørgensen (1992), Microsensor measurements of sulfate reduction and sulfide oxidation in compact microbial communities of aerobic biofilms, *Appl. Environ. Microbiol.*, *58*(4), 1164–1174.

Lawrence, N. S., J. Davis, and R. G. Compton (2000), Analytical strategies for the detection of sulfide: A review, *Talanta*, *52*(5), 771–784, doi:10.1016/S0039-9140(00)00421-5.

Ledin, M., and K. Pedersen (1996), The environmental impact of mine wastes—Roles of microorganisms and their significance in treatment of mine wastes, *Earth Sci. Rev.*, *41*(1–2), 67–108, doi:10.1016/0012-8252(96)00016-5.

Linde, N., A. Revil, A. Bolève, C. Dagès, J. Castermant, B. Suski, and M. Voltz (2007), Estimation of the water table throughout a catchment using self-potential and piezometric data in a Bayesian framework, *J. Hydrol.*, *334*(1–2), 88–98, doi:10.1016/j.jhydrol.2006.09.027.

Luther, G. W., III, T. F. Rozan, M. Taillefert, D. B. Nuzzio, C. Di Meo, T. M. Shank, R. A. Lutz, and S. C. Cary (2001), Chemical speciation drives hydrothermal vent ecology, *Nature*, *410*(6830), 813–816, doi:10.1038/35071069.

Maier, R. M., I. L. Pepper, and C. P. Gerba (2000), *Environmental Microbiology*, Academic, San Diego, Calif.

Minsley, B. J., J. Sogade, and F. D. Morgan (2007), Three-dimensional self-potential inversion for subsurface DNAPL contaminant detection at the Savannah River Site, South Carolina, *Water Resour. Res.*, *43*, W04429, doi:10.1029/2005WR003996.

Mirna, A. (1971), Sulfide analysis by the silver sulfide-silver electrode: Properties of the electrode, *Fresenius, Z. Anal. Chem.*, *254*(2), 114–116, doi:10.1007/BF01166126.

Naudet, V., A. Revil, J.-Y. Bottero, and P. Bégassat (2003), Relationship between self-potential (SP) signals and redox conditions in contaminated groundwater, *Geophys. Res. Lett.*, *30*(21), 2091, doi:10.1029/2003GL018096.

Naudet, V., A. Revil, E. Rizzo, J.-Y. Bottero, and P. Bégassat (2004), Groundwater redox conditions and conductivity in a contaminant plume from geoelectrical investigations, *Hydrol. Earth Syst. Sci.*, *8*(1), 8–22, doi:10.5194/hess-8-8-2004.

Ntarlagiannis, D., and A. Ferguson (2009), SIP response of artificial biofilms, *Geophysics*, *74*(1), A1–A5, doi:10.1190/1.3031514.

Ntarlagiannis, D., N. Yee, and L. Slater (2005a), On the low-frequency electrical polarization of bacterial cells in sands, *Geophys. Res. Lett.*, *32*, L24402, doi:10.1029/2005GL024751.

Ntarlagiannis, D., K. H. Williams, L. Slater, and S. Hubbard (2005b), Low-frequency electrical response to microbial induced sulfide precipitation, *J. Geophys. Res.*, *110*, G02009, doi:10.1029/2005JG000024.

21562202g, 2010, G3, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2009JG001157 by CochraneAustria, Wiley Online Library on [19/07/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Ntarlagiannis, D., E. A. Atekwana, E. A. Hill, and Y. Gorby (2007), Microbial nanowires: Is the subsurface "hardwired"?, *Geophys. Res. Lett.*, *34*, L17305, doi:10.1029/2007GL030426.

Nyquist, J. E., and C. E. Corry (2002), Tutorial—Self-potential: The ugly duckling of environmental geophysics, *Leading Edge*, *21*, 446–451, doi:10.1190/1.1481251.

Personna, Y. R., D. Ntarlagiannis, L. Slater, N. Yee, M. O'Brien, and S. Hubbard (2008), Spectral induced polarization and electrodic potential monitoring of microbially mediated iron sulfide transformations, *J. Geophys. Res.*, *113*, G02020, doi:10.1029/2007JG000614.

Prodan, C., F. Mayo, J. R. Claycomb, and J. H. Miller (2004), Low-frequency, low-field dielectric spectroscopy of living cell suspensions, *J. Appl. Phys.*, *95*(7), 3754–3756, doi:10.1063/1.1649455.

Ramsing, N. B., M. Kühl, and B. B. Jørgensen (1993), Distribution of sulfate-reducing bacteria, O_2, and H_2S in photosynthetic biofilms determined by oligonucleotide probes and microelectrodes, *Appl. Environ. Microbiol.*, *59*(11), 3840–3849.

Revil, A., D. Hermitte, M. Voltz, R. Moussa, J.-G. Lacas, G. Bourrié, and F. Trolard (2002), Self-potential signals associated with variations of the hydraulic head during an infiltration experiment, *Geophys. Res. Lett.*, *29*(7), 1106, doi:10.1029/2001GL014294.

Revsbech, N. P., B. B. Jørgensen, T. H. Blackburn, and Y. Cohen (1983), Microelectrode studies of the photosynthesis and O_2, H_2S, and pH profiles of a microbial mat, *Limnol. Oceanogr.*, *28*(6), 1062–1074, doi:10.4319/lo.1983.28.6.1062.

Revsbech, N. P., B. Madsen, and B. B. Jørgensen (1986), Oxygen production and consumption in sediments determined at high spatial resolution by computer simulation of oxygen microelectrode data, *Limnol. Oceanogr.*, *31*(2), 293–304, doi:10.4319/lo.1986.31.2.0293.

Rizzo, E., B. Suski, A. Revil, S. Straface, and S. Troisi (2004), Self-potential signals associated with pumping tests experiments, *J. Geophys. Res.*, *109*, B10203, doi:10.1029/2004JB003049.

Sato, M., and H. M. Mooney (1960), The electrochemical mechanism of sulfide self-potentials, *Geophysics*, *25*(1), 226–249, doi:10.1190/1.1438689.

Slater, L. D., and D. Lesmes (2002), IP interpretation in environmental investigations, *Geophysics*, *67*(1), 77–88, doi:10.1190/1.1451353.

Slater, L., D. Ntarlagiannis, N. Yee, M. O'Brien, C. Zhang, and K. H. Williams (2008), Electrodic voltages in the presence of dissolved sulfide: Implications for monitoring natural microbial activity, *Geophysics*, *73*(2), F65–F70, doi:10.1190/1.2828977.

Slater, L. D, F. D. Day-Lewis, D. Ntarlagiannis, M. O'Brien, and N. Yee (2009), Geoelectrical measurement and modeling of biogeochemical breakthrough behavior during microbial activity, *Geophys. Res. Lett.*, *36*, L14402, doi:10.1029/2009GL038695.

Stumm, W., and J. J. Morgran (1996), *Aquatic Chemistry, Chemical Equilibria and Rates in Natural Waters*, 3rd ed., John Wiley, New York.

Taillefert, M., G. W. Luther III, and D. B. Nuzzio (2000), The application of electrochemical tools for in situ measurements in aquatic systems, *Electroanalysis*, *12*(6), 401–412, doi:10.1002/(SICI)1521-4109(20000401)12:6<401::AID-ELAN401>3.0.CO;2-U.

Viollier, E., et al. (2003), Benthic biogeochemistry: State of the art technologies and guidelines for the future of in situ survey, *J. Exp. Mar. Biol. Ecol.*, *285–286*, 5–31, doi:10.1016/S0022-0981(02)00517-8.

Werkema, D. D., Jr., E. A. Atekwana, A. L. Endres, W. A. Sauck, and D. P. Cassidy (2003), Investigating the geoelectrical response of hydrocarbon contamination undergoing biodegradation, *Geophys. Res. Lett.*, *30*(12), 1647, doi:10.1029/2003GL017346.

Whitfield, M. (1971), A compact potentiometric sensor of novel design: In situ determination of pH, pS^{2-}, and Eh, *Limnol. Oceanogr.*, *16*(5), 829–837, doi:10.4319/lo.1971.16.5.0829.

Williams, K. H., D. Ntarlagiannis, L. D. Slater, A. Dohnalkova, S. S. Hubbard, and J. F. Banfield (2005), Geophysical imaging of stimulated microbial biomineralization, *Environ. Sci. Technol.*, *39*(19), 7592–7600, doi:10.1021/es0504035.

Williams, K. H., S. S. Hubbard, and J. F. Banfield (2007), Galvanic interpretation of self-potential signals associated with microbial sulfate-reduction, *J. Geophys. Res.*, *112*, G03019, doi:10.1029/2007JG000440.

Williams, K. H., A. Kemna, M. J. Wilkins, J. Druhan, E. Arntzen, A. L. N'Guessan, P. E. Long, S. S. Hubbard, and J. F. Banfield (2009), Geophysical monitoring of coupled microbial and geochemical processes during stimulated subsurface bioremediation, *Environ. Sci. Technol.*, *43*(17), 6717–6723, doi:10.1021/es900855j.

R. Doherty, Environmental Engineering Research Centre, School of Planning, Architecture and Civil Engineering, Queen's University of Belfast, Belfast BT9 5AG, UK.

D. Ntarlagiannis, L. Slater, and C. Zhang, Department of Earth and Environmental Sciences, Rutgers, State University of New Jersey, Newark, NJ 07102, USA. (chizhang@pegasus.rutgers.edu)

21562202g, 2010, G3, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2009GC001157 by CochraneAustria, Wiley Online Library on [19/07/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

01_张弛老师第一作者语言样本.pdf

文献 2/6

Spectral Induced Polarization Signatures of Hydroxide Adsorption and Mineral Precipitation in Porous Media

Chi Zhang; Lee Slater; George Redden; Yoshiko Fujita; Timothy Johnson; Don Fox

Environmental Science & Technology (2012); DOI: 10.1021/es204404e

证据用途：第一作者直接语言样本；张弛老师角色：第一作者/通讯作者；证据等级：A

Environmental Science & Technology

Article

pubs.acs.org/est

Spectral Induced Polarization Signatures of Hydroxide Adsorption and Mineral Precipitation in Porous Media

Chi Zhang,*[,†] Lee Slater,[†] George Redden,[‡] Yoshiko Fujita,[‡] Timothy Johnson,[§] and Don Fox[‡]

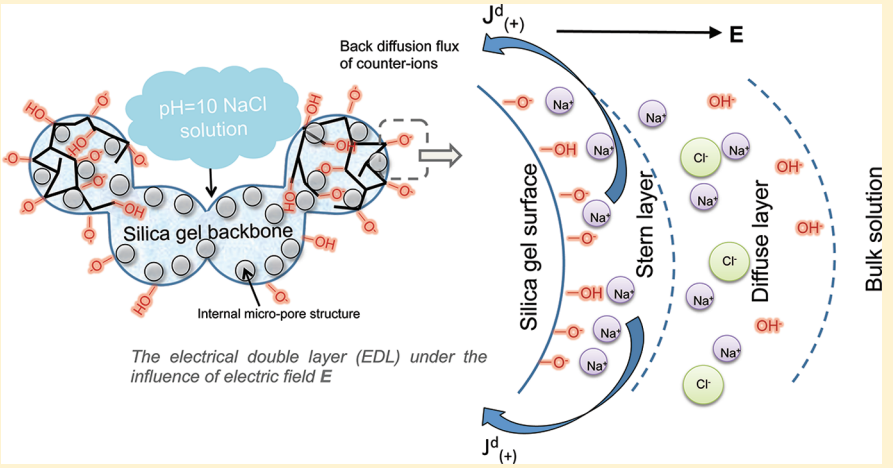
[†]Rutgers University-Newark, Newark, New Jersey 07102, United States

[‡]Idaho National Laboratory, Idaho Falls, Idaho 83415, United States

[§]Pacific Northwest National Laboratory, Richland, Washington 99352, United States

Supporting Information

Downloaded via ZHEJIANG UNIV on July 20, 2026 at 02:28:47 (UTC).
See https://pubs.acs.org/sharingguidelines for options on how to legitimately share published articles.



ABSTRACT: The spectral induced polarization (SIP) technique is a promising approach for delineating subsurface physical and chemical property changes in a minimally invasive manner. To facilitate the understanding of position and chemical properties of reaction fronts that involve mineral precipitation in porous media, we investigated spatiotemporal variations in complex conductivity during evolution of urea hydrolysis and calcite precipitation reaction fronts within a silica gel column. The real and imaginary parts of complex conductivity were shown to be sensitive to changes in both solution chemistry and calcium carbonate precipitation. Distinct changes in imaginary conductivity coincided with increased hydroxide ion concentration during urea hydrolysis. In a separate experiment focused on the effect of hydroxide concentration on interfacial polarization of silica gel and well-sorted sand, we found a significant dependence of the polarization response on pH changes of the solution. We propose a conceptual model describing hydroxide ion adsorption behavior in silica gel and its control on interfacial polarizability. Our results demonstrate the utility of SIP for noninvasive monitoring of reaction fronts, and indicate its potential for quantifying geochemical processes that control the polarization responses of porous media at larger spatial scales in the natural environment.

INTRODUCTION

Proposed approaches for in situ remediation of groundwater contamination include induced mineral precipitation schemes where contaminants are immobilized via coprecipitation or retarded by flowpath diversion. Successful application of such remediation strategies requires understanding of evolution of the relevant reaction fronts, and of changes in media properties due to precipitation. Given the transience of precipitation reaction fronts, the complexity of the reaction networks, and the challenge of accurately capturing potentially widely dispersed changes using traditional point sampling, novel monitoring approaches that sense both media property changes and temporal reaction fronts are needed to support both laboratory and field investigations.

Spectral induced polarization (SIP) is a promising geophysical technique for delineating variations in the properties of solid−fluid and fluid−fluid interfaces in porous media in a minimally invasively manner.[1−4] Recently SIP has been applied for spatiotemporal monitoring of biogeochemical processes in both laboratory and field settings.[2,3,5−7] Applications include monitoring of metal and nutrient cycling by sulfate-reducing microbes,[2,6,8] iron corrosion,[9,10] and calcite precipitation.[11−13] A recent review[4] describes how SIP has the potential to provide information on microbial growth, biofilm formation, and

Received: December 9, 2011
Revised: March 13, 2012
Accepted: March 15, 2012
Published: March 15, 2012

 © 2012 American Chemical Society

biomineralization in porous media. The strong polarization of the metal/electrolyte interface results in large SIP signatures when metallic mineral phases (e.g., FeS) are present.[2,3,6] However, the electrical properties of nonmetallic minerals and associated reaction fronts have been less studied. In one recent experiment,[11] significant SIP signals were recorded during calcite precipitation on glass beads induced by mixing of $CaCl_2$ and Na_2CO_3 solutions. These previous studies suggest that SIP may be a valuable technique for noninvasive monitoring of chemical reaction fronts associated with in situ mineral precipitation.

We describe an experiment to evaluate the ability of SIP to monitor the spatiotemporal evolution of processes associated with calcite precipitation driven by urea hydrolysis within a silica gel column, catalyzed by immobilized extracellular urease enzyme. This model system is an abiotic analog to an approach that has been proposed for remediation of contaminants that can be coprecipitated in calcite, where ureolysis is mediated by indigenous subsurface microorganisms.[14−16] The two primary reactions capable of changing interfacial electrical properties in this system are

$$(NH_2)_2CO(\text{urea}) + 3H_2O \xrightarrow{\text{urease}} 2NH_4^+ + HCO_3^- + OH^- \quad (1)$$

and

$$HCO_3^- + Ca^{2+} + OH^- \rightarrow H_2O + CaCO_3 \quad (2)$$

To aid interpretation of the urease column experimental results, we conducted an additional study focused on the reactivity of hydroxide with silica gel, and specifically to examine whether (1) hydroxide adsorption affects SIP signals, and/or (2) changes in SIP signals can be used to discern between varying reaction fronts at solid−solution interfaces. Our study could improve interpretation of geophysical data collected for monitoring of in situ mineral precipitation.

Spectral Induced Polarization. SIP measures the impedance magnitude and phase shift (φ) of a received sinusoidal voltage across a sample relative to the current waveform over a range of frequencies (typically <1 kHz). The magnitude and phase (φ) are converted to a complex conductivity ($\sigma^*(\omega)$) using the physical geometry of the sample. The dependence of $\sigma^*(\omega)$ on geologic matrix properties such as pore/grain size, permeability, and surface area has been extensively explored.[17−20]

The measured complex conductivity $\sigma^*(\omega)$ of a system can be expressed as

$$\sigma^*(\omega) = \sigma'(\omega) + i\sigma''(\omega) \quad (3)$$

The real conductivity (σ') represents conduction (energy loss), whereas imaginary conductivity (σ'') represents polarization (energy storage).

$$\sigma' = |\sigma|\cos\varphi \quad (4)$$

$$\sigma'' = |\sigma|\sin\varphi \quad (5)$$

where $|\sigma|$ is the conductivity magnitude calculated using the impedance magnitude and appropriate geometric factors.

At frequencies <1 kHz, $\sigma^*(\omega)$ of a saturated porous material in nonmetallic environments can be modeled as the sum of an electrolytic conduction (σ_{ele}) within the fluid-filled interconnected pore space and a surface conduction (σ^*_{surf}) occurring within an electrical double layer (EDL) at the grain−fluid interface.[21] It is usually assumed[22,23] that the electrolytic and surface conduction paths add in parallel such that

$$\sigma' = \sigma_{ele} + \sigma'_{surf} \quad (6)$$

and

$$\sigma'' = \sigma''_{surf} \quad (7)$$

The electrolytic conductivity σ_{ele} is expressed as $\sigma_{ele} = (1/F) \times \sigma_f$, where σ_f is fluid conductivity, and F is the electrical formation factor. Assuming electrical conduction is only through electrolyte-filled pores, then F describes the decrease in conductivity of the rock due to the presence of the insulating mineral grains.[24]

Mechanistic models describing σ^*_{surf} have recently been developed, and relationships between microgeometry (grain-size distributions, pore diameter, and pore-size distributions) and SIP signals are reported, mostly for a single rock formation.[25−27] However, the role of fluid chemistry was until recently overlooked. Vaudelet et al.[28] recently examined the SIP signals measured during advective flow of a $CuSO_4$ solution in a sand column, and developed a speciation model for adsorption of sodium and copper onto the silica grain surface. Skold et al.[29] introduced a similar model, but included a proton hopping process, which allowed predictions of SIP measurements made with varying pH and salinities. Such developments enable quantitative interpretation of specific sorption reactions from SIP data sets. However, these models are challenging to calibrate for complex geochemical systems.

Phenomenological models, such as the Cole−Cole model,[30] can instead be applied to observed SIP responses of porous media. Although these models are not directly parameterized in terms of microgeometric and/or geochemical parameters, parameters such as the chargeability and relaxation time constant are related to microgeometry[26,31] and pore fluid chemistry.[21] A popular variant of the Cole−Cole model can be written as[30,32]

$$\sigma^*(\omega) = \sigma_0\left[1 + m\left(\frac{(i\omega\tau)^c}{1 + (i\omega\tau)^c(1-m)}\right)\right] \quad (8)$$

where σ_0 is the DC-conductivity, m is the chargeability describing the magnitude of polarization associated with interfacial charge storage, τ is the characteristic relaxation time constant related to characteristic pore or grain size,[33] and c is a shape parameter (0.2−0.8 for unconsolidated soils). Integrating parameters from a Debye decomposition have also been studied to quantify the dependence of polarizability on pore fluid composition in sandstone samples.[34] The τ for polarization of the EDL surrounding a spherical grain of diameter d (m) is defined by Schwarz[35] as

$$\tau = \frac{d^2}{8D_s} = \frac{d^2}{8\beta_s kT/e} \quad (9)$$

where β_s is the effective ion mobility (in $m^2\ s^{-1}\ V^{-1}$), k is Boltzmann's constant (1.381×10^{-23} JK^{-1}), T is absolute temperature (K), and $D_s = \beta_s kT/e$ is the surface diffusion coefficient (in $m^2\ s^{-1}$). This model predicts that τ is proportional to square of grain size and inversely proportional to ion mobility.

MATERIALS AND METHODS

Experiment 1: Urease Mediated Mineral Precipitation. Silica gel (Sigma-Aldrich 236802, grade636) was used as the porous media due to its ability to immobilize urease enzyme on its external surface. The particle size was 250−500 μm, surface area was 480 m^2/g, intragranular porosity was 38%, and the internal pore diameter was 60 Å. Prior to packing, the silica gel was combined with 50 mM NaCl in a slurry, degassed under vacuum, and titrated with NaOH to neutral pH. Jackbean urease (Sigma-Aldrich, 091K7046) was adsorbed onto the silica gel by slowly stirring enzyme dissolved in 50 mM NaCl with the silica gel slurry for 1 h. The enzyme was expected to be at least twice the pore size[36] of the silica gel and therefore assumed to reside only on the external surface. The silica gel was packed into a polycarbonate (Lexan) column (48 cm, inner diameter 5.79 cm), with the urease-coated section restricted to a 16.2-cm interval midway through the column (Figure 1).

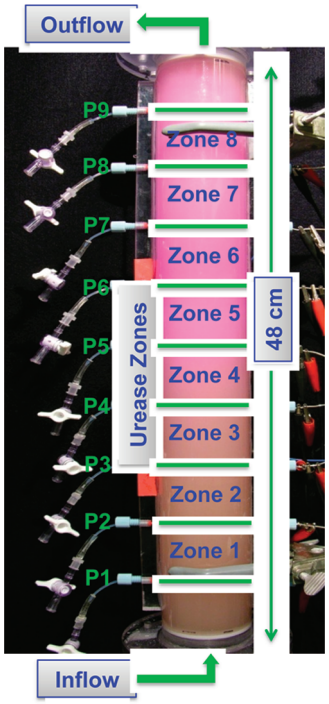


Figure 1. Experiment 1. Photograph of urea hydrolysis mediated mineral precipitation experiment. The urease enzyme has been immobilized 6 mm upstream of Zone 3 to 4 mm downstream of Zone 5. As the inflow carried urea into the column, the urea was hydrolyzed by urease and hydroxide ions were produced. The pH indicator turned pink where the pH was >8.2 in the urease and posturease zones.

The column was operated in an upflow configuration. It was first flushed with 50 mM NaCl background solution (all influent solutions mentioned hence contain 50 mM NaCl) for two pore volumes (PV; 1 PV = 916 mL). This was followed by a bromide tracer test (22.5 mM LiBr; 20 mL/min). After removal of the bromide tracer, a 10 mM urea solution was introduced into the column at 10 mL/min for 544 min (Stage 1). Next a solution containing urea (10 mM), $CaCl_2$ (10 mM), and $SrCl_2$ (0.1 mM) was injected at 10 mL/min for 4400 min (Stage 2). The pH indicator phenol red (0.005 mM) was added to all influent solutions to facilitate visual observation of pH changes using an automated image capture system. The dye transitions from yellow to red over pH 6.8 to 8.2, and becomes bright pink at pH >8.2.

Local solution chemistry was monitored by extracting fluid samples from ports (P1−P9) installed at 5-cm intervals along the column (Figure 1). The pH and fluid conductivity (σ_f) were measured immediately in collected samples. The Na^+, NH_4^+, and urea concentrations in acid-preserved samples were measured by ion chromatography, and the distribution of solid Ca and Sr at the end of Stage 2 was determined by inductively coupled plasma mass spectrometry (ICP-MS) analysis of excavated samples (digested in 10% HNO_3).

Nine Ag-AgCl potential electrodes were installed in electrolyte-filled chambers on the side of the column opposite the sampling ports. Spiral Ag-AgCl current electrodes were installed on both end-caps with the same inner diameter as the column to facilitate homogeneous electric flow. Real time SIP measurements were performed with a full waveform, multichannel broadband complex resistivity instrument, with an error of 0.1 mrad for φ and less than 1% for $|\sigma|$.[13,37] Nine measurement channels were available; one was designated to record the current waveform by measuring the response across a precision reference resistor in series with the column, and the remaining channels recorded the potential response on adjacent electrode pairs (Zone 1 to Zone 8; Figure 1). Zones 3, 4, and 5 were within the urease interval. SIP measurements were collected every 2 min without flow interruption using acquisition software described in Versteeg et al.[38] Measurements were acquired at 100 Hz for Stage 1, and over the frequency range 0.1−1 kHz for Stage 2, over a total duration of 1390 min, spanning Stage 1 and early Stage 2.

Experiment 2: Hydroxide Ion Adsorption Experiment. A Lexan column design similar to that for Experiment 1 was used in this experiment, except that 17 nonpolarizing Ag-AgCl electrodes (E1-E17) were installed at 2.5-cm intervals (Figure SI1). The first and last 15.7 cm of the column were packed with Ottawa sand (0.300−0.212 mm), whereas the intervening section (16.6 cm) was packed with the same type of silica gel (prepared as described previously; no urease) used in Experiment 1. The Ottawa sand/silica gel interfaces were centered on electrodes 6 and 12 (Figure SI1). The column was again operated in upflow mode and flushed with 50 mM NaCl (pH = 7, σ_f = 0.51 S/m) until stable pH and σ_f was achieved. Porosity of the sand was calculated to be 39.5% from wet and dry weight measurements. The effective porosity of the silica gel was estimated at 72.6% based on the Experiment 1 tracer test.

This experiment was conducted over 9 days with intermittent injection of 50 mM NaCl (constant σ_f = 0.51 S/m) at 10 mL/min. For the first 13 h on both Day 0 and Day 1, the injected solution was adjusted to pH 7. It was then replaced by the same solution adjusted to pH 8 on Day 2 for 8 h. From Day 3 until the end of the experiment (Day 8), the solution was adjusted to pH 10 and flowed through the column for a total of 40 h (6−8 h per day). Phenol red (0.005 mM) was included in all solutions, and pH adjustments were made using 0.3 M NaOH (all influent solutions were unbuffered). The influent solutions were housed in pyrex vessels closed with rubber stoppers fitted with ascarite traps to mitigate acidification by atmospheric CO_2. All tubing was wrapped with parafilm to reduce CO_2 diffusion. This experiment was performed at room temperature (22.6 ± 0.6 °C).

The σ_f and pH were monitored daily in the influent, effluent, and samples extracted from ports P1−P9. A two-channel

dx.doi.org/10.1021/es204404e | Environ. Sci. Technol. 2012, 46, 4357−4364

dynamic signal analyzer (National Instruments 4461) was employed to record SIP signatures over a frequency range from 0.1 Hz to 1 kHz at 100 logarithmic intervals on a daily basis. The measurement error is less than 1% for $|\sigma|$ and ≈0.1 mrad for φ below 100 Hz. The current was injected through the coiled Ag-AgCl electrodes at both ends of the column and the resulting potential differences were recorded between Ag-AgCl electrode pairs spanning both sand zones (E1−E3, E3−E5, E13−E15, and E15−E17) and the silica gel zone (E6−E9 and E9−E12). The magnitude and φ were measured relative to a known high-precision resistor on channel 1.

Spectral Induced Polarization Data Modeling. The SIP data sets from Experiment 2 were modeled with eq 8 using a two-step procedure. First, a Bayesian model using a Markov-chain Monte Carlo (MCMC) based method was used to obtain the medians of unknown parameters by starting from an arbitrary set of initial values. This step provides marginal probability distributions but not explicitly optimal solutions.[39] The medians of the parameter estimates were then used as the initial values for an iterative least-squares deterministic method with Marquardt regularization.[40] However, there were uncertainties and nonuniqueness in the parameter estimation. For example, the correlation matrix reveals that m is highly correlated with τ, the average correlation coefficient between these two parameters being 0.67 for measurements in silica gel. This problem is frequently experienced, especially for small measurement signals, and is not easily overcome.

RESULTS AND DISCUSSION

Experiment 1: Urease Mediated Mineral Precipitation. Scanning electron microscopy (SEM) images of solids excavated from another column with similar experimental conditions after the conclusion of the experiment (Figure SI2) showed that crystals ($CaCO_3$ composition inferred by energy dispersive spectroscopy) were produced at grain−grain contacts with a length scale of 100−200 μm. The ICP-MS analyses showed that mineralized Ca^{2+} reached 0.15 mmol/g in the urease zones, and that precipitation continued downstream, with decreased mineralized Ca^{2+} concentrations at the end of experiment (Figure 2).

Fluid Conductivity (σ_f) and Real Conductivity (σ'). The hydrolysis of urea, with accompanying changes in solution chemistry and the subsequent formation of carbonate precipitates, was reflected in the SIP measurements. The σ' and the σ_f data at different locations (Zones 1, 5, and 7) along the column are shown in Figure 3a, and the data for all zones (Zones 1−8) are plotted in Figure SI3. Note that for the individual zones, the measured pH and σ_f values at the two adjacent sampling ports were averaged to assist comparison with $\sigma^*(\omega)$.

In preurease zones (Zone 1 as representative), σ_f was steady at 0.531 S/m during Stage 1 and σ' also remained steady (≈ 0.33 S/m). During Stage 2, the σ_f increased to 0.7 S/m due to the introduction of Ca^{2+} and Sr^{2+} into the column, and remained elevated through the experiment. Similarly, the σ' also increased during Stage 2 to ≈0.42 S/m. The changes in σ' and σ_f in the preurease zone were directly proportional throughout the experiment, consistent with a dominant control of electrolytic conduction (σ_{ele}) on σ', and it resulted in a formation factor of 1.64.

In the urease zones ureolysis produces NH_4^+, HCO_3^-, and OH^- (eq 1), which will increase both σ_f and σ'. In Zone 5 (as representative of the urease zone), σ_f increased to 0.62 S/m but

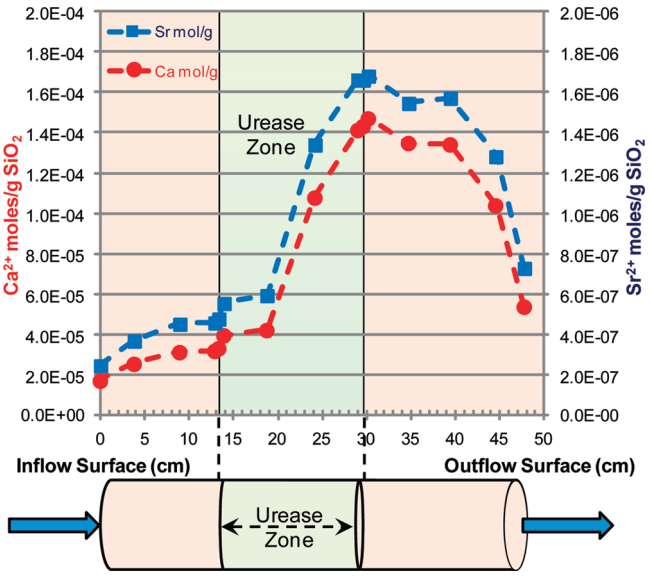


Figure 2. Experiment 1. Mineralized Ca and Sr in the column from destructive sampling of the solid media at the end of the experiment.

decreased later at the end of Stage 1 due to possible downstream migration of enzyme, which would have lowered the activity within the original urease zones. Urea arrival in Zone 5 during Stage 1 was evident in σ' data as a consequence of the production of ionic species at the beginning of Stage 1. Afterward, the σ' in Zone 5 increased significantly and reached a plateau at ≈0.52 S/m toward the end of Stage 1. With the onset of Stage 2, σ_f initially increased then dropped thereafter. The increase in σ' at the beginning of Stage 2 was associated with the arrival of Ca^{2+} and Sr^{2+}. Although changes in pore space and σ_f in the urease zones during Stage 2 were very small, apparent formation factor calculated from the ratio of σ_f to σ' increased after the precipitation, consistent with a decrease in σ^*_{surf} as confirmed by σ'' measurements. This was presumably due to lower surface charge and lower surface area of $CaCO_3$ precipitation relative to silica gel.[41]

In posturease zones (using Zone 7 as representative), the σ_f changes were consistent with those observed in urease zones during Stage 1 except that the σ_f values were higher (≈ 0.64 S/m) due to arrival of ions generated from upstream ureolysis. The correlation between σ_f and σ' is strong (R^2 = 0.96), and the σ' first increased when Stage 1 was initiated and remained relatively stable initially, although it started gradually increasing during the latter part of Stage 1. This may have been due to migration/spreading of the urease zone as a consequence of incomplete enzyme immobilization. With the onset of Stage 2, the σ' increase was much steeper before decreasing after 860 min. Qualitatively, the trends in σ_f, σ', and pH were consistent with the migration of ionic species (background electrolyte and reaction products) coupled with ion depletion by $CaCO_3$ precipitation.

The pH and Imaginary Conductivity (σ''). The σ'' data compared with pH changes in Zones 1, 5, and 7 are plotted in Figure 3b, and the data in all zones are shown in Figure SI3 (the φ data are also plotted in the same manner in Figure SI4). The pH in Zone 1 was steady around 7.5, and σ'' was ≈0.00072 S/m and remained steady during the urea injection. With Stage 2, the pH dropped in the preurease zones due to pH changes in the influent, and the σ'' increased slightly in Zone 1.

dx.doi.org/10.1021/es204404e | Environ. Sci. Technol. 2012, 46, 4357−4364

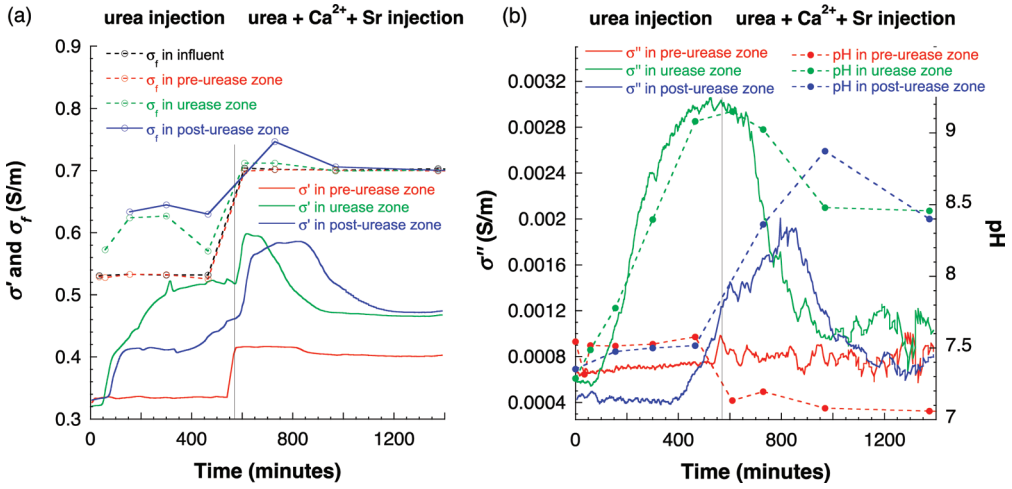


Figure 3. Experiment 1. (a) Real (σ') and fluid conductivities (σ_f) and (b) imaginary conductivities (σ'') and pH changes in the preurease (Zone 1 as representative), urease (Zone 5 as representative), and posturease zones (Zone 7 as representative) during ureolytically driven mineral precipitation experiment. The urea injection stage occurred from 0 to 544 min (Stage 1), followed by the coinjection of urea, Ca^{2+}, and Sr^{2+} (Stage 2) began until the end of the experiment, and the two stages are separated by the vertical line. The σ^*_{surf} data are from 100 Hz.

An important observation in the urease zones during Stage 1 is the correlation between pH and σ'' ($R^2 > 0.98$). The pH increases due to ureolysis were observed first in Zone 3, then 4 and 5, and the σ'' also increased in Zones 3 to 5 in that order. The polarization magnitude in urease zones was much higher than that in the pre- and posturease zones. The σ'' in Zone 5 increased to a peak of 0.003 S/m close to the end of Stage 1. Early in Stage 2, the pH decreased due to proton production from calcium carbonate precipitation. The σ'' dropped in all urease zones during Stage 2, coincident with the carbonate mineral precipitation. The variations of phase data (Figure SI4) were similar to σ'' data, and also associated with pH changes in urease zones.

In posturease zones, the pH was stable ($\approx$ 7.5) during Stage 1, but increased during Stage 2. The pH in Zone 7 peaked at 950 min and then decreased by the end of the experiment. This can be explained by the migration of OH^- formed during Stage 1 that continued after the start of stage 2. During Stage 1, the σ'' was much smaller ($\approx$ 0.0004 S/m) in the posturease zones than in the urease zones. It increased slightly in Zone 7 near the end of Stage 1. After Stage 2 started, the σ'' in Zones 7 first increased, but then fell again, although it was still higher than the initial Stage 1 value at the end of the experiment. The σ'' changes coincided with changes in σ' in these three posturease zones.

Polarization Mechanisms. In this highly porous and high surface area medium, adsorption of ions on the surface of silica gel is significant. Ion or charge accumulation at the surface would presumably contribute to an increase in surface polarization, and therefore the σ''. The OH^- adsorption, for example, would increase the negative surface charge, which would then attract counterions (i.e., H^+, Na^+, NH_4^+, Ca^{2+}, Sr^{2+}) into the Stern layer.

During Stage 1, σ^*_{surf} has risen as indicated by increases in σ''. This could be attributed to an accumulation of surface charge resulting from adsorption of OH^- or other ions. Similarly, both σ' and σ'' gradually reduced without significant changes in σ_f in the downstream direction during Stage 2, suggesting that σ^*_{surf} decreased as $CaCO_3$ precipitation occurred and pH decreased. These observations indicate that σ^*_{surf} (not σ_{ele}) primarily controls the SIP responses to $CaCO_3$ precipitation.

Mineral precipitation can result in changes in surface area, formation porosity, and pore throat diameter, with significant impact on SIP signals.[4] The electrical signals from precipitation of nonmetallic minerals such as calcite are small and associated with changes in pore volume/pore tortuosity and/or surface area/surface roughness.[4] However, both SEM image and tracer test suggest porosity change during Stage 2 was insignificant ($\approx$1%) as the precipitation of $CaCO_3$ was localized at the grain−grain contacting points, indicating the contribution of pore space to decreased σ^*_{surf} is negligible. Instead, the $CaCO_3$ precipitation has lower surface area relative to silica gel, and presumably reduced the total surface charge by forming a less negatively charged surface on the silica gel in the pH range 6−10.[41] In addition, calcium adsorption on silica gel is strong and would also lower the total surface charge.

Our observations can be compared to those of Wu et al.,[11] who conducted a column experiment with 3-mm glass beads and observed increased polarization responses during calcite precipitation. They suggested that the increase of σ'' during a discrete precipitation phase was caused by an increase in total surface area, due to the accumulation of small, well distributed, discrete calcite particles (average size <5 μm) relative to the glass beads. They specifically attributed the SIP response to polarization of the EDL surrounding calcite crystals. In our experiment, the porous matrix has much higher specific surface area than glass beads, and the surface area change caused by $CaCO_3$ precipitation in silica gel is likely negligible. However, $CaCO_3$ should have lower surface charge than silica gel,[41,42] resulting in a decrease of σ''. In a recent ureolytically driven $CaCO_3$ precipitation study using natural sediment materials,[12] an increase of σ'' was not observed during the precipitation, but a positive correlation between σ_f and σ'' was observed during early urea hydrolysis, and was attributed to production of NH_4^+ and adsorption of hydroxyl ions on the mineral surface.

Experiment 2: Hydroxide Ion Adsorption Experiment. *The pH Profile.* Figure 4 shows the pH profile along the column in Experiment 2. As the influent pH increased from 7 to 10, the pH of samples from the bottom sand section (P1−P3, Figure SI1) rose to values close to the influent pH values. The slightly lower pH in P1−P3 relative to the influent was likely due to buffering by the sand matrix. In the silica gel

dx.doi.org/10.1021/es204404e | Environ. Sci. Technol. 2012, 46, 4357−4364

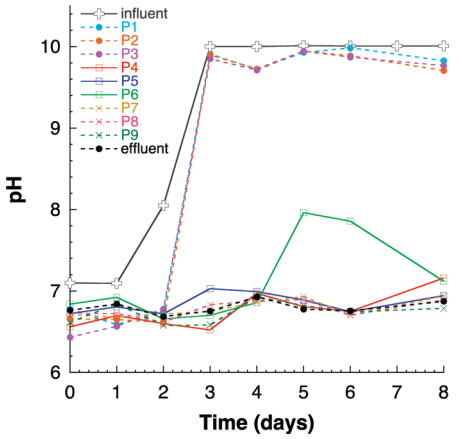


Figure 4. Experiment 2. The pH profile in the hydroxide adsorption experiment. The change of pH was monitored by extracting fluid samples at the 9 sampling ports as well as in the influent and effluent tubing.

section (P4–P6, Figure SI1), the pH remained at values similar to the initial influent throughout the experiment, except for the pH at P6 on Day 5 and 6. There was no observable pH change in the upper sand section (P7–P9, Figure SI1) as the influent pH increased. The pH profile in the column suggests that OH^- was buffered by (adsorbed by) the silica gel and the buffer capacity had not been exceeded by the end of the experiment.

Imaginary Conductivity (σ″) Dependence on pH. The spectral σ'' changes in silica gel and in the upstream sand section from each day are shown in Figure 5a. In the silica gel, the σ'' in the first two days were similar. When the influent pH increased on Day 2, σ'' also increased. The σ'' increased further when the influent pH increased to 10 on Day 3. By Day 4 the rate of increase had slowed and it reached a maximum near 1 Hz on Day 8. The characteristic critical frequencies (peak frequencies) for all of the SIP measurements were approximately 0.9 Hz.

The σ'' in the sand sections were much weaker (one magnitude lower compared to silica gel); only the signals from the upstream section are shown as the measurements in the downstream section were below the noise threshold (≈ 0.1 mrad). With the influent pH increase, the σ'' in the sand first increased, but then decreased on Days 7 and 8. There were no consistent variations in the characteristic critical frequency. The spectral phase (φ) changes and pH dependence in both silica gel and sand are shown in Figure SI5.

Figure 5b shows the σ'' changes in silica gel and sand over time at the characteristic critical frequency (0.9 Hz), normalized chargeability m_n ($m_n = m \times \sigma_0$) in silica gel over time and compared with influent pH. The σ'' was stable at 0.0009 S/m for the first 2 days, increasing by about 10% when the influent pH increased from 7 to 8. When the influent pH changed to 10, the σ'' increased and reached a plateau by Day 4, while the pH of the effluent stabilized around 6.8 throughout the rest of the experiment. The behavior of σ'' suggests that OH^- adsorbed onto silica gel after Day 4 did not contribute further to the polarization. The trend for σ'' (0.9 Hz) was very similar to the m_n changes over time in silica gel, but the value of m_n was 4–5 fold higher than σ'', since the σ'' here represents a single frequency whereas m_n is the integrated measurement of charge storage over the investigated frequency range. The increased adsorption of OH^- on the silica gel presumably caused the increase of m_n, which was apparent from Day 3 when the pH of the influent increased to 10. The m_n decreased on Day 6 and stayed relatively stable thereafter. The σ'' responses in the sand were much smaller than in the silica gel, and differences between the σ'' values over time were insignificant.

We attribute the much weaker signals observed in the sand to the much smaller total surface area to pore volume ratio compared to silica gel. With similar particle size (300–212 μm) but no internal porous structure, the sands have surface area <0.1 m^2/g. In contrast, the surface area of the silica gel is approximately 480 m^2/g, which includes internal porosity. The σ^*_{surf} is primarily controlled by the surface area, surface charge density, and surface ionic mobility.[18,19,21] In Experiment 2 with constant ionic strength and specific surface area, the surface charge density and effective mobility of ions in the Stern and diffuse layers of the EDL primarily determined the magnitude of the σ^*_{surf} responses. Lesmes and Frye[21] investigated the influence of pore fluid ionic strength and pH upon the SIP responses of sandstone. They reported that σ'' was dependent on pH, being reduced at pH 3, which is near the point of zero charge (pzc) of silica. In the case of silica minerals, the

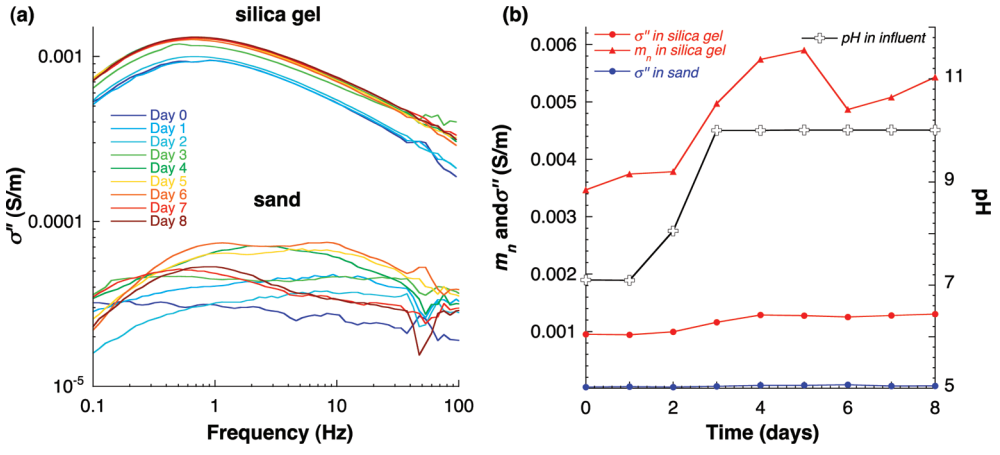


Figure 5. Experiment 2. (a) Spectral imaginary conductivities (σ'') from 0.1 to 100 Hz in both silica gel and sand over the experimental duration. (b) Cole–Cole model parameter normalized chargeability m_n for silica gel, the σ'' at critical frequency (0.9 Hz) for both silica gel and sand, and the pH of the influent. Data are from measurements between electrode pair 6 and 9 in the silica gel, and pair 1 and 3 in the sand.

dx.doi.org/10.1021/es204404e | Environ. Sci. Technol. 2012, 46, 4357–4364

experimental data as well as model predictions of Leroy et al.[43] indicated that the surface charge density increased with increasing pH (pH > pzc) at salinity between 10^{-1} and 10^{-3} M. Skold et al.[29] also showed increased σ'' as pH increased above the pzc in sand saturated by 0.01 M NaCl.

Cole–Cole Parameters m and τ. Figure SI6a shows the change of the m_n and τ over time in Experiment 2. The m_n was well resolved from the model in both sand and silica gel. The m_n for sand was significantly smaller than that for silica gel, and it tracked the change in σ'' as expected (both being measures of the polarization magnitude). The τ in the silica gel had a pattern similar to m_n, slowly increasing to a maximum on Day 5, decreasing thereafter. It is reasonable to interpret changes in τ to changes in β_s (eq 9). However, this would imply an increase in β_s after Day 5, which is unreasonable given continued accumulation of charge at the surface. Furthermore, the σ'' stayed stable after Day 5, consistent with a maximum surface charge density on the surface sites. Therefore, the significance of changes in τ is uncertain, and possibly complicated by the nonuniqueness of this parameter. In the following section, we propose a conceptual model for OH^- adsorption and polarization behavior in silica gel that could explain this plateau.

Model for Hydroxide Adsorption on Silica Gel. Figure SI7 is a conceptual representation for the formation of surface charge on silica gel resulting from reactions with OH^-. The silica gel matrix consists of silicon atoms joined by oxygen atoms in siloxane bonds. On contact with water, hydration of the silica surface initially produces multiple types of surface hydroxyl groups, and the surface of SiO_2 has a net negative change above pH 2.5–3. Addition of OH^- drives the deprotonation reaction forward and increases the concentration of "fixed" negative surface (often described as OH^- adsorption). Formation of these fixed charge results in the formation of the EDL on the surface of primary backbones of silica gel, which can be modeled as a consisting of a Stern layer of mobile counterions that move tangentially along the surface and an outer diffuse layer. The arrangement of surface charges in the EDL with varying mobilities contributes to the SIP effect in the presence of an external electric field. We neglect the specific adsorption of Cl^- anions at the surface since that usually occurs at pH < 3[43] where the net charge is close to zero, but where positively charged (protonated) sites can be present.

The OH^- ions are absorbed not only on the surface of the primary backbone, but possibly also in the numerous inner micropore structures. However, due to the small size of these pore structures and subsequent condensation reactions in silica gel, transport of Na^+ (a counterion of OH^-) may be restricted (Na^+ hydrated radius 0.79×10^{-9} m) and will have less impact on the SIP signatures. We assume that the accumulation of OH^- happened at a later stage (after Day 4) in our study, and this explains the plateau in σ'' since the excess negative charges are being stored in the internal structure. Because the chargeability m_n relates to the polarizable interfacial charge storage, the behavior of m_n (Figure 5b) is consistent with our conceptual model that the accumulation of OH^- at the later stage did not contribute to the polarizability.

Similarly to the observations in Experiment 1, we attribute the polarization responses observed in Experiment 2 to electrochemical polarization occurring at the EDL. The EDL polarization involves charge movement in both the fixed and diffused layers, and pH has an important influence on the polarization. The size of the mineral particle is fixed, so the relaxation time is determined primarily by charge density and charge mobility in the EDL.[17,35] We assert that the observed dependence of polarization on pH may reflect the relative contributions of charge density to the net surface conductivity response of the sample. Our conceptual model proposes that the increase of OH^-onto the primary silica gel surface will increase accumulation of Na^+, increasing the σ^*_{surf}. In reality, various surface conduction and competing physicochemical mechanisms likely combine to yield the net surface conductivity of the sample. To test the relative importance of the proposed OH^- adsorption behavior on the silica gel as illustrated in the conceptual model, further studies should analyze the inner structure of the material and characterize ion adsorption behavior.

We have shown that the SIP technique is sensitive enough to capture surface chemistry and associated interfacial properties of a solid–fluid surface. Our SIP measurements in the column experiments demonstrate the sensitivity of electrical geophysical methods to subtle and transient changes in porous media due to transport of ions and/or reactions such as precipitation and adsorption. The results suggest that pH controls the polarization response in silica gel. Although silica gel is a synthetic material, and the pH dependence of SIP is likely weaker for natural porous media, we believe our findings are still relevant to the assessment of methods for sensing and characterizing physical and chemical/biogeochemical events linked with subsurface remediation at larger spatial scales of natural environment. SIP can play an important role in improving the understanding of complex processes in porous media in both laboratory and field settings.

■ ASSOCIATED CONTENT

Supporting Information

Figures SI1–SI7. This information is available free of charge via the Internet at http://pubs.acs.org

■ AUTHOR INFORMATION

Corresponding Author

*E-mail: chizhang@pegasus.rutgers.edu.

Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

This work was funded by the Department of Energy under contract DE-AC07-05ID14517.

■ REFERENCES

(1) Atekwana, E. A.; Sauck, W. A.; Werkema, D. D. Investigations of geoelectrical signatures at a hydrocarbon contaminated site. *J. Appl. Geophys.* **2000**, *44* (2–3), 167–180.

(2) Personna, Y. R.; Ntarlagiannis, D.; Slater, L.; Yee, N.; O'Brien, M.; Hubbard, S. Spectral induced polarization and electrodic potential monitoring of microbially mediated iron sulfide transformations. *J. Geophys. Res.* **2008**, *113* (G2), G02020.

(3) William, K. H.; Kemna, A.; Wilkins, M. J.; Druhan, J.; Amtzen, E.; N'Guessan, A. L.; Hubbard, S. S.; Banfield, J. F. Geophysical monitoring of coupled microbial and geochemical processes during stimulated subsurface bioremediation. *Environ. Sci. Technol.* **2009**, *43* (17), 6717–6723.

(4) Atekwana, E. A.; Slater, L. D. Biogeophysics: A new frontier in Earth science research. *Rev. Geophys.* **2009**, *47* (4), RG4004.

(5) Williams, K. H.; Ntarlagiannis, D.; Slater, L. D.; Dohnalkova, A.; Hubbard, S. S.; Banfield, J. F. Geophysical Imaging of Stimulated

Microbial Biomineralization. *Environ. Sci. Technol.* **2005**, *39* (19), 7592−7600.
(6) Ntarlagiannis, D.; Williams, K.; Slater, L.; Hubbard, S. Low-frequency electrical response to microbial induced sulfide precipitation. *J. Geophys. Res.* **2005**, *110* (G2), 12.
(7) Davis, C. A.; Atekwana, E.; Atekwana, E.; Slater, L. D.; Rossbach, S.; Mormile, M. R. Microbial growth and biofilm formation in geologic media is detected with complex conductivity measurements. *Geophys. Res. Lett.* **2006**, *33* (18), L18403.
(8) Slater, L.; Ntarlagiannis, D.; Personna, Y. R.; Hubbard, S. Pore-scale spectral induced polarization signatures associated with FeS biomineral transformations. *Geophys. Res. Lett.* **2007**, *34* (21), L21404.
(9) Wu, Y.; D Slater, L.; Korte, N. Low Frequency Electrical Properties of Corroded Iron Barrier Cores. *Environ. Sci. Technol.* **2006**, *40* (7), 2254−2261.
(10) Wu, Y.; Slater, L. D.; Korte, N. Effect of Precipitation on Low Frequency Electrical Properties of Zerovalent Iron Columns. *Environ. Sci. Technol.* **2005**, *39* (23), 9197−9204.
(11) Wu, Y.; Hubbard, S.; Williams, K. H.; Ajo-Franklin, J. On the complex conductivity signatures of calcite precipitation. *J. Geophys. Res.* **2010**, *115*, G00G04.
(12) Wu, Y.; Ajo-Franklin, J.; Spycher, N.; Hubbard, S.; Zhang, G.; Williams, K.; Taylor, J.; Fujita, Y.; Smith, R. Geophysical monitoring and reactive transport modeling of ureolytically-driven calcium carbonate precipitation. *Geochem. Trans.* **2011**, *12* (1), 7.
(13) Wu, Y.; Versteeg, R.; Slater, L.; LaBrecque, D. Calcite precipitation dominates the electrical signatures of zero valent iron columns under simulated field conditions. *J. Contam. Hydrol.* **2009**, *106* (3−4), 131−143.
(14) Warren, L. A.; Maurice, P. A.; Parmar, N.; Ferris, F. G. Microbially Mediated Calcium Carbonate Precipitation: Implications for Interpreting Calcite Precipitation and for Solid-Phase Capture of Inorganic Contaminants. *Geomicrobiol. J.* **2001**, *18* (1), 93−115.
(15) Fujita, Y.; Ferris, F. G.; Lawson, R. D.; Colwell, F. S.; Smith, R. W. Calcium Carbonate Precipitation by Ureolytic Subsurface Bacteria. *Geomicrobiol. J.* **2000**, *17* (4), 305−318.
(16) Fujita, Y.; Taylor, J. L.; Gresham, T. L. T.; Delwiche, M. E.; Colwell, F. S.; McLing, T. L.; Petzke, L. M.; Smith, R. W. Stimulation Of Microbial Urea Hydrolysis In Groundwater To Enhance Calcite Precipitation. *Environ. Sci. Technol.* **2008**, *42* (8), 3025−3032.
(17) Lesmes, D. P.; Morgan, F. D. Dielectric spectroscopy of sedimentary rocks. *J. Geophys. Res.* **2001**, *106* (B7), 13329−13346.
(18) Slater, L.; Lesmes, D. P. Electrical-hydraulic relationships observed for unconsolidated sediments. *Water Resour. Res.* **2002**, *38* (10), 1213.
(19) Slater, L.; Glaser, D. Controls on induced polarization in sandy unconsolidated sediments and application to aquifer characterization. *Geophysics* **2003**, *68* (5), 1547−1558.
(20) Börner, F. D.; Schön, J. H. A relation between the quadrature component of electrical conductivity and the specific surface area of sedimentary rocks. *Log Anal.* **1991**, *32*, 612−613.
(21) Lesmes, D. P.; Frye, K. M. Influence of pore fluid chemistry on the complex conductivity and induced polarization responses of Berea sandstone. *J. Geophys. Res.* **2001**, *106* (B3), 4079−4090.
(22) Waxman, M. H.; Smits, L. J. Electrical conductivities in oil-bearing shaly sands. *Trans. Am. Inst. Miner., Metall. Pet. Eng.* **1968**, *243*, 107−122.
(23) Vinegar, H. J.; Waxman, M. H. Induced polarization of shaly sands. *Geophysics* **1984**, *49* (8), 1267−1287.
(24) Archie, G. E. The electrical resistivity log as an aid in determining some reservoir characteristics. *Trans. Am. Inst. Miner., Metall. Pet. Eng.* **1942**, *146*, 54−62.
(25) Scott, J. B. T.; Barker, R. D. Determining pore-throat size in Permo-Triassic sandstones from low-frequency electrical spectroscopy. *Geophys. Res. Lett.* **2003**, *30* (9), 1450.
(26) Binley, A.; Slater, L. D.; Fukes, M.; Cassiani, G. Relationship between spectral induced polarization and hydraulic properties of saturated and unsaturated sandstone. *Water Resour. Res.* **2005**, *41* (12), W12417.
(27) Scott, J. B. T.; Barker, R. D. Characterization of sandstone by electrical spectroscopy for stratigraphical and hydrogeological investigations. *Q. J. Eng. Geol. Hydrogeol.* **2005**, *38* (2), 143−154.
(28) Vaudelet, P.; Revil, A.; Schmutz, M.; Franceschi, M.; Bégassat, P. Induced polarization signatures of cations exhibiting differential sorption behaviors in saturated sands. *Water Resour. Res.* **2011**, *47* (2), W02526.
(29) Skold, M.; Revil, A.; Vaudelet, P. The pH dependence of spectral induced polarization of silica sands: Experiment and modeling. *Geophys. Res. Lett.* **2011**, *38* (12), L12304.
(30) Cole, K. S.; Cole, R. H. Dispersion and Absorption in Dielectrics I. Alternating Current Characteristics. *J. Chem. Phys.* **1941**, *9* (4), 341−351.
(31) Titov, K.; Komarov, V.; Tarasov, V.; Levitski, A. Theoretical and experimental study of time domain-induced polarization in water-saturated sands. *J. Appl. Geophys.* **2002**, *50* (4), 417−433.
(32) Pelton, W. H.; Ward, S. H.; Hallof, P. G.; Sill, W. R.; Nelson, P. H. Mineral discrimination and removal of inductive coupling with multifrequency IP. *Geophysics* **1978**, *43* (3), 588−609.
(33) Kruschwitz, S.; Binley, A.; Lesmes, D.; Elshenawy, A. Textural controls on low-frequency electrical spectra of porous media. *Geophysics* **2010**, *75* (4), WA113−WA123.
(34) Weller, A.; Breede, K.; Slater, L.; Nordsiek, S. Effect of changing water salinity on complex conductivity spectra of sandstones. *Geophysics* **2011**, *76* (5), F315−F327.
(35) Schwarz, G. A theory of the low-frequency dielectric dispersion of colloidal particles in electrolyte solution1,2. *J. Phys. Chem.* **1962**, *66* (12), 2636−2642.
(36) Benini, S.; Ciurli, S.; Rypniewski, W. R.; Wilson, K. S.; Mangani, S. Crystallization and preliminary high-resolution X-ray diffraction analysis of native and β-mercaptoethanol-inhibited urease from *Bacillus pasteurii*. *Acta Crystallogr., D* **1998**, *54* (3), 409−412.
(37) Wu, Y.; Slater, L.; Versteeg, R.; LaBrecque, D. A comparison of the low frequency electrical signatures of iron oxide versus calcite precipitation in granular zero valent iron columns. *J. Contam. Hydrol.* **2008**, *95* (3−4), 154−167.
(38) Versteeg, R. J.; Richardson, A. N.; Rowe, T. Web-Accessible Scientific Workflow System for Performance Monitoring. *Environ. Sci. Technol.* **2006**, *40* (8), 2692−2698.
(39) Chen, J.; Kemna, A.; Hubbard, S. S. A comparison between Gauss-Newton and Markov-chain Monte Carlo-based methods for inverting spectral induced-polarization data for Cole-Cole parameters. *Geophysics* **2009**, *74* (4), Y7−Y7.
(40) Kemna, A. *Tomographic Inversion of Complex Resistivity—Theory and Application*; Der Andere Verlag: Osnabrück, Germany, 2000.
(41) Eriksson, R.; Merta, J.; Rosenholm, J. B. The calcite/water interface: I. Surface charge in indifferent electrolyte media and the influence of low-molecular-weight polyelectrolyte. *J. Colloid Interface Sci.* **2007**, *313* (1), 184−193.
(42) Bolt, G. H. Determination of the Charge Density of Silica Sols. *J. Phys. Chem.* **1957**, *61* (9), 1166−1169.
(43) Leroy, P.; Revil, A.; Kemna, A.; Cosenza, P.; Ghorbani, A. Complex conductivity of water-saturated packs of glass beads. *J. Colloid Interface Sci.* **2008**, *321* (1), 103−117.
(44) Revil, A. Spectral induced polarization of shaly sands: Influence of the electrical double layer. *Water Resour. Res.* **2012**, *48* (2), W02517.

01_张弛老师第一作者语言样本.pdf

文献 3/6

Complex Dielectric Properties of Sulfate-Reducing Bacteria Suspensions

Chi Zhang; Lee Slater; Camelia Prodan

Geomicrobiology Journal (2013); DOI: 10.1080/01490451.2012.719997

证据用途：第一作者直接语言样本；张弛老师角色：第一作者/通讯作者；证据等级：A

Geomicrobiology Journal (2013) **30**, 490–496
Copyright © Taylor & Francis Group, LLC
ISSN: 0149-0451 print / 1521-0529 online
DOI: 10.1080/01490451.2012.719997

Taylor & Francis
Taylor & Francis Group

Complex Dielectric Properties of Sulfate-Reducing Bacteria Suspensions

CHI ZHANG[1,2*], LEE SLATER[1], and CAMELIA PRODAN[3]

[1]*Department of Earth and Environmental Sciences, Rutgers University-Newark, Newark, New Jersey, USA*
[2]*Department of Geophysics, Colorado School of Mines, Golden, Colorado, USA*
[3]*Department of Physics, New Jersey Institute of Technology, Newark, New Jersey, USA*

Received March 2012, Accepted August 2012

Sulfate-reducing bacteria (SRB) can potentially enhance the remediation of heavy metals in the subsurface. Previous geophysical research has demonstrated the sensitivity of electrical measurements to SRB-mediated mineral transformation in porous media. However, the inherent dielectric properties of SRB and their direct contribution to the electrical properties of porous media are poorly understood. We studied the complex dielectric properties of SRB (*Desulfovibrio vulgaris*) suspensions at different concentrations and at different growth stages using a two-electrode dielectric spectroscopy measurement over the frequency range of 20 Hz to 1 MHz. Our results show higher dielectric responses (relative dielectric permittivity, real and imaginary conductivity) occurred with higher bacteria concentration at frequencies <10 kHz. Additionally, permittivity and conductivity both decreased as cells aged from mid-log phase to late stationary phase. Our results suggest that dielectric spectroscopy measurements can be used to noninvasively monitor biomass and various growth stages of SRB. Our work advances the interpretation of electrical signals associated with SRB observed in the subsurface.

Keywords: biogeophysics, dielectric properties, dielectric spectroscopy, monitoring bioremediation, sulfate-reducing bacteria

Introduction

The problem of metal (e.g., lead, cadmium, zinc, etc.) and radionuclide (e.g., uranium) contamination in the subsurface has promoted the exploitation of microbial mediated reduction of redox-sensitive metals (Valls and de Lorenzo 2002). Sulfate-reducing bacteria (SRB) have been extensively studied for their ability to reduce and accumulate heavy metals and radionuclides either enzymatically (Lovley et al. 1993) or through precipitation of metal sulfide minerals. Stimulated SRB activity and associated immobilization of heavy metals is a promising remediation approach as SRB regulate the cycling of essential elements such as iron and sulfur. To validate successful *in situ* bioremediation by SRB, long-term and noninvasive monitoring of SRB activity and associated biomineralization at sufficiently high spatial resolution over large domains is required.

Recent biogeophysics research has demonstrated the sensitivity of electrical geophysical techniques to subsurface microbial processes and microbial-induced alterations to geologic media (Atekwana and Slater 2009). Among these techniques, spectral induced polarization (SIP) has shown potential for 1) detection of bacterial cells and microbial growth (Ntarlagiannis et al. 2005; Abdel Aal et al. 2010), 2) characterizing microbial-mineral alterations at mineral-fluid interfaces (Personna et al. 2008; Williams et al. 2005), and 3) improving understanding of metal and nutrient cycling driven by SRB (Ntarlagiannis et al. 2005; Personna et al. 2008; Slater 2007). Recent studies (Flores Orozco et al. 2011; Williams et al. 2009) also show that the electrical geophysical methods could potentially be used to monitor bioremediation by SRB at the field-scale.

Both SIP and dielectric spectroscopy (DS) measure the complex dielectric permittivity (or complex electrical conductivity) of a material over a frequency range. Whereas SIP is a geophysical technique for studying porous media in both laboratory and field settings, DS is a bench-top technique widely used in many areas including biophysics and pharmacology. Previous geophysical studies focused on sand columns inoculated with SRB that were monitored using SIP at frequencies less than a kilohertz (Ntarlagiannis et al. 2005; Personna et al. 2008; Williams et al. 2005). However, the intrinsic broadband dielectric properties of SRB and their direct contribution to the electrical properties of a porous medium are not well studied. Such decomposition of electrical signals from complex microbial sulfate reduction processes in porous media is required to better constrain interpretation of SIP data from laboratory and field settings (Williams et al. 2009).

*Address correspondence to Chi Zhang, Department of Geophysics, Colorado School of Mines, Golden, CO 80401, USA; Email: chi.zhang15@gmail.com

In this study, we investigated the complex dielectric properties of *Desulfovibrio vulgaris*, a model SRB well studied for its bioremediation capacity, in cellular suspensions with varying concentrations and at different growth stages from 20 Hz to 1 MHz. We show that DS can precisely capture the intrinsic macroscopic dielectric function of the bacterial suspensions, and distinguish the variations in bacterial concentrations and life stages. Both dielectric permittivity and complex conductivity were enhanced at low frequencies with increased bacteria concentration, but were reduced when bacteria aged from mid-log to late stationary growth stages.

Materials and Methods

Bacteria and Growth Conditions

Subculturing of *Desulfovibrio vulgaris* (strain *Hildenborough*/ATCC #35115) was minimized by using −80°C *D. vulgaris* stocks as 9% inocula for 5 ml of a defined lactate sulfate medium [LS4D medium (Mukhopadhyay et al. 2006)]. These starter cultures were then used as 9% inocula for 5 ml production cultures, and incubated in centrifuge tubes in an anaerobic chamber. During *D. vulgaris* growth, increasing cell numbers and precipitation of metal sulfides (mainly iron sulfide) will increase optical density (OD). Measuring the OD of growing cultures is a common method to quantify the cell density instead of counting the live/dead cell numbers under microscope. The OD and cell density readings can be related by a set of calibration curves, and the OD reading increases as the cell density increases.

We use OD measurements at a wavelength of 600 nm with a Genesys 10S Vis spectrophotometer as a quick approximation of cell abundance in the bacterial growth suspension. When the desired ODs for DS measurements were reached, the cultures were pelleted at 1500 × *g* for 5 min. After removal of the supernatant, the pellet was resuspended and homogenized by pipetting into 20 ml of 5 mM HEPES solutions (pH = 7.2 and conductivity = 0.01 S/m at 24°C), which is commonly used to maintain physiological pH of cells and has no interference with metal substrates (Panitz et al. 1985). Each DS measurement lasted less than 7 min, ensuring the viability of *D. vulgaris* cells in the buffer during the measurement (Fournier et al. 2003). The cultures with ODs of 0.28 and 0.37 grown in LS4D for 48 hours were prepared for DS measurements to evaluate the effect of cell concentrations on dielectric signals, and the cultures with ODs of 0.38 at 26, 49, and 73 h of growth were prepared in the same way for testing the impact of cell aging on the dielectric response.

Samples grown for 48 h were collected for scanning electron microscopy (SEM) and X-ray absorption spectroscopy, and the preparation of samples followed previous studies (Personna et al. 2008; Twining et al. 2003). For SEM analysis, centrifuged pellets were immersed overnight in 2.5% glutaraldehyde in phosphate buffer solution in the anaerobic chamber, rinsed three times in phosphate buffer, dehydrated in an ethanol-water mixture series (10%, 25%, 50%, 75%, 90%, and subsequently in triplicate in 100% ethanol for 1 h each), and dried in ambient air.

The morphology of *D. vulgaris* cells was characterized with a LEO 1530 VP SEM system under operating voltage of 2 kV. For X-ray adsorption spectroscopy, the cells were preserved with 0.25% glutaraldehyde immediately prior to centrifugation. The centrifuged pellets were then rinsed with Milli-Q water and were mounted on Au grids coated with a thin film and the samples were kept in anaerobic chamber prior to transfer to X-ray sample chambers. X-ray absorption spectra were measured in fluorescence mode at the National Synchrotron Light Source beamlines X11B (Fe K-edge, room temperature) at Brookhaven National Laboratory. The detailed description of X-ray absorption spectra measurements and the results will be presented and further discussed in another study.

Dielectric Spectroscopy (DS) Measurement

DS measurements were performed with a Solartron 1260 impedance/gain-phase analyzer from 20 Hz to 1 MHz. The experimental setup is described in Prodan et al. (2004) and Prodan and Bot (2009) and illustrated in Figure 1. The suspension to be measured was placed between two parallel circular steel electrodes (radius $R = 28.5$ mm) enclosed in a glass cylinder. The distance between the two electrode plates was controlled by a micrometer, and was set to be 3 mm. The applied voltage was 0.15 V. The complex dielectric function ε^* is expressed as

$$\varepsilon^* = \varepsilon + \sigma'/i\omega\varepsilon_0, \tag{1}$$

where ε and σ' represent measured relative dielectric permittivity and electrical conductivity (real part) of the sample, and ω is the angular frequency. These two terms are calculated from the impedance measurements,

$$\varepsilon = \frac{-Z''}{Z'^2 + Z''^2} \cdot \frac{d}{\omega\varepsilon_0 \pi R^2}, \tag{2}$$

$$\sigma' = \frac{Z'}{Z'^2 + Z''^2} \cdot \frac{d}{\pi R^2}, \tag{3}$$

where Z' and Z'' are real and imaginary part of measured complex impedance Z^*, ε_0 is the vacuum permittivity, and d is the length of the sample. The measured imaginary conductivity (σ'') can be calculated from

$$\sigma'' = \varepsilon\varepsilon_0\omega, \tag{4}$$

Therefore, measured complex conductivity σ^* is given by,

$$\sigma^* = \sigma' + \sigma'' = \sigma' + i\varepsilon\varepsilon_0\omega \tag{5}$$

Two electrode DS measurements can reliably record the complex dielectric properties when the phase angle ($\varphi = tan^{-1}(\sigma'/\sigma'')$) is small. The phase for the measured *D. vulgaris* cell suspension was always lower than 0.2 mrad at frequencies <10 kHz, which is much weaker than mineralized rock and sediments (equal or greater than the order of 10 mrads), and is below the noise threshold of four electrode SIP measurements. The two electrode DS drastically exceeds the resolution and accuracy of contemporary SIP, and this advantage of DS could

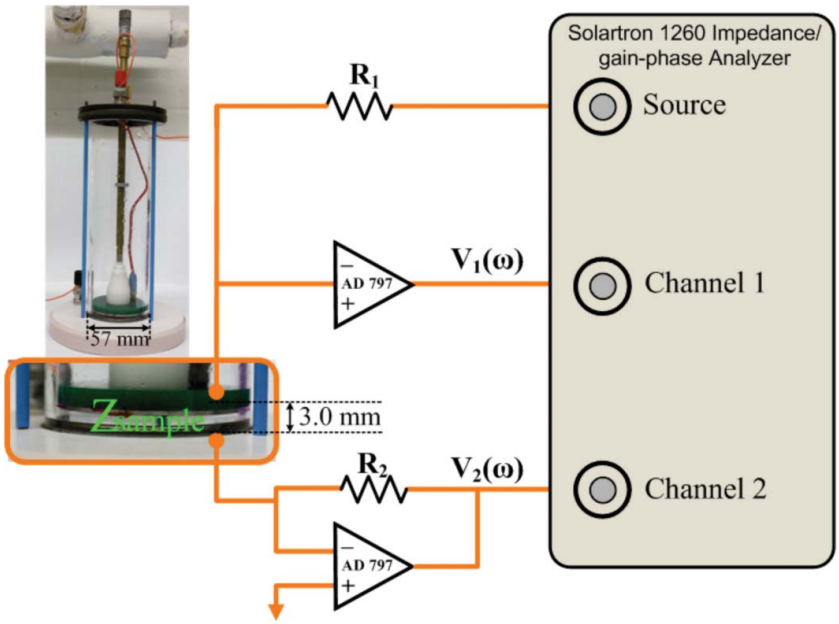


Fig. 1. The experimental setup and the electronic circuitry used in this experiment. The bacterial suspensions were placed in a glass cylinder with two parallel round stainless steel electrodes. The distance between the electrodes was controlled by a micrometer. A Solartron impedance analyzer provided a voltage on the top electrode, while the bottom was kept at virtual ground through the negative input of a preamplifier AD797 (color figure available online).

make it ideal for examining the polarization characteristics of SRB.

Electrode Polarization Error Removal

In two-electrode DS measurements, electrode polarization appears at the interface between electrolyte and electrode due to the buildup of charges in the presence of water molecules and hydrated ions, and introduces frequency-dependent contributions to the measured sample impedance. The electrode polarization becomes apparent below 1 kHz and significant below 100 Hz, and. can cause large errors at low frequencies. The full-frequency behavior of the dielectric response could be modeled by a series combination of the true sample impedance (Z_c^*) and the electrode impedance errors (Z_p^*), i.e., the measured impedance of bacterial suspensions (Z_s^*) is the sum of Z_c^* and Z_p^*. Prodan and Bot (2009) proposed a substitution method to reduce Z_p^* for colloidal suspensions, in which the cells are removed by gentle centrifugation and the resulting supernatant is used to measure Z_p^*.

First, the experimental data for the impedance of the supernatant (Z_m^*) was fitted with theoretical value, by giving a large weight to the high frequency data and a low, or almost zero weight to the low frequency data and extrapolate the fit through the full frequencies. Then, compute the Z_p^* as the difference between Z_m^* and the fitted data. Following their approach, we analyzed the Z_p^* at full frequency range, and corrected large Z_p^* at low frequencies to obtain Z_c^*. Figure 2 illustrates how Z_p^* was corrected using the imaginary part of impedance at full frequency (as one can see, the Z_p^* becomes negligible at high frequency).

The imaginary impedance for the bacterial suspensions (Z_s'') is close to the imaginary impedance for the medium or supernatant (Z_m''), except that the former is higher at low frequencies due to the presence of SRB interfacial polarization. This evidenced that the supernatant has the same ionic properties as the medium surrounding the cells in suspension, so the Z_p^* should be similar for Z_m^* *and* Z_s^*. The true imaginary impedance of the sample (Z_c'') was then determined by subtracting Z_p'' from Z_s''. Such error analysis and calibration is important for resolving subtle differences in ε and σ^* responses of a solution due to the presence of *D. vulgaris* bacteria.

Results

The SEM image of *D. vulgaris* is shown in Figure 3. A single rod shape bacteria cell is in the middle of the image with length about 2 μm. Figure 4a and b show the measured and intrinsic ε of *D. vulgaris* bacterial suspensions at two different concentrations (OD 0.28 and OD 0.37), each obtained by averaging three independent measurements as indicated by error bars. The error bars (one error bar in each decimal was shown) can determine the fluctuations induced by our two-step electrode polarization error removal technique and are small enough to show the resolution of two different concentrations. After removing the effects of Z_p^*, the subtle differences in ε curves between the two suspensions became more obvious. At low frequencies (<10 kHz), ε increases as the OD increases. However, ε decreases as the concentration of bacteria increases at higher frequencies, as expected.

For direct comparison with SIP data, Figure 4c and d shows σ' and σ'' for *D. vulgaris* bacterial suspensions of ODs at 0.28

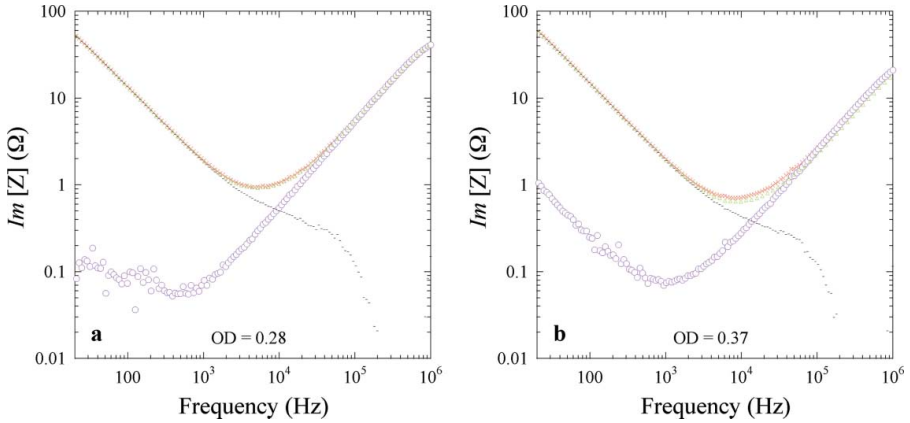


Fig. 2. Imaginary parts of complex impedances as a function of frequency: *D. vulgaris* cell suspensions (crosses, Z_s''), supernatant (triangles, Z_m''), polarization error (dashed line, Z_p''), and corrected cell suspensions after polarization removal (open circles, Z_c'') for different concentrations of *D. vulgaris* cell suspensions OD 0.28 (subplot a) and OD 0.37 (subplot b) (color figure available online).

and 0.37 after Z_p^* is removed. The σ' increased from 0.0075 S/m to 0.013 S/m as ODs increased from 0.28 to 0.37. The σ'' in OD 0.37 is one order of magnitude larger than it in the OD 0.28 cultures at low frequencies (<10 kHz). However, at higher frequencies, the σ'' decreases slightly with increased OD.

The variations in the dielectric responses of bacterial suspensions for different growth stages are shown in Figure 5. The *D. vulgaris* growth response to LS4D indicates that 26 h of growth represents the mid-log phase, 49 h of growth represents the early stationary phase, and 73 h of growth represents the late stationary phase (Mukhopadhyay et al. 2006; He et al. 2010). The ε for bacteria suspensions (OD = 0.38) decreased as the bacteria aged from 26 h to 73 h of growth at frequencies <10 kHz. At higher frequencies, suspensions at earlier growth stage show lower ε. Figure 5b shows σ ' decreased from 0.017 S/m to 0.011 S/m as bacteria continuously grew over the entire measurement frequency range.

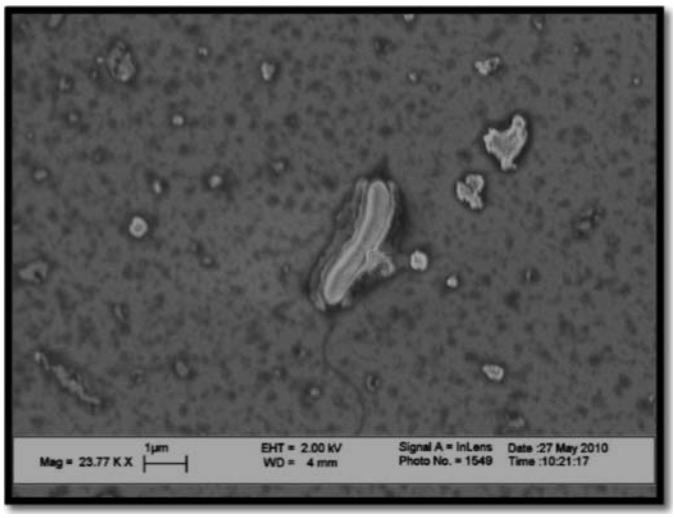


Fig. 3. Scanning electron microscopy (SEM) of *D. vulgaris* grown in LS4D medium in anaerobic chamber at 48 h.

Discussion

By varying *D. vulgaris* bacterial concentrations in suspensions and growing them at different stages, we have demonstrated that DS measurements can sense changes in cell concentration and different growth stages. Unlike previous biogeophysical studies (Abdel Aalet al. 2010; Ntarlagiannis et al. 2005; Williamset al. 2005), we investigated the polarization properties of the microbes themselves and associated growth stages. We excluded the effects of stimulated sulfate reduction in growth solution and/or sulfide biomineral transformations in porous media.

The dielectric response of live cells is known to result from movement of accumulated mobile charges at the membrane surfaces under an oscillating electric field at low frequencies (typically <10 kHz, known as α-dispersion), or as a consequence of polarization at the interface between the cell membrane and the external medium, giving rise to a Maxwell–Wagner effect (known as β-dispersion) (Foster and Schwan 1989). Most experimental dielectric analysis of biological cell suspensions is limited to the β–dispersion in biophysical studies, and almost all biogeophysical research has been restricted to frequencies lower than 10 kHz. In this study, we are the first to explore the dielectric properties of SRB bacteria suspensions over a relatively broad range of frequencies, and determine the dependence of dielectric responses on SRB bacteria concentration and metabolism stages. The recorded changes in ε and σ^* reflect variations in not only bacteria concentration, but also bacteria metabolism stages.

Examining Figures 4b and 5a, we identify a very steep rise in ε curves at frequencies <10 kHz. In this range, the ε of live *D. vulgaris* suspensions is as high as 10^6, being 6- to 10-fold higher in the OD 0.37 suspensions relative to the OD 0.28 suspensions. The significant α-dispersion depends not only on the conductivities of the extracellular medium and cell cytoplasm, but also on the cell size, shape, and biomass concentration (Di Biasio and Cametti 2007; Schwan 1957; Yang et al. 1999). Previous work (Bot and Prodan 2009) shows that living *E. coli* cells in pure water have ε of 10^5 and increasing with cell

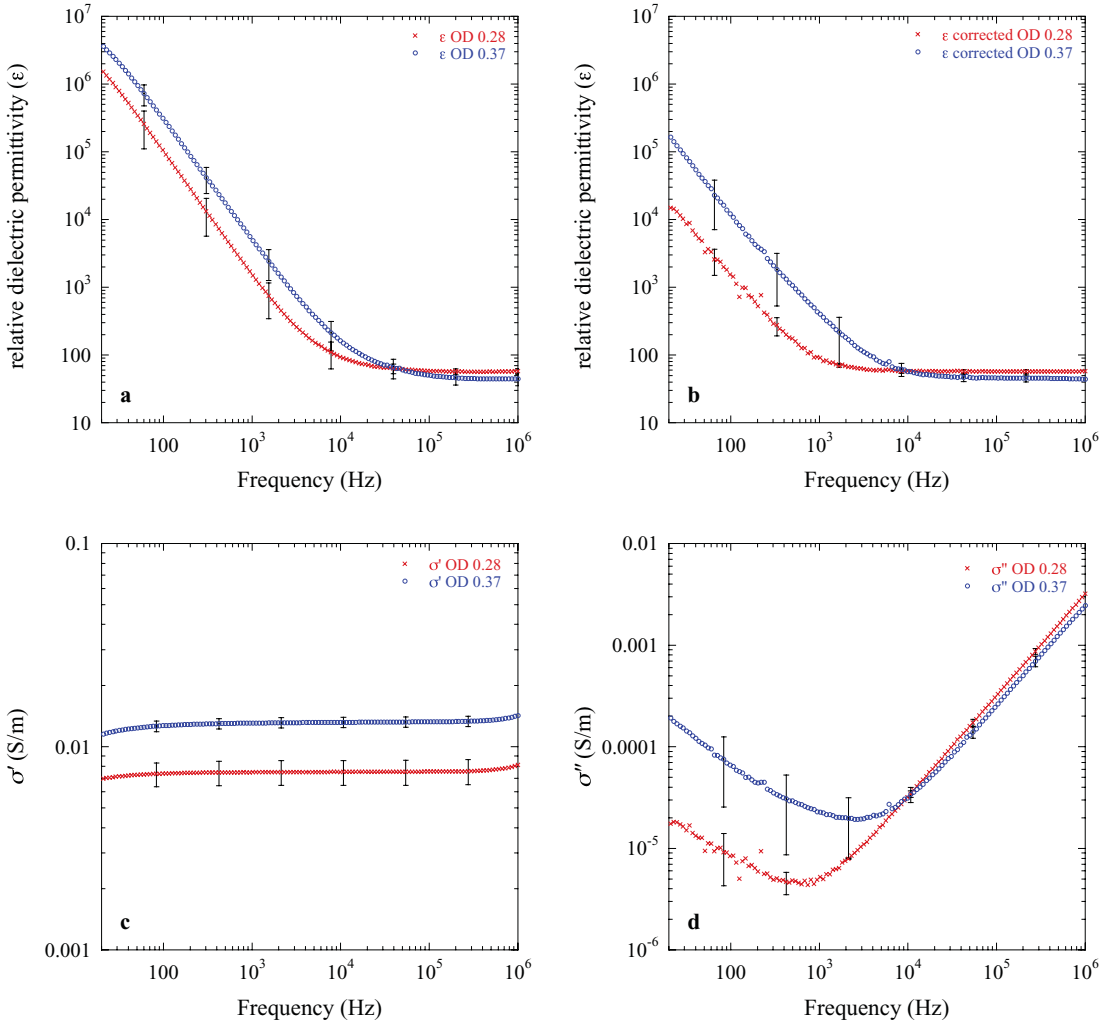


Fig. 4. Raw (a) and corrected (b) relative dielectric permittivity ε of *D. vulgaris* cell suspensions with different cell concentration–OD 0.28 (cross) and OD 0.37 (open circle); (c) Real conductivity (σ') and (d) imaginary conductivity (σ'') of *D. vulgaris* cell suspensions with different cell concentration (color figure available online).

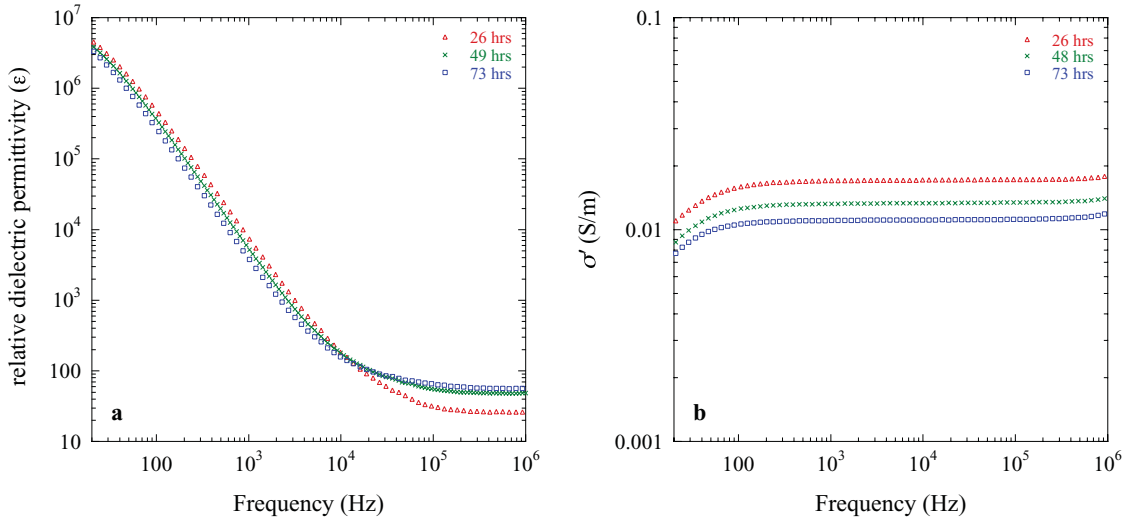


Fig. 5. Relative dielectric permittivity (a) and real conductivity (b) of suspensions with the same *D. vulgaris* cell concentration (OD = 0.38) but harvested at different growth period: 26 h of growth (triangles), 49 h of growth (crosses), and 73 h of growth (open squares). As the cell ages, the relative dielectric permittivity decreases and the conductivity decreases (color figure available online).

concentration from 10 Hz to 10 kHz. Thus the polarization of *D. vulgaris* is significantly greater than *E. coli.* Both *D. vulgaris* and *E. coli* are curved rod-shape bacteria with similar cell dimensions [*D. vulgaris* is $\sim$2 μm long from our SEM images, whereas the average length of *E. coli* is reported to be 2 μm (Trueba and Woldringh 1980)].

Therefore, the morphological difference between the two cells may not explain our observations. Given the relatively abundance of Fe^{2+} in the growth solution, the iron sulfide minerals would be the major metal sulfide species precipitate at the bacterial surface. Indeed, nanostructured FeS_2 was found to assemble on the membranes of *D. vulgaris* from Fe K-edge X-ray near edge absorption spectra. In contrast, *E. coli* cells do not precipitate metal sulfide minerals on the bacterial membrane. This metal sulfide could contribute to larger polarization of *D. vulgaris* cells relative to *E. coli* cells.

The reversal in the dependence of ε on OD at frequencies >10 kHz was also observed in studies on *E. coli* (Bot and Prodan 2009). This observation was theoretically verified for prolate spheroids (Di Biasio et al. 2010). The same study showed the higher frequency dispersion could be affected by different strengths of α-dispersions resulting from variations in the surface conductivity.

The α and β-dispersions are not clearly separated in our dielectric curves as the two dispersions influence each other heavily and both are very sensitive to the conductivity of the medium. The β-dispersion would shift to lower frequencies as conductivity of the medium decreases (E Prodan et al. 2008). In our experiment, we used 5 mM HEPES as the medium for the DS measurements, which has a much lower conductivity compared to other experimental and/or theoretical work where the β-dispersion occurred from 10 kHz to 1 MHz (Bai et al. 2006). Therefore, the β-dispersion should be shifted to lower frequencies in our study, thereby being integrated with the α-dispersion.

We clearly found a significant increase in ε, as well as in both σ' and σ'', due to higher SRB concentrations at frequencies <10 kHz. The SIP study by Abdel Aal et al. (2010) showed no effect on σ' with varying concentrations of both live and dead cells of *P. aeruginosa* in sand columns, but a linear increase in σ'' (maximum 40%) as live cell concentration increased. In contrast, we found a maximum 10-fold increase in both σ' and σ'' as *D. vulgaris* concentration increased by 30%. The dielectric signal has previously been reported to be linear with biomass concentration and attributable to the increased number of polarized membranes (Davey et al. 1996). The DS measurement could therefore serve as a nonintrusive SRB biomass sensor that could be applied in more complex environments.

The polarization of individual microbial cell surfaces is related to the dynamic chemical composition of the cell membrane. The intracellular cation concentrations are functions of the age of the culture, the pH of the medium, and the accumulation of metabolic products. We observed that as bacteria aged from earlier growth stage (26 h) to later growth stage (73 h), the ε decreased at frequencies <10 kHz, while the σ' also decreased over the entire measured frequency range. The metabolism of living biological cells has been monitored by DS (Bot and Prodan 2010; Gómez et al. 2002; Sarrafzadeh et al. 2005), and it shows that the transfer of ions across the cell membrane of living cells results in a higher conductivity in the suspension surrounding the cells than in an equivalent suspension containing dead cells from which all loosely bound ionic species were removed. Various mechanisms exist for live cells to control osmotic pressure of the cytoplasm, such as the release or uptake of K^+, which is attracted by the negative charges inside the cell (Gómez et al. 2002). However, the cell membrane becomes more permeable when the cell ages, resulting in the decrease of the membrane potential (Bot and Prodan 2010). Ions (e.g., K^+) that are forced to concentrate inside of the viable cell by this potential should be able to diffuse back to the surrounding medium, but the concentration of such ions decreases as the cell ages, so the σ' of bacteria suspended in buffer consequently decreases. The decrease in both ε and σ' as cell ages is consistent with recent observations in Bot and Prodan (2010).

Conclusions

DS can provide an accurate and effective way to distinguish between SRB suspensions with different volume fractions, as well as at different growth stages. The ε, as well as σ' and σ'' were shown to rise with increased *D. vulgaris* concentrations, but decrease at frequencies <10 kHz as cells aged. Our results suggest that the electrical signatures obtained in previous biogeophysical studies on SRB are not only due to the bacterial modification of the geological media, but may also be directly due to SRB and underlying SRB metabolism processes. Furthermore, DS measurements may afford unique opportunities for non-invasively monitoring SRB bacteria in column-scale experiments on porous media.

Acknowledgments

Support for this research was provided by Subsurface Biogeochemical Research, U.S. Department of Energy, under grant ER64855. We thank Ashwini Bendiganavale (New Jersey Institute of Technology) and Dr. Dimitrios Ntarlagiannis (Rutgers University –Newark) for SEM images, and Dr. Trevor Tyson (New Jersey Institute of Technology) for synchrotron X-ray absorption spectroscopy. We thank Dr. Andre Revil (Colorado School of Mines) for helpful discussions, and two referees for constructive edits and suggestions.

References

Abdel Aal GZ, Atekwana EA, Rossbach S, Werkema DD. 2010. Sensitivity of geoelectrical measurements to the presence of bacteria in porous media. J Geophys Res 115(G3):G03017.

Atekwana EA, Slater LD. 2009. Biogeophysics: A new frontier in earth science research. Rev Geophys 47(4):RG4004.

Bai W, Zhao KS, Asami K. 2006. Dielectric properties of E. coli cell as simulated by the three-shell spheroidal model. Biophys Chem 122(2):136–142.

Bot CT, Prodan C. 2009. Probing the membrane potential of living cells by dielectric spectroscopy. Euro Biophys J 38(8):1049–1059.

Bot CT, Prodan C. 2010. Quantifying the membrane potential during E. coli growth stages. Biophys Chem 146(2–3):133–137.

Davey HM, Davey CL, Woodward AM, Edmonds AN, Lee AW, Kell DB. 1996. Oscillatory, stochastic and chaotic growth rate fluctuations in permittistatically controlled yeast cultures. Biosystems 39(1):43–61.

Di Biasio A, Ambrosone L, Cametti C. 2010. The dielectric behavior of nonspherical biological cell suspensions: An analytic approach. Biophys J 99(1):163–174.

Di Biasio A, Cametti C. 2007. Effect of shape on the dielectric properties of biological cell suspensions. Bioelectrochemistry 71(2):149–156.

Flores Orozco AN, Williams KH, Long PE, Hubbard SS, Kemna A. 2011. Using complex resistivity imaging to infer biogeochemical processes associated with bioremediation of an uranium-contaminated aquifer. J Geophys Res 116(G3):G03001.

Foster KR, Schwan HP. 1989. Dielectric properties of tissues and biological materials: A critical review. Crit Rev Biomed Eng 17(1):25–104.

Fournier M, Zhang Y, Wildschut JD, Dolla A, Voordouw JK, Schriemer DC, Voordouw G. 2003. Function of oxygen resistance proteins in the anaerobic, sulfate-reducing bacterium desulfovibrio vulgaris hildenborough. J Bacteriol 185(1):71–79.

Gómez R, Bashir R, Bhunia AK. 2002. Microscale electronic detection of bacterial metabolism. Sens Actuat B: Chem 86(2–3):198–208.

He Q, et al. 2010. Impact of elevated nitrate on sulfate-reducing bacteria: A comparative study of desulfovibrio vulgaris. ISME J 4(11):1386–1397.

Lovley D, Roden E, Phillips E, Woodward J. 1993. Enzymatic iron and uranium reduction by sulfate-reducing bacteria. Mar Geol 113(1–2):41–53.

Mukhopadhyay A, He Z, Alm EJ, Arkin AP, Baidoo EE, Borglin SC, Chen W, et al. 2006. Salt stress in desulfovibrio vulgaris hildenborough: An integrated genomics approach. J Bacteriol 188(11):4068–4078.

Ntarlagiannis D, Williams K, Slater L, Hubbard S. 2005. Low-frequency electrical response to microbial induced sulfide precipitation. J Geophys Res 110(G2):12.

Panitz JA, Andrews CL, Bear DG. 1985. A deposition technique for the imaging and analysis of protein interactions with metal and semiconductor surfaces. J Elect Microsc Technol 2(4):285–292.

Personna YR, Ntarlagiannis D, Slater L, Yee N, O'Brien M, Hubbard S. 2008. Spectral induced polarization and electrodic potential monitoring of microbially mediated iron sulfide transformations. J Geophys Res 113(G2):G02020.

Prodan C, Bot C. 2009. Correcting the polarization effect in very low frequency dielectric spectroscopy. J Phys D: Appl Phys 42(17): 175505.

Prodan C, Mayo F, Claycomb JR, Miller JM, Benedik MJ. 2004. Low-frequency, low-field dielectric spectroscopy of living cell suspensions. J Appl Phys 95(7):3754–3756.

Prodan E, Prodan C, Miller J. 2008. The dielectric response of spherical live cells in suspension: An analytic solution. Biophys J 95(9):4174–4182.

Sarrafzadeh MH, Belloy L, Esteban G, Navarro JM, Ghommidh C. 2005. Dielectric monitoring of growth and sporulation of Bacillus thuringiensis. Biotechnol Lett 27(7):511–517.

Schwan H. 1957. Electrical properties of tissue and cell suspensions. Adv Biol Med Phys 5:147–209.

Slater L. 2007. Near surface electrical characterization of hydraulic conductivity: From petrophysical properties to aquifer geometries—A review. Surv Geophys 28(2):169–197.

Trueba FJ, Woldringh CL. 1980. Changes in cell diameter during the division cycle of Escherichia coli. Bacteriol 142(3):869–878.

Twining BS, Baines SB, Fisher NS, Maser J, Vogt S, Jacobsen C, Tovar-Sanchez A, Sañudo-Wilhelmy SA. 2003. Quantifying trace elements in individual aquatic protist cells with a synchrotron x-ray fluorescence microprobe. Anal Chem 75(15):3806–3816.

Valls M, de Lorenzo V. 2002. Exploiting the genetic and biochemical capacities of bacteria for the remediation of heavy metal pollution. FEMS Microbiol Rev 26(4):327–338.

Williams KH, Ntarlagiannis D, Slater LD, Dohnalkova A, Hubbard SS, Banfield JF. 2005. Geophysical imaging of stimulated microbial biomineralization. Environ Sci Technol 39(19):7592–7600.

Williams KH, Kemna A, Wilkins MJ, Druhan J, Amtzen E, N'Guessan AL, Hubbard SS, Banfield JF. 2009. Geophysical monitoring of coupled microbial and geochemical processes during stimulated subsurface bioremediation. Environ Sci Technol 43(17): 6717–6723.

Yang J, Huang Y, Wang X, Wang X-B, Becker FF, Gascoyne PRC. 1999. Dielectric properties of human leukocyte subpopulations determined by electrorotation as a cell separation criterion. Biophys J 76(6):3307–3314.

Copyright of Geomicrobiology Journal is the property of Taylor & Francis Ltd and its content may not be copied or emailed to multiple sites or posted to a listserv without the copyright holder's express written permission. However, users may print, download, or email articles for individual use.

01_张弛老师第一作者语言样本.pdf

文献 4/6

Quadrature conductivity: A quantitative indicator of bacterial abundance in porous media

Chi Zhang; André Revil; Yoshiko Fujita; Junko Munakata-Marr; George Redden

Geophysics (2014); DOI: 10.1190/GEO2014-0107.1

证据用途：第一作者直接语言样本；张弛老师角色：第一作者；证据等级：A

GEOPHYSICS, VOL. 79, NO. 6 (NOVEMBER-DECEMBER 2014); P. D363–D375, 14 FIGS., 3 TABLES.
10.1190/GEO2014-0107.1

Quadrature conductivity: A quantitative indicator of bacterial abundance in porous media

Chi Zhang[1], André Revil[2], Yoshiko Fujita[3], Junko Munakata-Marr[4], and George Redden[5]

ABSTRACT

The abundance and growth stages of bacteria in subsurface porous media affect the concentrations and distributions of charged species within the solid-solution interfaces. Therefore, spectral induced polarization (SIP) measurements can be used to monitor changes in bacterial biomass and growth stage. Our goal was to gain a better understanding of the SIP response of bacteria present in a porous material. Bacterial cell surfaces possess an electric double layer and therefore become polarized in an electric field. We performed SIP measurements over the frequency range of 0.1–1 kHz on cell suspensions alone and cell suspensions mixed with sand at four pore water conductivities. We used *Zymomonas mobilis* at four different cell densities (including the background). The quadrature conductivity spectra exhibited two peaks, one around 0.05–0.10 Hz and the other around 1–10 Hz. Because SIP measurements on bacterial suspensions are typically made at frequencies greater than 1 Hz, these peaks have not been previously reported. In the bacterial suspensions in growth medium, the quadrature conductivity at peak I was linearly proportional to the density of the bacteria. For the case of the suspensions mixed with sands, we observed that peak II presented a smaller increase in the quadrature conductivity with the cell density. A comparison of the experiments with and without sand grains illustrated the effect of the porous medium on the overall quadrature conductivity response (decrease in the amplitude and shift of the peaks to the lower frequencies). Our results indicate that for a given porous medium, time-lapse SIP has potential for monitoring changes in bacterial abundance within porous media.

INTRODUCTION

Bacteria play an important role in environmental applications such as the remediation of nitrates and organic contaminants in groundwater (Toledo et al., 2006), the bioleaching of ores (e.g., Eisele and Gabby, 2014), managed aquifer recharge (Cockett and Pidlisecky, 2014), and the natural attenuation of leachates from landfills (Naudet et al., 2004). In addition, there is a growing interest in understanding the role of indigenous anaerobic bacteria in biodegraded oil fields (Grabowski et al., 2005) as well as in situ bioaugmentation and biostimulation of contaminated groundwater (Ginn et al., 2002). In addition, in microbial-enhanced oil recovery (in which microorganisms are injected into a reservoir), quantitative information on bacterial abundance and spatiotemporal knowledge of microorganism locations are essential (Heenan et al., 2013).

Minimally invasive (for instance involving shallow wells for crosswell tomography) diagnostic techniques are needed to provide nearly real time information of the spatiotemporal distribution of bacteria in porous media for calibrating and validating predictive models and for monitoring bacterial growth in situ. In the last decade, the remote detection of processes associated with bacterial activity in porous media has been demonstrated using various active and passive geophysical methods including electric and seismic techniques (e.g., Naudet and Revil, 2005; Ntarlagiannis et al.,

Manuscript received by the Editor 10 March 2014; revised manuscript received 8 July 2014; published online 16 September 2014.
[1]Formerly Colorado School of Mines, Department of Geophysics, Golden, Colorado, USA and Idaho National Laboratories, Idaho Falls, Idaho, USA; presently Rutgers University, Department of Earth and Environmental Science, Newark, New Jersey, USA. E-mail: chi.zhang15@gmail.com.
[2]Colorado School of Mines, Department of Geophysics, Golden, Colorado, USA and Université de Savoie, ISTerre, CNRS, UMR, Le Bourget du Lac, France. E-mail: arevil@mines.edu.
[3]Idaho National Laboratory, Idaho Falls, Idaho, USA and University of Copenhagen, Department of Chemistry, Nanogeoscience Group, Copenhagen, Denmark. E-mail: yfujita@nano.ku.dk.
[4]Colorado School of Mines, Department of Civil and Environmental Engineering, Golden, Colorado, USA. E-mail: jmmarr@mines.edu.
[5]University of Copenhagen, Department of Chemistry, Nanogeoscience Group, Copenhagen, Denmark. E-mail: gredden@alumni.stanford.edu.
© 2014 Society of Exploration Geophysicists. All rights reserved.

2005b; Williams et al., 2005; Davis et al., 2006; Linde and Revil, 2007; Personna et al., 2008; Zhang et al., 2010). Under some circumstances, bacterial activity can also be detected passively using the self-potential method (Naudet et al., 2003, 2004; Revil et al., 2010; Risgaard-Petersen et al., 2012, 2014). This passive technique is used to detect remotely the occurrence of electric current associated with the displacement of electrons through biotic electronic conductors. It is very similar, in essence, to the electroencephalographic method used to detect remotely the electric current generated by the opening of ionic channels at the synapses between neurons (Grech et al., 2008). Bacteria can form biogeobatteries in the subsurface of the Earth. In areas of strong redox potential gradients, their activity can be therefore remotely detected using the resulting electric field (Revil et al., 2010). However, this method does not provide any information regarding the density of bacterial populations in porous media and its changes over time.

Induced polarization is an active geophysical method investigating the reversible storage of electric charges in porous media. In other words, it investigates the capacitance of porous media at low frequencies of the exciting electric field (typically between 1 mHz and 10 kHz). Recent reviews on this geophysical method, the associated mechanisms, and applications in environmental geosciences can be found in Kemna et al. (2012) and Revil et al. (2012b). The in-phase component of the complex conductivity describes the electromigration of the charge carriers in the material, whereas its quadrature component describes the polarization. The induced polarization method has been qualitatively shown to be sensitive to the presence of bacteria in porous media (Abdel Aal et al., 2004, 2010; Ntarlagiannis et al., 2005a; Davis et al., 2006; Albrecht et al., 2011; Revil et al., 2012a; Zhang et al., 2012). The low-frequency polarization of bacteria in porous media has been recently modeled by Revil et al. (2012a) with a special emphasis on Gram-positive bacteria. Revil et al. (2012a) use a polarization model of the double layer surrounding the bacterial cells. They upscaled their results to a representative elementary volume of a porous material with a composite model taking into account the presence of the bacteria and the grains.

In the present study, we investigated the polarization properties of a typical Gram-negative bacteria *Zymomonas mobilis* whose surface is surrounded by an electric double layer. This organism was selected for the experiments because it produces ethanol from glucose, so that neither the substrate nor product is charged (avoiding ion effects on electric conductivity; see Gunasekaran et al., 1986). In addition, precipitation of mineral products, which could impact porosity or affect the formation factor, was not expected as a result of its metabolism. We investigated the spectral induced polarization (SIP) signatures of colloidal suspensions of this bacterium in growth media. Then, we mixed the cell suspensions with sand grains. Our goals were (1) to provide clear data showing the influence of the abundance of the bacteria upon the quadrature conductivity and (2) to determine if the relationship between the quadrature conductivity and the bacterial cell density in porous media could be quantitatively defined.

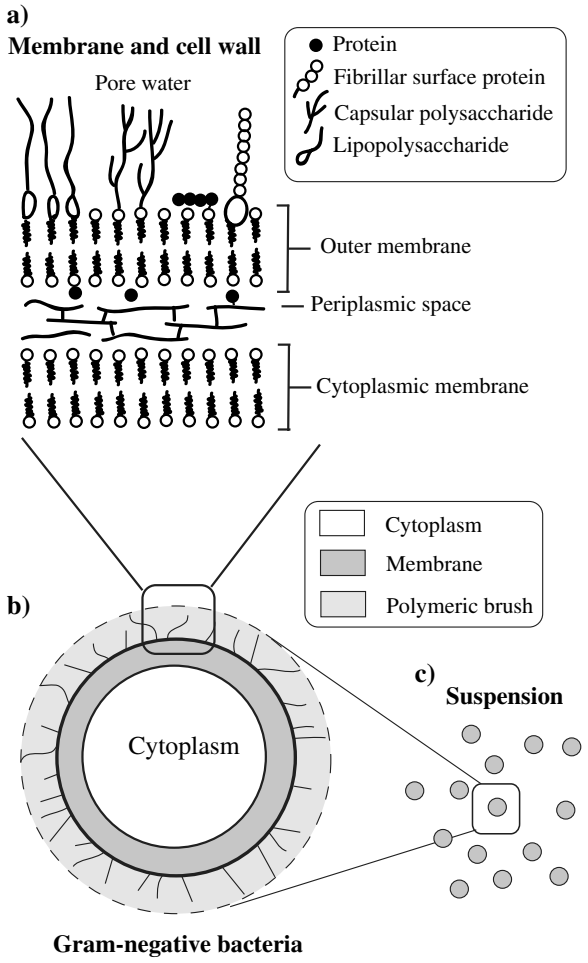


Figure 1. Conceptual model of Gram-negative bacteria. (a) The surface of Gram-negative bacteria with the membrane. This surface has a negative charge at a pH of greater than 3. (b) A single cell is considered to consist of the polymeric brush coating the surface of the bacteria, the membrane, and the cytoplasm. (c) We are interested in the properties of a suspension of bacteria in which each bacterium has an electric double layer comprising the Stern and diffuse layers.

THEORETICAL BACKGROUND

General considerations

Bacterial cells consist of a cytoplasm enclosed by an insulating membrane (Figure 1). Generally speaking, at pH values greater than 3–4, the cell external surface or cell wall is negatively charged as a result of protonation/deprotonation of surface ionizable groups (amine groups, carboxylic acid groups, hydroxyl groups, phosphodiesters, and phosphomonoester; see Figure 2a) (e.g., Willis, 1980; Revil et al., 2012a). This negative charge on the cell surface is compensated by cations that are either sorbed on the bacterial surface (Stern layer) or present in the diffuse layer (Figure 2b).

Under an oscillating electric field at low frequencies (<100 kHz), the electric double layer of bacteria can become polarized. Revil et al. (2012a) provide a model of this polarization, which is known as α-polarization in colloidal chemistry (e.g., Foster and Schwan, 1989). In the next sections, we will consider three cases of polarization: (1) a hypothetical bacterial colloidal suspension in a brine solution (Figure 3a), (2) a suspension of bacteria in a growth solution (called the *reconstitution medium* (RM); see Figure 3b) and (3) a mixture of the (bacterial + RM) colloidal suspension in the pore space between sand grains (Figure 3c). The model we use below is valid for dilute or concentrated suspensions and for low and high porosities. For instance, it has been used successfully to under-

stand the complex conductivity of low-porosity gas shales by Revil et al. (2013) and Woodruff et al. (2014).

Low-frequency polarization of a colloidal suspension of bacteria

We summarize below the model of Revil et al. (2012a) for a suspension of bacteria (Figure 3a), focusing on the relationship between the magnitude of the quadrature conductivity and the bacterial abundance, rather than modeling the bell-shape frequency dependence of the quadrature conductivity. The complex conductivity of the bacteria is represented by $\sigma_B{}^*$. The in-phase (real) component is related to electromigration of the charge carriers occurring in the bulk pore water and in the diffuse layer surrounding the surface of the bacteria (Figure 2b). The quadrature (imaginary) conductivity is related to the polarization of the Stern layer shown in Figure 2b. The in-phase and quadrature conductivity of a suspension of bacteria are given by

$$\sigma_B{}^* = \sigma_B{}' + i\sigma_B{}'', \tag{1}$$

where

$$\sigma_B{}' \approx \sigma_w + \bar{C}_B\,\bar{V}_B\,\rho_B\,(1-f)\,\beta_{(+)}\,\mathrm{CEC}_B \tag{2}$$

and

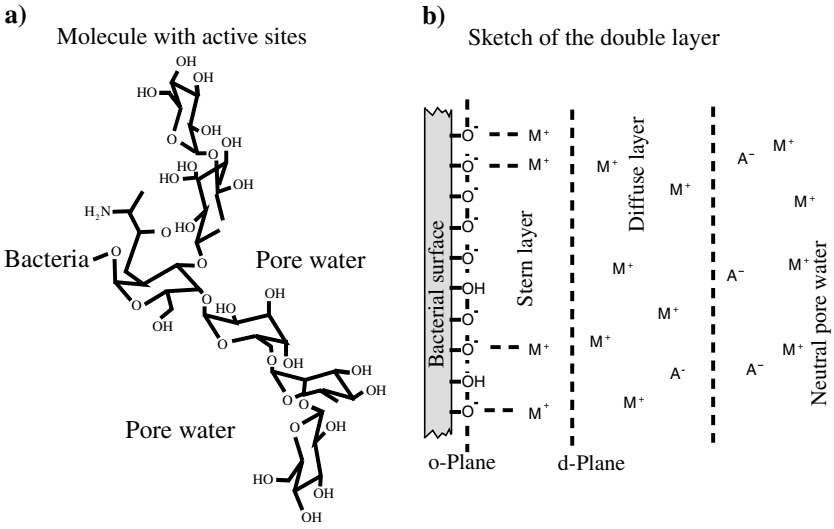


Figure 2. Structure of the electric double layer at the surface of a bacterium. (a) Typical organic molecule at the surface of Gram-negative bacteria (lipopolysaccharide component of the cell wall) showing ionizable groups >OH and $>NH_2$ in their uncharged forms. (b) Structure of the electric double layer showing the distinction between the Stern layer of weakly sorbed counterions and the diffuse layer. The symbol M^+ represents a metal cation, whereas A^- represents an anion.

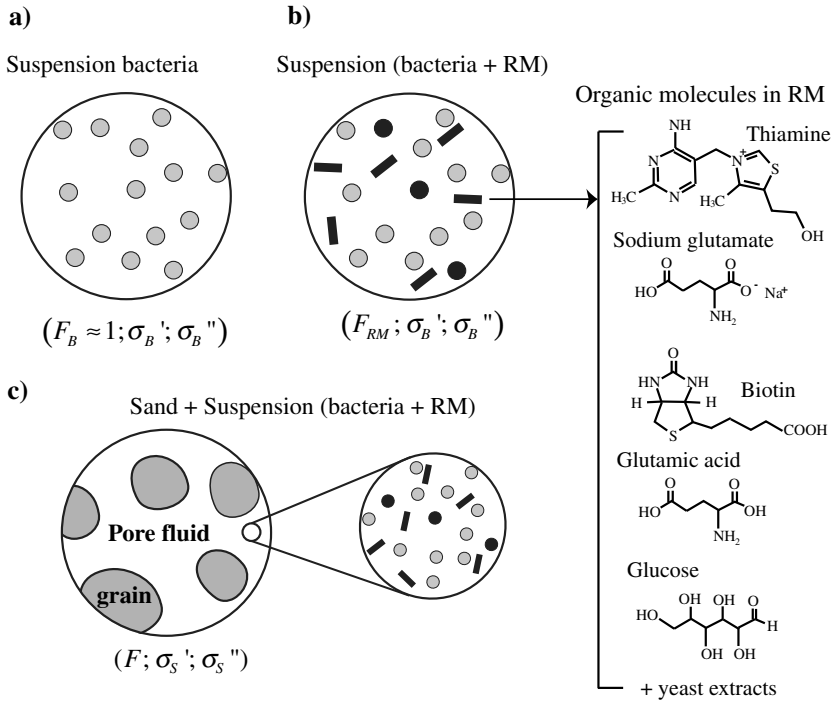


Figure 3. Representative elementary volumes of the suspensions and porous media. (a) Colloidal suspension of bacteria. (b) Colloidal suspension of bacteria with the RM including (black) all the nutrients in the suspension (for a dilute suspension of bacteria, the formation factor associated with the bacteria is equal to 1). These nutrients include various large organic molecules and yeast extracts. This soup of organic molecules and bacteria is characterized by its own formation factor $F_{RM} \approx 1$ and porosity ($\approx$1). (c) Granular material with its pore fluid, which can be either an electrolyte like NaCl or KCl or the colloidal RM with or without the bacteria.

$$\sigma_B'' \approx -\bar{C}_B\,\bar{V}_B\,\rho_B\,f\beta^S_{(+)}\,\mathrm{CEC}_B, \tag{3}$$

and where i ($i^2 = -1$) denotes the pure imaginary number, f denotes the fraction of the counterions in the Stern layer around the bacteria and therefore $(1–f)$ denotes the fraction of counterions in the diffuse layer; $\beta_{(+)}$ and $\beta^S_{(+)}$ (in $m^2\,s^{-1}\,V^{-1}$) denote the mobility of the counterions in the diffuse and Stern layers, respectively; σ_w ($S\,m^{-1}$) denotes the conductivity of the pore water, ρ_B ($kg\,m^{-3}$) is the mass density of the bacteria; $\bar{C}_B = N_B/V$ denotes the number of bacteria per unit pore volume (expressed usually in cell m^{-3} in SI units but generally in cell mL^{-1}) where N_B denotes the total number of bacteria and V is the volume of the colloidal suspension; $\bar{V}_B$ is the specific volume of a bacterium; and CEC_B ($C\,kg^{-1}$) is the cation exchange capacity (CEC) of the bacteria. This CEC represents, at a given pH and solution composition, the number of exchangeable sites on the external wall of the bacteria (see Revil et al. [2012a] for a collection of references regarding the CEC of bacteria).

The second term of equation 2 corresponds to the conductivity of the bacteria themselves (the surface conductivity of the bacteria $\sigma_B^S{}'$ and the conductivity of a single bacterium is given by $\sigma_B = (1-f)\,\beta_{(+)}\,\rho_B\,\mathrm{CEC}_B$) (Revil et al., 2012a). Note that alternatively, we could express the conductivity of a bacterium in terms of its radius and its surface conductance produced by the diffuse layer (see, for instance, Revil and Skold [2011] for sand grains). Both approaches are valid, but the CEC is usually the reported parameter describing the electrochemical properties of the bacteria surface, not the surface conductance. Equation 3 indicates that the quadrature conductivity of the suspension of bacteria is directly proportional to the concentration of the bacteria $\bar{C}_B$, an intuitive result. The mobility of the counterions (cations) populating the Stern layer is typically $\beta^S_{(+)} \approx (4.7 \pm 0.8) \times 10^{-10}\ m^2\,s^{-1}\,V^{-1}$ at 25°C according to Revil et al. (2012a), whereas the mobility of the counterions in the diffuse layer $\beta_{(+)}$ is equal to the mobility of the same ions in the bulk pore water.

Revil et al. (2012a) conclude that all the CECs of Gram-negative bacteria contribute to surface conductivity and quadrature conductivity and are likely to be distinctly different from the underlying substrate mineral, especially if time-lapse measurements are used to distinguish between the time-dependent effect associated with the growth of bacteria from the static value associated with the mineral grains. In contrast, only a fraction of the CEC of Gram-positive bacteria contributes to surface conductivity and polarization. Indeed, according to the surface speciation of Gram-positive bacteria (see an extensive discussion in Revil et al., 2012a), this behavior is due to the pH dependence of the amino surface groups of the protein and peptidoglycan molecules on the surface of the Gram-positive bacteria. These sites can accept counterions only at pH >8. On Gram-negative cells, lipopolysaccharides replace peptidoglycan molecules. They contain charged phosphate and carboxyl groups but no amino surface groups.

In general, for Gram-negative bacteria, the value of CEC_B is in the order of 1.8 $mMol\,g^{-1}$ (see Table 1 of Revil et al., 2012a) but can vary considerably between species. We have not been able to find a CEC value specific to *Z. mobilis* in the literature. In the following, we will see that our data are consistent with assuming $\mathrm{CEC}_B \approx 2.0$ $mMol\,g^{-1}$ for *Z. mobilis* therefore close to the mean value reported for Gram-negative bacteria in Revil et al. (2012a). The mass density of active and healthy bacteria is from $\rho_B \sim 1010$ to $1020\ kg\,m^{-3}$ (Bratbak and Dundas, 1984). Finally, $\bar{V}_B$, the volume of a single bacterium, is typically in the range 0.5–2.0 μm^3.

Now, we consider that the previous suspension of bacteria is mixed with the RM. The RM is considered to be a dilute colloidal suspension with its own formation factor F_{RM}, which is close to unity if the suspension is dilute enough (Figure 3b). Note that high ionic strength may destroy the cell through osmotic shocks. So, the complex conductivity of the RM could be defined as

$$\sigma_{\mathrm{RM}}{}^* \equiv \sigma'_{\mathrm{RM}} + i\sigma_{\mathrm{RM}}{}'' = \frac{1}{F_{\mathrm{RM}}}\,\sigma_B{}^* \approx \sigma_B{}^*. \tag{4}$$

Consequently, in our model, the in-phase and quadrature conductivity of the bacteria + RM suspension are those of the suspension of the bacteria itself. The slope of the quadrature conductivity of the suspension (bacteria plus the RM) versus the bacteria concentration can be written as

$$\sigma_{\mathrm{RM}}{}'' \approx -a_S\,\bar{C}_B, \tag{5}$$

where

$$a_S \approx \bar{V}_B\rho_B\,f\beta^S_{(+)}\,\mathrm{CEC}_B. \tag{6}$$

Equation 6 will be verified later in this paper. Taking, for instance, $\beta^S_{(+)} = 5\times10^{-10}\,m^2\,s^{-1}\,V^{-1}$ at 25°C, $\mathrm{CEC}_B = 1$ $m\,Mol\,g^{-1}$ ($\approx 10^5\ C\,kg^{-1}$), $f = 0.8$ (see the discussion of these values in Revil et al., 2012a), $\rho_B = 10^3\ kg\,m^{-3}$, and $\bar{V}_B = 1.0\ \mu m^3$ ($10^{-18}\ m^3$), we obtain $a_S = 4.0\times10^{-14}$ (in $m\,S^{-1}\,mL\,cell^{-1}$) with $\bar{C}_B$ expressed in cells mL^{-1}. For a cell density of 1×10^9 cell mL^{-1}, the quadrature conductivity would be $\sigma_{\mathrm{RM}}{}'' = -4.0\times10^{-5}\ S\,m^{-1}$. The surface conductivity would be $\sigma_B^S{}' = \bar{C}_B\,\bar{V}_B\rho_B\,(1-f)\,\beta_{(+)}\,\mathrm{CEC}_B = 1.0\times10^{-3}\ S\,m^{-1}$. This provides a first-order evaluation of the surface and quadrature conductivities that we can expect in this study.

Polarization of porous materials with bacteria and reconstitution medium solution

The complex conductivity of a porous material $\sigma* = \sigma' + i\sigma''$ without the bacteria and/or the RM fluid is given by

$$\sigma^* = \frac{1}{F}\sigma_w + \sigma_g{}^*, \tag{7}$$

where F denotes the (intrinsic) formation factor of the porous material saturated by an electrolyte ($F = \phi^{-m}$ according to Archie's law, where φ denotes the connected porosity) and without the bacteria and $\sigma_g{}^* = \sigma_g{}' + i\sigma_g{}''$ denotes the complex-valued surface conductivity associated with electromigration ($\sigma_g{}'$) and polarization ($\sigma_g{}''$) in the electric double layer coating the surface of the mineral grains. Replacing the pore water electrolyte by the suspension of bacterial cells (within the RM suspension) in the porous material is therefore equivalent to replacing σ_w in equation 7 by the complex conductivity of the suspension $\sigma_{\mathrm{RM}}{}^*$. This yields

$$\sigma^* = \frac{1}{F}\sigma_{\mathrm{RM}}{}^* + \sigma_g{}^*. \tag{8}$$

The in-phase and quadrature conductivity of the porous material with the colloidal suspension as pore water are related to the complex conductivity σ^* by $\sigma^* = \sigma' + i\sigma''$. Therefore, we have

$$\sigma' = \frac{1}{F}(\sigma_w + \bar{C}_B \bar{V}_B \rho_B (1-f)\beta_{(+)}\mathrm{CEC}_B) + \sigma_g' \quad (9)$$

and

$$\sigma'' = -\frac{1}{F}\bar{C}_B \bar{V}_B \rho_B f \beta^S_{(+)}\mathrm{CEC}_B + \sigma_g''. \quad (10)$$

The quadrature conductivity is the sum of the contribution associated with the grains (background polarization) and the contribution associated with the bacteria, dependent on the abundance of the bacteria but affected by the formation factor of the porous material. In other words, the total polarization of the material is not simply the sum of the polarization of the grains and the polarization of the bacteria. The in-phase conductivity can be written as

$$\sigma' = \frac{1}{F}\sigma_w + \sigma_S', \quad (11)$$

where σ_S' denotes the total surface conductivity of the porous material including the effect of the bacteria and the mineral grains as

$$\sigma_S' = \frac{1}{F}(\bar{C}_B \bar{V}_B \rho_B (1-f)\beta_{(+)}\mathrm{CEC}_B) + \sigma_g'. \quad (12)$$

The in-phase conductivity is not expected to be strongly dependent on the bacterial concentration because the surface conductivity of the bacteria is expected to be small with respect to the conductivity of the pore water σ_w. In contrast, the quadrature conductivity is expected to be directly proportional to the concentration of bacteria with a small contribution of the background quadrature conductivity (Revil et al., 2012a). We will investigate below what is the relative contribution of the quadrature conductivity due to the bacteria with respect to the grains. As discussed above, the slope of the quadrature conductivity versus the bacterial concentration is given by a_S for a suspension. In the case of a porous material, it is given by the following expression:

$$\sigma'' = -(a\bar{C}_B + b), \quad (13)$$

where

$$a = \frac{1}{F}\bar{V}_B \rho_B f \beta^S_{(+)}\mathrm{CEC}_B \quad (14)$$

and

$$b = -\sigma_g''. \quad (15)$$

Therefore, according to our model, the slope of the trend between the quadrature conductivity and the concentration of bacteria will depend on the formation factor of the material and the CEC of the bacteria. We expect therefore to have smaller sensitivity to change in the concentration of bacteria when the porosity decreases and the formation factor increases.

MATERIAL AND METHODS

Bacterial culture preparation

Z. mobilis is a rod-shaped Gram-negative bacterium with an average length of about 2.5 μm that grows optimally under anaerobic conditions but is an aerotolerant, motile fermenter with a highly active ethanol fermentation pathway (Grube et al., 2012). The optimum pH range for the growth of *Z. mobilis* is 5.5–7.0 and the optimum temperature range is 25–31°C (Gunasekaran et al., 1986). Cultures of *Z. mobilis* (strain ZM4 [CP4]/ATCC, American Type Culture Collection, #31821) were started by using −80°C *Z. mobilis* stocks as 9% inocula for 5 mL of a defined RM based on ATCC medium 1341, which included glucose 20 g L^{-1}, yeast extract 10 g L^{-1}, and KH_2PO_4 2 g L^{-1} (pH adjusted to 6). The electric conductivity of this original RM suspension was 0.277 S m^{-1} (initial conductivity of the medium with the 9% inoculum of cells). After 24 h of growth, this original culture was used as 9% inocula for 100 mL "production" cultures. The production cultures were then incubated in 250-mL Erlenmeyer flasks with foam stoppers (allowing air to circulate) at 30°C in a Fisher Isotemp incubator without shaking. The purity of the cell cultures was checked regularly by plating on Luria Bertani agar.

The growth curves of optical density at 600 nm (OD_{600}) versus incubation time for *Z. mobilis* in RM from two tests are shown in Figure 4. The OD_{600} measurements were made using a Genesys 10Vis spectrophotometer. *Z. mobilis* reached the stationary phase and a maximum OD_{600} (OD = 0.787) only after ~8.5 h (Figure 4), remained stable until ~18.5 h, and then the OD started to drop until the end of the measurement period (1500 min). The correlation of OD_{600} with actual cell numbers was determined by performing

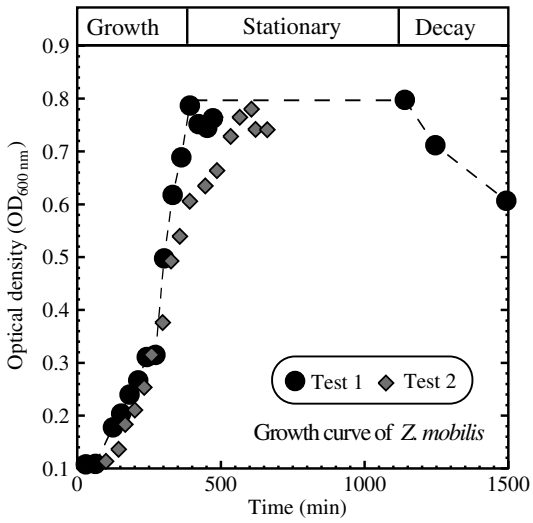


Figure 4. Growth curves. Plot for the growth of *Z. mobilis* measured at 30°C in RM showing the growth, stationary, and decay phases. *Z. mobilis* is characterized by a very fast growth rate. The two tests demonstrate the repeatability of the measurements. To maintain consistency across experiments in our study, we harvested cells in the growth phase of the curve for SIP measurements.

direct microscopic counts using a Petroff-Hausser counting chamber (1/400 mm^2, 0.02 mm) under phase contrast microscopy. The culture grown for 12 h (test 1, Figure 4) was collected and diluted with 0.85% NaCl solution for cell counting. Dilutions from the initial inocula showed a linear relationship between the OD measurements and the number of cells (Figure 5). To construct the calibration curve, seven observed ODs (the minimum is 0.177, and the maximum is 0.752) were chosen and the corresponding calculated cell numbers were plotted against the OD. A linear relationship exists between the cell number (density) and OD_{600} up to approximately $OD_{600} = 0.6$ (Figure 5). Then, the cultures were diluted in RM for three desired OD (three different dilutions) for SIP measurements. A second test (test 2, Figure 4) was performed to check the reproducibility of the growing phase.

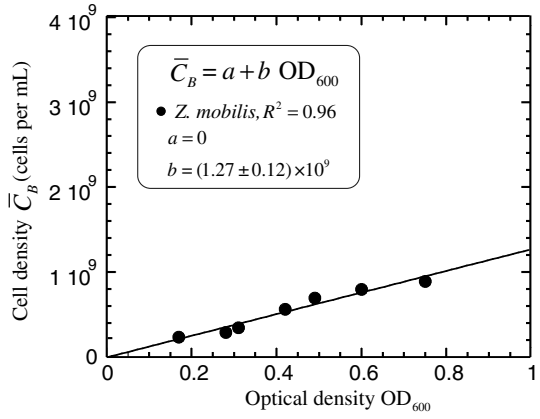


Figure 5. Plot of the observed OD_{600} against the direct count cell numbers per milliliter and associated linear regression fit for *Z. mobilis*.

The bacterial concentrations were estimated from the plot of cell numbers per milliliter against the corresponding OD_{600} readings (Figure 5). Three cell densities of 0.41×10^9, 0.35×10^9, and 0.23×10^9 cells mL^{-1} were used in the SIP experiments, and the corresponding fluid conductivities of these three cell suspensions were 0.025, 0.0165, and 0.0118 S m^{-1}. All samples were suspended in phosphate buffer at pH 5.75 prior to use in the experiments with sand.

Experimental procedure for induced polarization

The equipment used for the SIP measurement is shown in Figure 6. One hundred twenty-five ml wide-mouth cylindrical plastic jars (VWR polypropylene autoclavable bottles with screw closures, diameter: 5.02 cm) were modified for use as the sample holders (Figure 6a and 6b). Four pellets of biomedical, nonpolarizing Ag-AgCl electrodes with a spacing of 1.5 cm from each other were installed vertically along one side of the jar. The outer two electrodes (A and B) were used for current injection/retrieval, and the two middle electrodes (M and N) were used to record the potential (Wenner array, Figure 6a). For a Wenner array in a half-space, the geometric factor is given by $g = 2\pi s$, where s is the distance between the electrodes. In our case, this yields $g = 9.4$ cm for an ideal half-space electrode array. We performed a calibration of the geometric factor of the cell with a KCl salt solution at a known conductivity (0.025 S m^{-1}), and the calibrated geometric factor was $g = 5.6$ cm, which is the geometric factor used in our calculations. This shows how important such calibration is to get the correct value of the geometric factor.

To prevent electrode polarization that could obscure the polarization contribution from the bacteria, we use a four-electrode approach separating the voltage electrodes from the current electrodes. Only the current electrodes would display polarization because of the

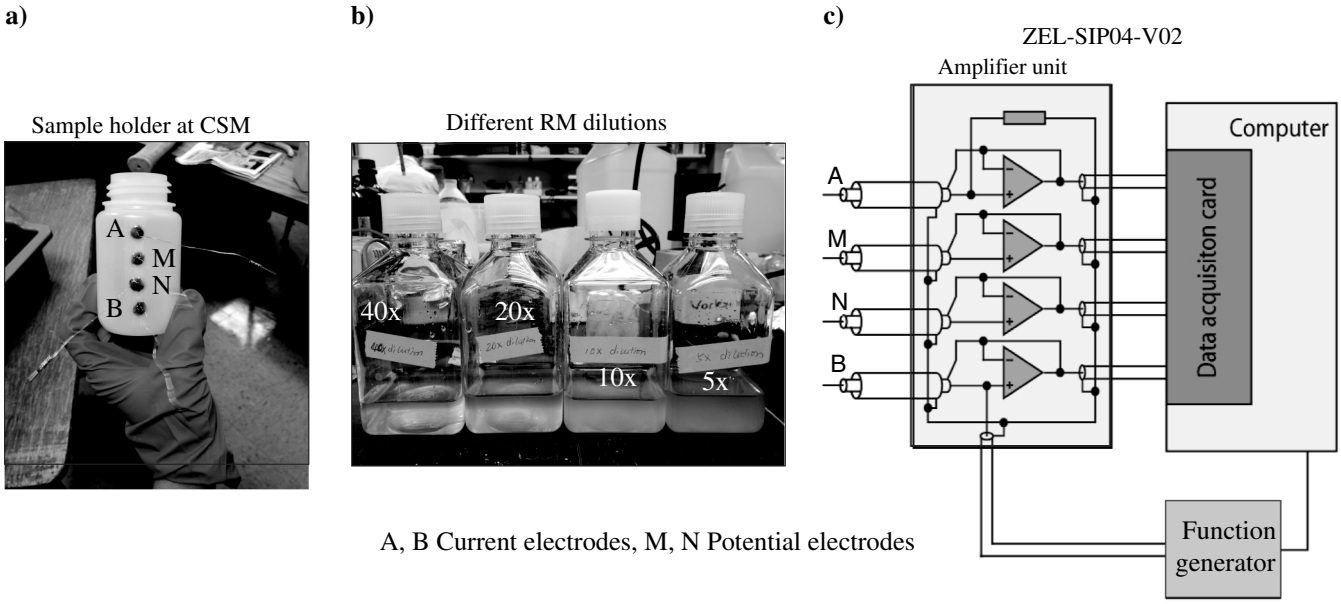


Figure 6. Experimental setup used for the SIP measurements. (a) Sample holder used with the ZEL-SIP04-V02 impedance meter showing the position of the 4 Ag/AgCl electrodes (Wenner array). (b) RM solutions with bacteria at four different dilutions (the most concentrated is on the right side). The geometric factors for the sample holder were calibrated using a solution of known conductivity. The factors 40× to 5× denote the different dilutions as explained in the main text. (c) ZEL-SIP04-V02 impedance meter.

change of the nature of the charge carriers at their surface (Kemna, 2000). Several benchmark tests of the impedance meter used in this study can be found in Revil and Skold (2011). The complex conductivity response obtained with water alone, as expected, does not display polarization effects (see Figure 5 of Revil and Skold, 2011).

The SIP measurements for the *Z. mobilis* experiments were taken using the ZEL-SIP04-V02 impedance meter (Zimmermann et al. [2008], Figure 6c). The accuracy on the phase is 0.1 mrad below 100 Hz (Zimmermann et al., 2008). The data acquisition system can operate in the frequency range from 1 mHz to 45 kHz with a phase accuracy of 0.1 mrad below 100 Hz (see Revil and Skold [2011] for various benchmark tests, including tests performed with electric circuits with a known resistance in parallel with a known capacitance).

For the SIP measurements, 125 mL of bacterial suspension at different cell concentrations was prepared, and each measurement was taken from 0.1 Hz to 10 kHz with 2 V input voltage. The time that would have been required to do low-frequency measurements, below 0.05 Hz, was too long with respect to the risk of lysis of the cells. The pH and fluid conductivity of each cell suspension sample were measured before and after the SIP measurements to make sure the physicochemical properties of the suspension remained the same. Then, 180 g of #70 autoclaved silica sand (physical properties provided in Table 1) was gently added to the 62.5 mL of each bacterial suspension sample to fill the bottles. We use a wet packing procedure to avoid trapping air bubbles and to facilitate homogeneous distribution of the sand without damaging the cells. Constant and gentle tapping on the container was applied during the introduction of the mixture of bacterial suspension and sand into the sample holders. The total volume of saturated sands and bacterial suspension was calculated and recorded for each packing. The SIP measurements were conducted at room temperature (21 ± 2°C) on the bacteria and sand mixtures.

Control experiments

We performed two sets of control experiments. In the first set of control experiments, the complex conductivity (in-phase and quadrature conductivity) of the sand was determined using NaCl solutions. The second set of control experiments was conducted using the same growth medium (RM) used for the biotic experiments but in the absence of cells. These control samples were incubated in the isothermal chamber at 30°C for the same time as for the cell samples.

EXPERIMENTAL RESULTS

Control experiments

The results of the control experiments with the NaCl solutions are shown in Figure 7a and 7b and with the RM solution without the cells in Figure 7c and 7d. The electric conductivity of the RM suspension indicates a formation factor for the RM suspension of $F_{RM} = 1.1$ (not shown); therefore, it is very close to the value for an infinite dilution (1.0). The electric conductivity of the sand with the NaCl solutions or the RM colloidal suspensions (no bacteria) at the four dilutions of the initial RM solution yields a formation factor of $F = 3.7$–3.8 (Figure 7a and 7c and reported in Table 1). The pore water conductivities of the RM solutions are

Table 1. Properties of the sand.

Properties	Sand grains
Material	Silica
Porosity ϕ	0.41[3]
Grain size d_{50}	200 μm
Formation factor F	3.7 ± 0.1[3]
Hydraulic conductivity Ks	1.4×10^{-4} m s^{-1}[3]
Surface conductivity,[1] σ_g'	1.2×10^{-4} S m^{-1}
[2]Quadrature conductivity,[2] σ_g''	$-(0.5 - 1.0 \times 10^{-5}$ S m$^{-1})$

[1]Bolève et al. (2007) for similar silica materials in terms of grain diameter.
[2]At 0.1 Hz.
[3]From Ikard et al. (2012).

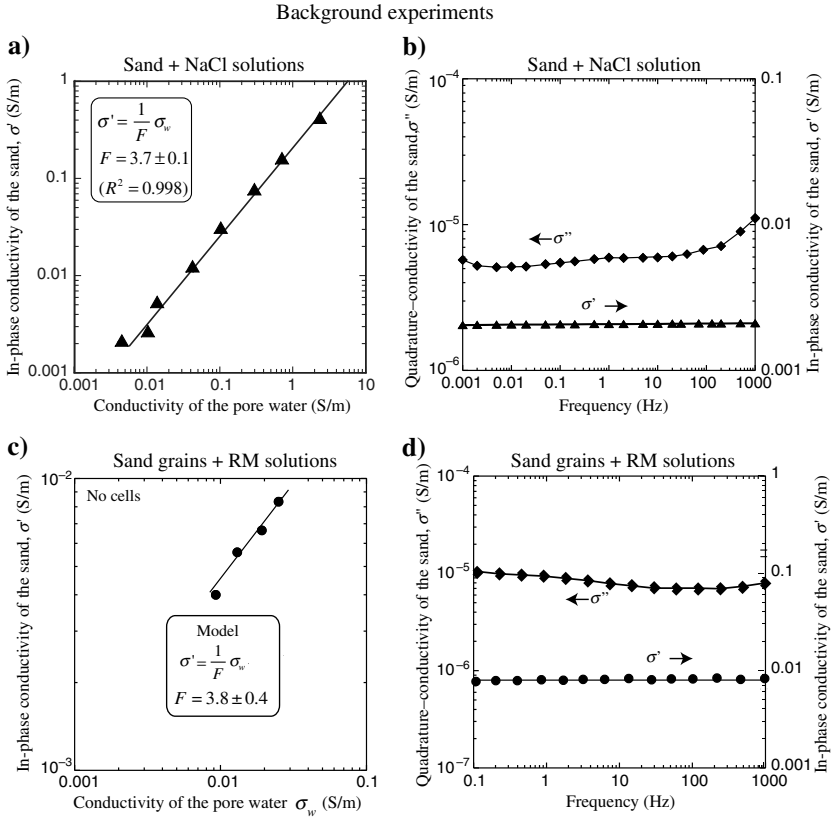


Figure 7. Control experiments: (a) Direct determination of the intrinsic formation factor $F = 3.7$ for the sand saturated by a NaCl solution. (b). In-phase and quadrature conductivities versus frequency for the sand saturated by an NaCl solution (0.0075 S m^{-1} at 25°C, pH = 7.1). (c) In-phase conductivity versus pore water conductivity for the sand saturated with the RM solution. (d) In-phase and quadrature conductivities versus frequency for the sand saturated by the RM solution.

reported in Table 2. The background quadrature conductivity for the sand filled with the RM colloidal suspensions without the cells is $-0.8 \pm 0.2 \times 10^{-5}$ S m^{-1}, which corresponds to the quadrature conductivity of the sand grains without the cells σ_g'' (Figure 7b and 7d, Table 1). These data provide baselines (background in-phase and quadrature conductivities) for the following experiments that will show the effect of the bacteria on the complex conductivity response.

Polarization of suspensions of bacteria

We now describe the results of the biotic experiments starting with the suspensions of bacterial cells. Figure 8a–8c shows the quadrature conductivity whereas Figure 8d shows the in-phase conductivity data for *Z. mobilis* colloidal suspensions at three different densities. The amplitude of the quadrature conductivity increased significantly as the cell concentration increased (Figure 8a–8c). For the bacterial suspensions, the quadrature conductivity spectra showed two distinct low-frequency dispersions: one peaking around 0.1 Hz (denoted as peak I in Figure 8a–8c) and a second one peaking around 5–10 Hz (denoted as peak II in Figure 8a–8c). At about 100 Hz, we started to see the high-frequency Maxwell-Wagner polarization (see the discussion in Revil et al., 2012a). The quadrature conductivity in the presence of bacteria (up to ~9.4×10^{-5} S m^{-1} for OD 0.338) was significantly higher than in the case of the control experiments (~1.0×10^{-5} S m^{-1}, see Figure 7d).

Table 2. Control experiments.

Control	OD_{600}	σ_w(S m^{-1})
1	0.338	0.0250
2	0.294	0.0192
3	0.189	0.0130
4	0.200	0.0093

In Figure 9, we plot the quadrature conductivity of different suspensions of *Z. mobilis* versus the density of the bacterial cells estimated from the OD_{600} measurements (see Figure 5). Taking $F_{RM} = 1$, $\beta^S_{(+)} = 5 \times 10^{-10}$ m^2 s^{-1} V^{-1} at 25°C, $f = 0.8$, $\rho_B = 10^3$ kg m^{-3}, and $\bar{V}_B = 1.0$ μm^3 (1×10^{-18} m^3), and the observed value of $a_S = 8.3 \times 10^{-14}$ m S^{-1} mL cell^{-1} (from the slope shown in Figure 9), we can determine the value of the CEC of the bacteria given from equation 6 by

$$CEC_B = \frac{a_S}{\bar{V}_B \rho_B f \beta^S_{(+)}}. \quad (16)$$

This yields $CEC_B = 2.1$ mMol g^{-1} (reported in Table 3). This value is consistent with the mean value for Gram-negative bacteria reported in Table 1 of Revil et al. (2012a): $CEC_B = 1.80$ mMol g^{-1}.

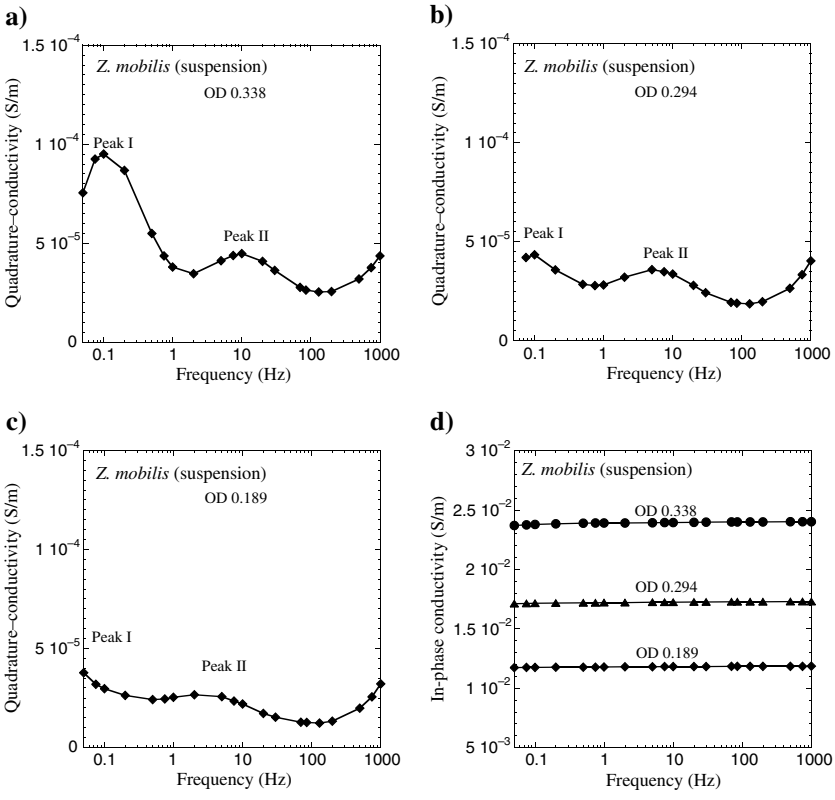


Figure 8. In-phase and quadrature conductivities for *Z. mobilis* cell suspensions at three different concentrations of bacteria (pH 5.75). Quadrature conductivities for (a) OD 0.338, (b) OD 0.294, (c) OD 0.189, and (d) in-phase conductivity for the three bacterial densities.

Suspensions in porous media

The quadrature and in-phase conductivities for the three different concentrations of *Z. mobilis* mixed with silica sand (properties summarized in Table 1) are shown in Figure 10a–10c. The in-phase conductivity increased from 0.0051 S m^{-1} to 0.0082 S m^{-1} (Figure 10d) as the OD increased from 0.189 to 0.338. The magnitude of the quadrature conductivity was also greater for higher cell concentrations (Figure 10a–10c). We still recognize two polarization peaks, but localized at relatively lower frequencies than in the suspensions (for OD 0.338 and OD 0.294). This observation will be discussed in more detail below. Note that in Figure 10, the measurements at the lowest frequencies (below 0.05 Hz) could be affected by cell lysis during the long measurement period.

The evolution of the quadrature conductivity at 0.1 and 5 Hz with the density of bacterial cells is shown in Figure 11a and 11b for peaks I and II, respectively. Our model predicts the following value: $a = a_S/F = 2.2 \times 10^{-5}$ m S^{-1} using $a_S = 8.3 \times 10^{-14}$ m S^{-1} mL cell^{-1} and $F = 3.7$. This value remains, however, smaller than the observed values $a_S = 3.4 \times 10^{-14}$ m S^{-1} mL cell^{-1} for peak II. A higher value is reached for peak I ($a_S = 10 \times 10^{-14}$ m S^{-1} mL cell^{-1}). Therefore, our model fails to accurately predict the dependence of the quadrature conductivity with the density of bacteria.

DISCUSSION

Some comments regarding the experimental results

Our model has been successful with respect to two points: (1) The CEC required to explain the experimental data is consistent with the CEC of Gram-negative bacteria and (2) the quadrature conductivity is proportional to the abundance of bacteria in the solution or in the porous material. We focus our discussion below on the effect of the porous material regarding the relationship between the quadrature conductivity and the abundance of bacteria in the pore water phase.

In Figure 12, we plot the quadrature conductivity of the colloidal suspension of *Z. mobilis* at OD 0.338 with and without the sand grains. The first observation is related to the presence of the two peaks in the quadrature conductivity, for the suspensions and for the suspensions in the porous materials. One possibility is that peak I corresponds to the polarization of the bacteria and peak II corresponds to the polarization of the protruding organic molecules around the bacteria themselves (Figures 1 and 2). In this case, the situation would be similar to what Leroy et al. (2008) observes for glass beads with the low-frequency peak related to the polarization of the grains and a higher frequency peak associated with the roughness of the grains (both corresponding to α-polarization mechanisms associated with the polarization of the electric double layer). Indeed, the roughness of the grains or the protruding organic molecules around the bacteria corresponds to smaller polarizable length scales by comparison with the sand grains or the bacteria, respectively, and is therefore expected to polarize at higher frequencies. For instance, micrometer-scale heterogeneities on a glass bead produce polarization in the range 1–100 Hz (Leroy et al., 2008).

In Figure 12, we clearly see both a reduction in the magnitude of the amplitude of the quadrature conductivity as well as a shift toward lower frequencies (approximately a factor of four in the presence of sand). This may indicate that the diffusion coefficient in the definition of the relaxation time may need a correction by the bulk tortuosity of the pore space. Indeed, the effective diffusion coefficient of a salt or an anionic species should be corrected at the macroscale of a representative elementary volume by the tortuosity of the pore network (see Leroy et al. [2006]; see Binley et al. [2010] for the SIP problem). The bulk tortuosity also affects the surface and quadrature conductivities directly according to the model developed by Revil (2013a). We expect indeed that the pathways for the migration of the ions need to be corrected for the tortuosity effect around the grains.

Figure 13 shows the high-frequency parts of the spectra for the suspension of *Z. mobilis*. We note that the high-frequency part of the spectra is dominated by a pure dielectric constant $\varepsilon_\infty \approx \varepsilon_r \varepsilon_0$ where ε_0 denotes the dielectric constant of a vacuum and $\varepsilon_r \approx 300$ represents the relative permittivity for the suspension. This high value (compared with the relative permittivity of water ~80) may be due to Maxwell-Wagner polarization (β-polarization). Because of the scaling with frequency, this polarization cannot be an α-polarization mechanism associated with, for instance, the polarization of the electric double layer around the protruding organic molecules around the cell of the bacteria.

In Figure 14, we plot the surface conductivity versus the quadrature conductivity of *Z. mobilis* and the sand used in this study. Our results agree with the data set discussed by Weller et al. (2013) for clayey media and are consistent with the prediction of the model of Revil (2013a, 2013b). This indicates, as suggested by Revil et al. (2012a), that the mobility of the counterions of the bacteria is very close to the mobility of the counterions for clay minerals. Using equations 2 and 3 and the data displayed in Figure 14, the ratio of the surface conductivity to the quadrature conductivity is given by

$$\frac{\sigma_B^{S\prime}}{|\sigma_B''|} = \frac{(1-f)\beta_{(+)}}{f\beta_{(+)}^S} \approx 23.8. \tag{17}$$

Assuming $\beta_{(+)}^S = 5 \times 10^{-10}$ m^2 s^{-1} V^{-1} (Revil et al., 2012a) and $\beta_{(+)} = 5 \times 10^{-8}$ m^2 s^{-1} V^{-1} (mobility of Na^+ in water), we obtain $f = 0.81$. This probably represents the fraction of counterions located in the Stern layer (~80% in this case), in close agreement with the discussion provided by Revil et al. (2012a). Figure 14 also indicates that the surface conductivity and quadrature conductivity of *Z. mobilis* are on the same order of magnitude as the surface and quadrature conductivities of minerals. Additional data will be needed to confirm this finding. Therefore, only time-lapse studies can be used to decipher, in the electric signals, any changes associated with growth and decay of bacterial populations.

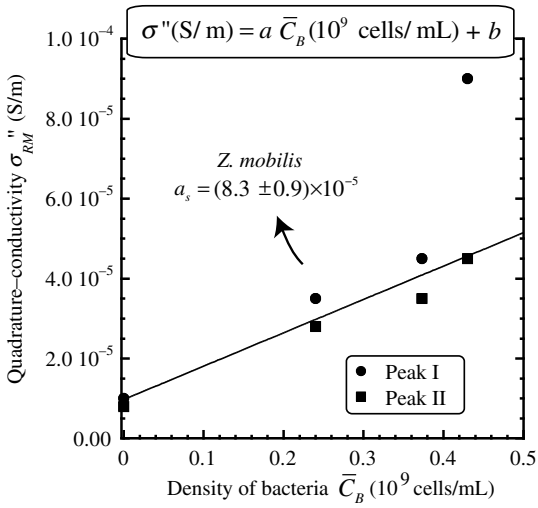


Figure 9. Quadrature conductivity versus bacteria density for different concentrations of bacteria. The data are taken at the two peak frequencies. Fitted parameters $a = (8.3 \pm 0.9) \times 10^{-5}$ S m^{-1} and $b = (1.0 \pm 0.3) \times 10^{-5}$ S m^{-1} ($R^2 = 0.934$).

Table 3. Properties of *Z. mobilis*.

Properties	*Z. mobilis*
Stern later mobility $\beta_{(+)}^S$	5×10^{-10} m^2 s^{-1} V^{-1}
Cation exchange capacity CEC_B	2×10^5 C kg^{-1}
Partition coefficient f	0.8
Mass density ρ_B	10^3 kg m^{-3}
Elementary volume $\bar{V}_B$	1.0 μm^3 (10^{-18} m^3)

Potential applications and future works

If our goal is to understand the SIP response during the monitoring of bacterial activity in the shallow subsurface associated with remediation activities such as oil spills, then we need to understand the various contributions to the electric response. We believe that there are three major contributions to the changes observed in SIP signatures during oil biodegradation. The first is the contribution associated with the presence of the oil itself. As shown for instance by Schmutz et al. (2010), Revil et al. (2011), and Abdel Aal and Atekwana (2014), oil saturation and wettability play a central role in the SIP response of oil-bearing media. Note that the wettability of oil can change over time in an oil spill due to the formation of biopolymers that change the surface tension of oil.

The second contribution is due to the presence of bacteria. Their growth and decay can be described by Monod kinetics. Revil et al. (2012a) provide a model for this contribution (especially in the case of Gram-positive bacteria), and the present work extends this research to Gram-negative bacteria. The present work can be used to predict semiquantitatively the change in the quadrature conductivity associated with the presence of bacteria. As our data suggest, the magnitude of the quadrature conductivity increases with cell density. This provides us with a rough approximation of what we can observe in field conditions according to our model. Kostka et al. (2011) studied beach sands contaminated by the Deepwater Horizon oil spill in the Gulf Coast of Mexico. They found a concentration of oil-degrading bacteria of 2.4×10^{16} cells m^{-3} (24×10^{9} cells mL^{-1}) in oil-contaminated sands, whereas the background concentration was on the order of 2 to 9×10^{12} cells m^{-3} ($0.02-0.09 \times 10^{9}$ cells mL^{-1}). Our work implies that for sand, the variation in quadrature conductivity $\Delta\sigma''$ associated with a variation in bacterial cells concentration $\Delta\bar{C}_B$ is given by $\Delta\sigma'' = a\Delta\bar{C}_B$ (at 0.1 Hz) with $a = 10^{-4}$ S mL m^{-1} $cells^{-1}$ and with the concentration of bacteria expressed in 10^9 cells mL^{-1}. Therefore, a change in bacteria concentration of 24×10^9 cells mL^{-1} would correspond to a highly detectable change of 2.4×10^{-3} S m^{-1} (at 0.1 Hz) in the quadrature conductivity (assuming that the phase is greater than the detectable value of 0.5 mrad in field conditions).

The third contribution is the one associated with the presence of semiconductors such as pyrite and magnetite that can be produced by the bacteria themselves (e.g., Mewafy et al. [2013], in the case of magnetite). Despite the fact that this contribution has been clearly demonstrated in the laboratory, challenges remain with respect to understanding its nature and the effect of the redox conditions in the pore water. Future works will be dedicated to better understanding this contribution by expanding the work of Wong (1979). The ultimate goal is to predict the SIP response of bacterial activity over time by integrating the three contributions in a unified mechanistic framework.

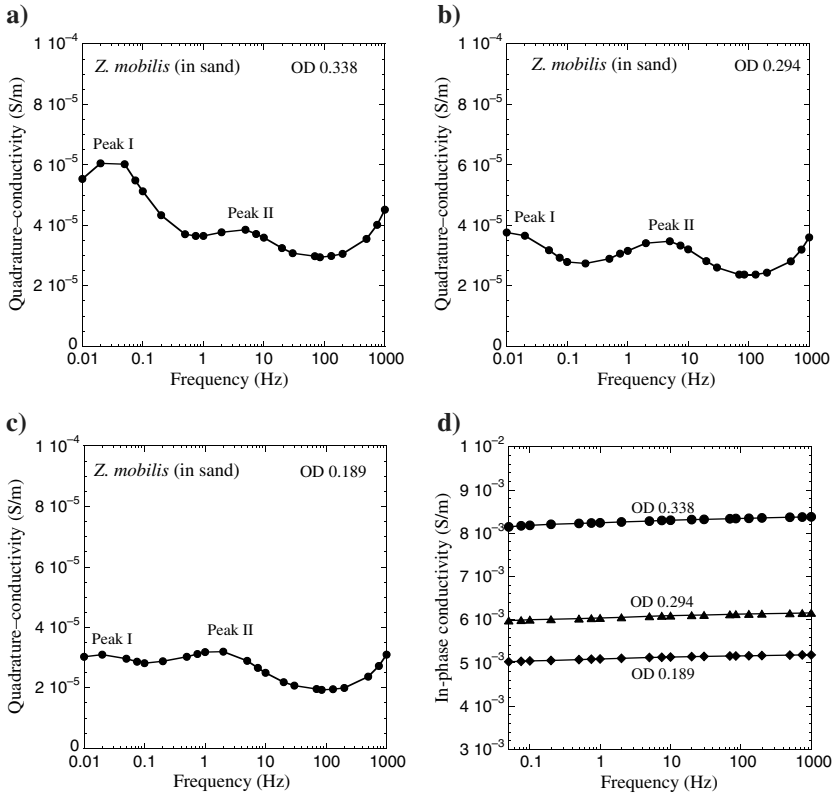


Figure 10. The in-phase σ' and quadrature conductivity σ'' for *Z. mobilis* cell suspensions in sand at three different densities of bacteria. Quadrature conductivities for (a) OD 0.338, (b) OD 0.294, (c) OD 0.189, and (d) in-phase conductivity for the three bacterial cell densities.

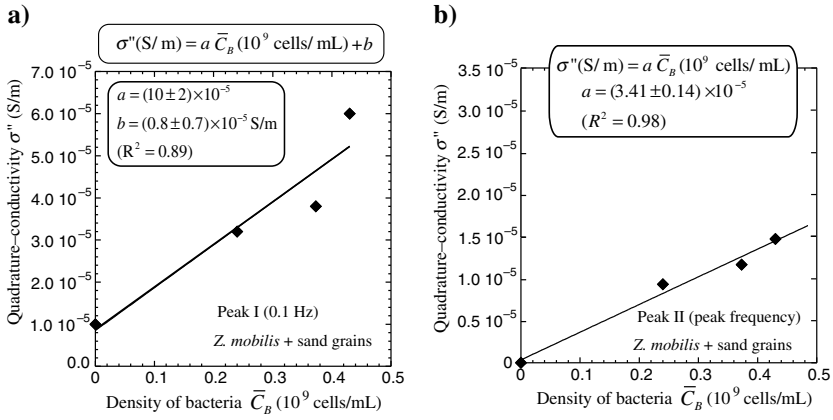


Figure 11. Quadrature conductivity versus the density of bacteria at (a) 0.1 Hz and (b) 5 Hz for *Z. mobilis* with sand. The data are fitted by equation 13. The slope between the quadrature conductivity and the density of bacteria is substantially smaller than in the case of the suspension of the same bacteria in the absence of the solids (Figure 9). From the value of b, we have the value of the quadrature conductivity of the grains: $\sigma_g'' = -0.8 \times 10^{-5}$ S m^{-1} for the sand grains.

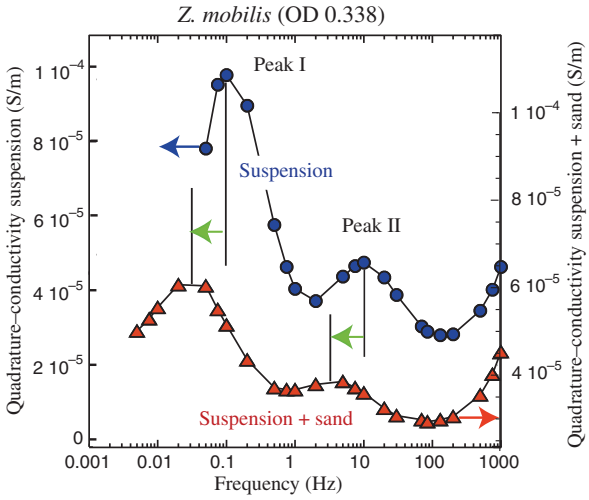


Figure 12. Comparison between the quadrature conductivity for a suspension of *Z. mobilis* (OD 0.338) with and without sand grains. The presence of the sand grains decreases the amplitude of the polarization and also displaces the polarization peaks I and II to lower frequencies (see green arrows).

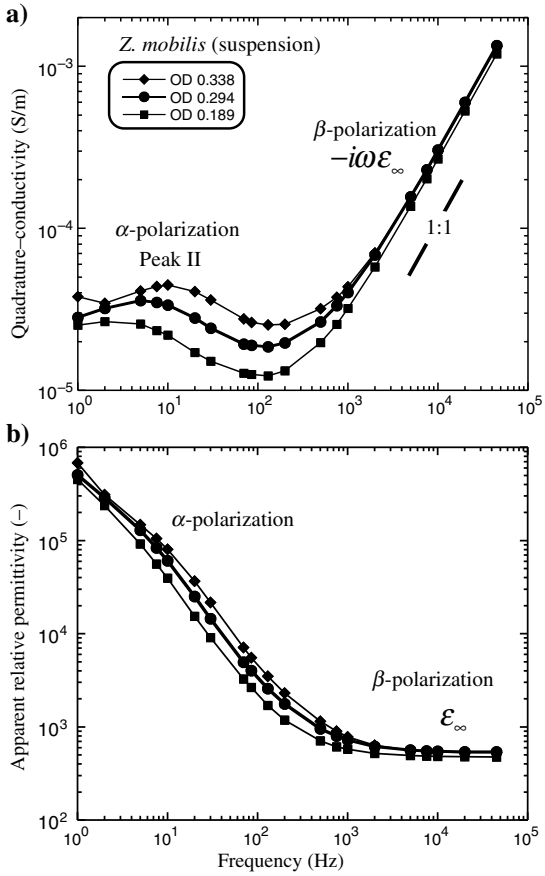


Figure 13. High-frequency part of the quadrature conductivity spectra for *Z. mobilis* (colloidal suspension with RM). The influence of the bacteria is clearly much smaller above 1 kHz. The α- and β-polarizations correspond to the polarization of the electric double layer and Maxwell-Wagner polarization, respectively.

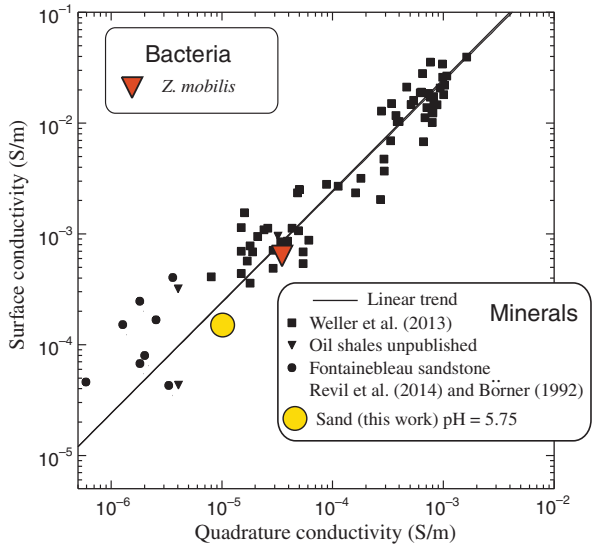


Figure 14. Comparison of the relationship determined by Weller et al. (2013) for clayey materials and additional data including oil shales with very low porosity and samples of Fontainebleau sandstones (Revil et al. [2014] and sample F3 from Börner, 1992). As discussed by Revil et al. (2012), the bacteria seem to be characterized by a low mobility for the counterions that is similar to that of clays. In both the cases, the surface conductivity was determined from the conductivity data at different salinities and the quadrature conductivity is reported at the peak frequency of peak I.

CONCLUSIONS

We have performed complex conductivity measurements using the Gram-negative bacterium *Z. mobilis* at different concentrations of bacterial cells. The measurements were done with suspensions of cells and by mixing these suspensions with sand grains. The following conclusions have been reached:

1) The quadrature conductivity spectra exhibit two peaks, one around 0.05–0.5 Hz (peak I) and another around 1–10 Hz (peak II). For peaks I and II, the experiments show that the amplitude of the quadrature conductivity depends on the density of cells for the suspensions of bacteria and the mixtures of sand with these cells.
2) The trend between the quadrature conductivity of the cell suspension and the density of the bacteria can be qualitatively reproduced with a simple polarization model of the electric double layer. However, the quantitative relationship between the quadrature conductivity and the density of bacteria in porous media in these experiments was not well predicted by our model. Further studies, e.g., with different bacteria and with different porous media, are needed to evaluate this relationship.
3) The ratio between the surface conductivity of the bacteria and their quadrature conductivity seems to be consistent with the ratio found for the minerals, indicating that the mobility of the counterions in the Stern layer is substantially smaller than the mobility of the counterions in the diffuse layer and that approximately 80% of the counterions are located in the Stern layer.

ACKNOWLEDGMENTS

This work is funded by Idaho National Laboratory Scientific Focus Area, the Office of Science, and Chevron Energy Technology

Company (grant #CW852844). We thank J. Henriksen for advice regarding the choice of bacterial species for the experiments and G. Abdel Aal and E. Atekwana for their comments on a previous version of the manuscript. Y. Fujita and G. Redden would also like to express their deep gratitude to the NanoGeoScience program at Copenhagen University for facilitating their contributions to the preparation of this manuscript, and especially to the National Bank of Denmark for helping to make their residence with Copenhagen University possible. We also thank the three referees for the constructive reviews and the associate editor E. Saenger for handling our manuscript.

REFERENCES

Abdel Aal, G., E. Atekwana, L. Slater, and E. Atekwana, 2004, Effects of bacterial processes on electrolytic and interfacial electrical properties of unconsolidated sediments: Geophysical Research Letters, **31**, L12505doi: 10.1029/2004GL020030.

Abdel Aal, G. Z., and E. A. Atekwana, 2014, Spectral induced polarization (SIP) response of biodegraded oil in porous media: Geophysical Journal International, **196**, 804–817, doi: 10.1093/gji/ggt416.

Abdel Aal, G. Z., E. A. Atekwana, S. Rossbach, and D. D. Werkema, 2010, Sensitivity of geoelectrical measurements to the presence of bacteria in porous media: Journal of Geophysical Research, **115**, G03017, doi: 10.1029/2009JG001159.

Albrecht, R., J. C. Gourry, M.-O. Simonnot, and C. Leyval, 2011, Complex conductivity response to bacterial growth and biofilm formation on phenanthrene spiked medium: Journal of Applied Geophysics, **75**, 558–564, doi: 10.1016/j.jappgeo.2011.09.001.

Binley, A., S. Kruschwitz, D. Lesmes, and N. Kettridge, 2010, Exploiting the temperature effects on low frequency electrical spectra of sandstone: A comparison of effective diffusion path lengths: Geophysics, **75**, no. 6, A43–A46, doi: 10.1190/1.3483815.

Bolève, A., A. Crespy, A. Revil, F. Janod, and J. L. Mattiuzzo, 2007, Streaming potentials of granular media: Influence of the Dukhin and Reynolds numbers: Journal of Geophysical Research, **112**, B08204, doi: 10.1029/2006JB004673.

Börner, F. D., 1992, Complex conductivity measurements of reservoir properties: *in* Chardaire-Riviere, C., and P. F. Worthington, eds., Proceedings of the Third European Core Analysis Symposium, Harwood Academic, 359–386.

Bratbak, G., and I. Dundas, 1984, Bacterial dry matter content and biomass estimations: Applied and Environmental Microbiology, **48**, 755–757.

Cockett, A. R., and A. Pidlisecky, 2014, Simulated electrical conductivity response of clogging mechanisms for managed aquifer recharge: Geophysics, **79**, no. 2, D81–D89, doi: 10.1190/geo2013-0054.1.

Davis, C. A., E. Atekwana, L. D. Slater, S. Rossbach, and M. R. Mormile, 2006, Bacterial growth and biofilm formation in geologic media is detected with complex conductivity measurements: Geophysical Research Letters, **33**, L18403, doi: 10.1029/2006GL027312.

Eisele, T. C., and K. L. Gabby, 2014, Review of reductive leaching of iron by anaerobic bacteria: Mineral Processes and Extractive Metallurgy Review, **35**, 75–105, doi: 10.1080/08827508.2012.703627.

Foster, K. R., and H. P. Schwan, 1989, Dielectric properties of tissues and biological materials: A critical review: Critical Reviews in Biomedical Engineering, **17**, 25–104.

Ginn, T. R., B. D. Wood, K. E. Nelson, T. D. Scheibe, E. M. Murphy, and T. P. Clement, 2002, Processes in bacterial transport in the natural subsurface: Advances in Water Resources **25**, 1017–1042, doi: 10.1016/S0309-1708(02)00046-5.

Grabowski, A., O. Nercessian, F. Fayolle, D. Blanchet, and C. Jeanthon, 2005, Bacterial diversity in production waters of a low-temperature biodegraded oil reservoir: FEMS Microbiology Ecology, **54**, 427–443, doi: 10.1016/j.femsec.2005.05.007.

Grech, R. T., J. Cassar, K. P. Muscat, S. G. Camilleri, M. Fabri, P. Zervakis, V. Xanthopoulos, B. Sakkalis, and B. Vanrumste, 2008, Review on solving the inverse problem in EEG source analysis: Journal of NeuroEngineering and Rehabilitation, **5**, 25, doi: 10.1186/1743-0003-5-25.

Grube, M., R. Rutkis, M. Gavare, Z. Lasa, I. Strazdina, N. Galinina, and U. Kalnenieks, 2012, Application of FT-IR spectroscopy for fingerprinting of *Zymomonas mobilis* respiratory mutants: Spectroscopy: An International Journal, **27**, 581–585, doi: 10.1155/2012/163712.

Gunasekaran, P., T. Karunakaran, and M. Kasthuribai, 1986, Fermentation pattern of *Zymomonas mobilis* strains on different substrates — A comparative study: Journal of Bioscience, **10**, 181–186, doi: 10.1007/BF02703475.

Heenan, J., A. Porter, D. Ntarlagiannis, L. Young, D. Werkema, and L. Slater, 2013, Sensitivity of the spectral induced polarization method to microbial enhanced oil recovery processes: Geophysics, **78**, no. 5, E261–E269, doi: 10.1190/geo2013-0085.1.

Ikard, S. J., A. Revil, A. Jardani, W. F. Woodruff, M. Parekh, and M. Mooney, 2012, Saline pulse test monitoring with the self-potential method to non-intrusively determine the velocity of the pore water in leaking areas of earth dams and embankments: Water Resources Research, **48**, W04201, doi: 10.1029/2010WR010247.

Kemna, A., 2000, Tomographic inversion of complex resistivity: Dissertation, Ruhr Universität Bochum.

Kemna, A., A. Binley, G. Cassiani, E. Niederleithinger, A. Revil, L. Slater, K. H. Williams, A. Flores Orozco, A. F.-H. Haegel, A. Hördt, S. Kruschwitz, V. Leroux, K. Titov, and E. Zimmermann, 2012, An overview of the spectral induced polarization method for near-surface applications: Near Surface Geophysics, **10**, 453–468, doi: 10.3997/1873-0604.2012027.

Kostka, J. E., O. Prakash, W. A. Overholt, S. J. Green, G. Freyer, A. Canion, and M. Huettel, 2011, Hydrocarbon-Degrading Bacteria and the Bacterial Community Response in Gulf of Mexico Beach Sands Impacted by the Deepwater Horizon Oil Spill: Applied and Environmental Microbiology, **77**, no. 22, 7962–7974, doi: 10.1128/AEM.05402-11.

Leroy, P., A. Revil, and D. Coelho, 2006, Diffusion of ionic species in bentonite: Journal of Colloid and Interface Science, **296**, 248–255, doi: 10.1016/j.jcis.2005.08.034.

Leroy, P., A. Revil, A. Kemna, P. Cosenza, and A. Ghorbani, 2008, Spectral induced polarization of water-saturated packs of glass beads: Journal of Colloid and Interface Science, **321**, 103–117, doi: 10.1016/j.jcis.2007.12.031.

Linde, N., and A. Revil, 2007, Inverting self-potential data for redox potentials of contaminant plumes: Geophysical Research Letters, **34**, L14302, doi: 10.1029/2007GL030084.

Mewafy, F. M., D. D. Werkema, E. A. Atekwana, L. D. Slater, G. Abdel Aal, A. Revil, and D. Ntarlagiannis, 2013, Evidence that bio-metallic mineral precipitation enhances the complex conductivity response at a hydrocarbon contaminated site: Journal of Applied Geophysics, **98**, 113–123, doi: 10.1016/j.jappgeo.2013.08.011.

Naudet, V., and A. Revil, 2005, A sandbox experiment of the relationship between redox and self-potential and its application to the interpretation of self-potential data over contaminant plumes: Geophysical Research Letters, **32**, L11405, doi: 10.1029/2005GL022735.

Naudet, V., A. Revil, J.-Y. Bottero, and P. Bégassat, 2003, Relationship between self-potential (SP) signals and redox conditions in contaminated groundwater: Geophysical Research Letters, **30**, 2091, doi: 10.1029/2003GL018096.

Naudet, V., A. Revil, E. Rizzo, J.-Y. Bottero, and P. Bégassat, 2004, Groundwater redox conditions and conductivity in a contaminant plume from geoelectrical investigations: Hydrology and Earth System Sciences (HESS), **8**, 8–22, doi: 10.5194/hess-8-8-2004.

Ntarlagiannis, D., K. Williams, L. Slater, and S. Hubbard, 2005b, Low-frequency electrical response to bacterial induced sulfide precipitation: Journal of Geophysical Research, **110**, G02009, doi: 10.1029/2005JG000024.

Ntarlagiannis, D., N. Yee, and L. Slater, 2005a, On the low-frequency electrical polarization of bacterial cells in sands: Geophysical Research Letters, **32**, L24402, doi: 10.1029/2005GL024751.

Personna, Y. R., D. Ntarlagiannis, L. Slater, N. Yee, M. O'Brien, and S. Hubbard, 2008, Spectral induced polarization and electrodic potential monitoring of bacterially mediated iron sulfide transformations: Journal of Geophysical Research, **113**, G02020, doi: 10.1029/2007JG000614.

Pethig, R., and G. H. Markx, 1997, Applications of dielectrophoresis in biotechnology: Trends in Biotechnology, **15**, 426–432, doi: 10.1016/S0167-7799(97)01096-2.

Revil, A., 2013a, Effective conductivity and permittivity of unsaturated porous materials in the frequency range 1 mHz–1 GHz: Water Resources Research, **49**, 306–327, doi: 10.1029/2012WR012700.

Revil, A., 2013b, On charge accumulations in heterogeneous porous materials under the influence of an electrical field: Geophysics, **78**, no. 4, D271–D291, doi: 10.1190/geo2012-0503.1.

Revil, A., E. Atekwana, C. Zhang, A. Jardani, and S. Smith, 2012a, A new model for the spectral induced polarization signature of bacterial growth in porous media: Water Resources Research, **48**, W09545, doi: 10.1029/2012WR011965.

Revil, A., M. Karaoulis, T. Johnson, and A. Kemna, 2012b, Review: Some low-frequency electrical methods for subsurface characterization and monitoring in hydrogeology: Hydrogeology Journal, **20**, 617–658, doi: 10.1007/s10040-011-0819-x.

Revil, A., P. Kessouri, and C. Torres-Verdín, 2014, Electrical conductivity, induced polarization, and permeability of the Fontainebleau sandstone: Geophysics, **79**, no. 5, D301–D318, doi: 10.1190/geo2014-0036.1.

Revil, A., C. A. Mendonça, E. Atekwana, B. Kulessa, S. S. Hubbard, and K. Bolhen, 2010, Understanding biogeobatteries: Where geophysics meets microbiology: Journal of Geophysical Research, **115**, G00G02, doi: 10.1029/2009JG001065.

Revil, A., M. Schmutz, and M. L. Batzle, 2011, Influence of oil wettability upon spectral induced polarization of oil-bearing sands: Geophysics, **76**, no. 5, A31–A36, doi: 10.1190/geo2011-0006.1.

Revil, A., and M. Skold, 2011, Salinity dependence of spectral induced polarization in sands and sandstones: Geophysical Journal International, **187**, 813–824, doi: 10.1111/j.1365-246X.2011.05181.x.

Revil, A., W. F. Woodruff, C. Torres-Verdín, and M. Prasad, 2013, Complex conductivity tensor of hydrocarbon-bearing shales and mudrocks: Geophysics, **78**, no. 6, D403–D418, doi: 10.1190/geo2013-0100.1.

Risgaard-Petersen, N., L. R. Damgaard, A. Revil, and L. P. Nielsen, 2014, Mapping electron sources and sinks of microbial generated electric currents in marine sediment: Journal of Geophysical Research, doi: 10.1002/2014JG002673.

Risgaard-Petersen, N., A. Revil, P. Meister, and L. P. Nielsen, 2012, Sulfur, iron-, and calcium cycling associated with natural electric currents running through marine sediment: Geochimica et Cosmochimica Acta, **92**, 1–13, doi: 10.1016/j.gca.2012.05.036.

Schmutz, M., A. Revil, P. Vaudelet, M. Batzle, P. Femenía Viñao, and D. D. Werkema, 2010, Influence of oil saturation upon spectral induced polarization of oil bearing sands: Geophysical Journal International, **183**, 211–224 doi: 10.1111/j.1365-246X.2010.04751.x.

Toledo, F. L., C. Calvo, B. Rodelas, and J. Gonzales-Lopez, 2006, Selection and identification of bacteria isolated from waste crude oil with polycyclic aromatic hydrocarbons removal capacities: Systematic and Applied Microbiology, **29**, 244–252, doi: 10.1016/j.syapm.2005.09.003.

Weller, A., L. Slater, and S. Nordsiek, 2013, On the relationship between induced polarization and surface conductivity: Implications for petrophysical interpretation of electrical measurements: Geophysics, **78**, no. 5, D315–D325, doi: 10.1190/geo2013-0076.1.

Williams, K. H., D. Ntarlagiannis, L. D. Slater, A. Dohnalkova, S. S. Hubbard, and J. F. Banfield, 2005, Geophysical imaging of stimulated bacterial biomineralization: Environmental Science & Technology, **39**, 7592–7600, doi: 10.1021/es0504035.

Willis, M. R., 1980, Dielectric and electronic properties of biological materials, *in* R. Pethig, ed., Biochemical education: John Wiley & Sons, vol. 8, 31.

Wong, J., 1979, An electrochemical model of the induced-polarization phenomenon in disseminated sulfide ores: Geophysics, **44**, 1245–1265, doi: 10.1190/1.1441005.

Woodruff, W. F., A. Revil, and C. Torres-Verdín, 2014, Laboratory determination of the complex conductivity tensor of unconventional shales: Geophysics, **79**, no. 5, E183–E200, doi: 10.1190/GEO2013-0367.1.

Zhang, C., D. Ntarlagiannis, L. Slater, and R. Doherty, 2010, Monitoring bacterial sulfate reduction in porous media using multipurpose electrodes: Journal of Geophysical Research, **115**, G00G09, doi: 10.1029/2009JG001157.

Zhang, C., L. Slater, and C. Prodan, 2013, Complex dielectric properties of sulfate-reducing bacteria suspensions: Geomicrobiology Journal, **30**, 490–496, doi: 10.1080/01490451.2012.719997.

Zimmermann, E., A. Kemna, J. Berwix, W. Glaas, H. M. Munch, and J. A. Huisman, 2008, A high-accuracy impedance spectrometer for measuring sediments with low polarizability: Measurement Science and Technology, **19**, 105603, doi: 10.1088/0957-0233/19/10/105603.

01_张弛老师第一作者语言样本.pdf

文献 5/6

A critical mini-review on the low-field nuclear magnetic resonance investigation of pore coupling effects in near-surface environments

Francisca Soto Bravo; Chi Zhang

Frontiers in Water (2023); DOI: 10.3389/frwa.2023.1059128

证据用途：第一作者直接语言样本；张弛老师角色：共同第一/通讯作者；证据等级：A

frontiers | Frontiers in Water

TYPE Mini Review
PUBLISHED 05 May 2023
DOI 10.3389/frwa.2023.1059128

Check for updates

OPEN ACCESS

EDITED BY
Alex Furman,
Technion Israel Institute of Technology, Israel

REVIEWED BY
Jianchao Cai,
China University of Petroleum, China
Stephan Costabel,
Federal Institute For Geosciences and Natural Resources, Germany
Ran A. N.,
Wuhan University of Science and Technology, China

*CORRESPONDENCE
Chi Zhang
✉ chi.zhang@univie.ac.at

†These authors have contributed equally to this work and share first authorship

RECEIVED 30 September 2022
ACCEPTED 11 April 2023
PUBLISHED 05 May 2023

CITATION
Soto Bravo F and Zhang C (2023) A critical mini-review on the low-field nuclear magnetic resonance investigation of pore coupling effects in near-surface environments. *Front. Water* 5:1059128. doi: 10.3389/frwa.2023.1059128

COPYRIGHT
© 2023 Soto Bravo and Zhang. This is an open-access article distributed under the terms of the Creative Commons Attribution License (CC BY). The use, distribution or reproduction in other forums is permitted, provided the original author(s) and the copyright owner(s) are credited and that the original publication in this journal is cited, in accordance with accepted academic practice. No use, distribution or reproduction is permitted which does not comply with these terms.

A critical mini-review on the low-field nuclear magnetic resonance investigation of pore coupling effects in near-surface environments

Francisca Soto Bravo[1†] and Chi Zhang[1,2*†]

[1]Environmental Geophysics, Institute of Meteorology and Geophysics, Faculty of Earth Sciences, Geography and Astronomy, University of Vienna, Vienna, Austria, [2]State Key Laboratory of Geomechanics and Geotechnical Engineering, Institute of Rock and Soil Mechanics, Chinese Academy of Sciences, Wuhan, China

The assessment and monitoring of groundwater resources is of increasing importance to ensure the continuous supply of fresh water for human activity and endangered ecosystems. These groundwater resources include fully saturated aquifers, water in unsaturated soil, and water trapped as rock moisture in weathered bedrocks. Low-field nuclear magnetic resonance (NMR) is a method with unique sensitivity to pore water, as it is based on the magnetization and relaxation behavior of the spin magnetic moment of hydrogen atoms forming water molecules. It is a cost-effective and minimally-invasive technology that can help characterize the pore structures and the groundwater distribution and transport in different types of subsurface materials. However, the interpretation of NMR data from samples with complex bimodal or multimodal porous geometries requires the consideration of pore coupling effects. A pore-coupled system presents significant magnetization exchange between macro- and micropores within the measurement time, making the independent characterization of each pore environment difficult. Developing a better understanding of pore coupling is of great importance for the accurate estimation of hydrogeological parameters from NMR data. This mini-review presents the state-of-art in research exploring the two factors controlling pore coupling: surface geochemistry and network connectivity, summarizes existing experimental and numerical modeling approaches that have been used to study pore coupling and discusses the pore coupling effects in fully and partially saturated conditions. At the end of this review, we outline major knowledge gaps and highlight the research needs in the vadose zone.

KEYWORDS

low-field NMR, pore coupling effects, near-surface environment, hydrogeophysics, critical zone, unsaturated condition, vadose zone

1. Introduction

The exploration and management of groundwater resources is of increasing importance in the context of climate change. Monitoring aquifers and understanding the transport of water and contaminants through soil is vital to elucidate critical zone functions, forecast ecosystem processes, and ensure the sustainable exploitation of groundwater resources.

The low-field nuclear magnetic resonance (NMR) method, based on the relaxation behavior of protons within external magnetic fields, is a non-invasive geophysical method gaining popularity for near-surface research, especially for hydrogeologic investigations. Due to its unique direct sensitivity to pore water, NMR is used to characterize the structure of porous media and to quantitatively estimate water contents and distributions. The deployment of low-field NMR measurements can be done in the laboratory (labNMR) with benchtop instrument for core samples, as a logging tool in boreholes (bNMR), or at Earth's surface using Earth's magnetic field (sNMR) (Behroozmand et al., 2015; Binley et al., 2015). Numerous laboratory studies and field applications encouraged the development of low-field NMR techniques including theory, modeling, inversion, and field instrumentation (Behroozmand et al., 2015). Besides groundwater exploration and aquifer characterization, other NMR applications include the monitoring of infiltration processes in controlled flood events (e.g., Walsh et al., 2014), the delineation of weathering profiles in fractured rocks (e.g., Flinchum et al., 2019), the study of frost damage (e.g., Ding et al., 2020) and freeze-thaw processes in general (e.g., An et al., 2022), as well as the monitoring of melting soil (e.g., Watanabe and Wake, 2009) and permafrost (e.g., Parsekian et al., 2013; Kass et al., 2017; Sullivan, 2022). Further applications for near surface characterization, as well as detailed explanations on the working principles of NMR, are reviewed in Behroozmand et al. (2015).

The key assumption that each pore environment relaxes independently fails when interpreting NMR signals in scenarios presenting pore coupling effects. Pore coupling distorts the NMR-gained relaxation time distributions, characterizing an averaged version of the pore network instead of identifying individual pore environments (e.g., Grunewald and Knight, 2009). This represents a non-negligible effect in accurately applying NMR tools in heterogeneous subsurface materials with dual or multimode pore systems, where magnetisation can be exchanged between pore environments during the NMR measurement. Pore coupling effects were first described by Brownstein and Tarr (1979) and have been observed in fully-saturated rocks and unconsolidated material through laboratory measurements (e.g., Toumelin et al., 2003; Anand and Hirasaki, 2007; Grunewald and Knight, 2009) and numerical simulation studies (e.g., Mohnke and Klitzsch, 2010; Chi and Heidari, 2015; Mitchell et al., 2019). It is important to understand and quantify pore coupling effects and the factors that drive their occurrence and severity.

Although magnetisation exchange can take place in multiple settings beyond the critical zone, e.g., in organic rich shale (Washburn and Cheng, 2017), and research on the application of NMR in unconventional reservoirs under multi-phase and pore coupling conditions (e.g., Singer et al., 2020; Elsayed et al., 2022) has greatly contributed to the understanding and interpretation of NMR data and fluid relaxation, this mini-review focuses on water interactions between the pore environments in near-surface conditions. We summarize the approaches used to investigate pore coupling effects using NMR, discuss the connectivity between pore environments and the pore wall's surface geochemistry as two factors controlling the degree of coupling, and highlight the most pressing open questions for further research opportunities, particularly for application in the vadose zone.

2. Basic concepts of NMR

NNR measurements are based on the behavior of the magnetic spin of protons inside changing magnetic fields. In the presence of an applied constant magnetic field, the spins of hydrogen atoms inside geologic material containing pore water align along the field direction at their characteristic Larmor frequency. A secondary radio frequency pulse perturbs this alignment by exciting the spins, causing increased magnetization in the system. After removing this secondary pulse, the hydrogen atoms precess and decay back to equilibrium in a process called relaxation. The initial magnitude of the relaxation decay is proportional to water content. Relaxation is enhanced when the protons, diffusing inside the pore space, encounter the pore's wall. This process is called surface relaxation ($T_{1S,2S}$). Besides surface relaxation, Equation (1) outlines other relaxation mechanisms for both longitudinal (T_1) and transverse (T_2) relaxation times—bulk relaxation ($T_{1B,2B}$) takes place spontaneously in the fluid, depending on its physical properties and without the influence of the pore wall, while diffusion relaxation (T_{2D}) occurs in inhomogeneous magnetic fields and only affects transverse relaxation (T_2).

$$\frac{1}{T_{\text{relax}}} = \frac{1}{T_{1,2}} = \frac{1}{T_{1B,2B}} + \frac{1}{T_{1S,2S}} + \frac{1}{T_{2D}} \tag{1}$$

Both bulk and diffusion relaxation are considerably slower compared to surface relaxation and can often be neglected. Though they do not provide information about the pore's structure, in certain scenarios (e.g., in systems with large pore spaces or inside strong magnetic gradients) they must be considered for a correct interpretation (e.g., Dlugosch et al., 2013; Costabel et al., 2018, 2022).

Brownstein's parameter μ differentiates between fast, intermediate and slow diffusion regimes and is defined by Equation (2), where L_2 is the average distance traveled by a proton during the NMR measurement and D is the self-diffusion coefficient of water. Fast diffusion, with a μ value below 0.1 (Ryu, 2009), is a precondition for the tradition interpretation of NMR signals (Brownstein and Tarr, 1979). Further considerations regarding to diffusion regimes can be found e.g., in Müller-Petke et al. (2015).

$$\mu = \frac{\text{relaxation rate}}{\text{diffusion rate}} = \frac{\rho/L_2}{D/L_2^2} = \frac{\rho L_2}{D} \tag{2}$$

Other scenarios that requires extra careful interpretation of the NMR signal are pore-coupled systems. Here, the diffusing protons are able to move between pore environments during the NMR measurement time, leading to distorted relaxation time distributions where the peaks are no longer representing the distinct pore environments but an average of the whole sample (e.g., Grunewald and Knight, 2009). In pore-coupled systems, the average pore size is smaller than the diffusion length scale, l, defined by the Einstein equation for self-diffusion (see Equation 3) where D is the self-diffusion coefficient of water (a temperature dependant constant) and T is the time scale of the NMR measurement.

$$l = \sqrt{6DT} \tag{3}$$

For a system in the fast diffusion regime and without pore coupling effects, Equation (1) can be written as Equation (4), where S_{por} stands for the pore's surface area-to-volume ratio. Thus, the relaxation time (T_1 or T_2) can be used to estimate pore size distribution, irreducible and movable liquid saturation, surface area-to-volume ratios, and hydraulic conductivity under saturated condition (e.g., Coates et al., 1999).

$$\frac{1}{T_{1,2}} = \frac{1}{T_{1S,2S}} = \rho S_{por} \tag{4}$$

While the violation of the fast diffusion regime and the pore coupling effects are only of relevance in specific scenarios and for certain type of materials, they can lead to serious misinterpretations of the NMR data if they are ignored or overlooked.

3. Pore coupling effects

As NMR gains popularity in new fields of research and starts to be deployed in new conditions or to study more complex materials (Behroozmand et al., 2015; Schmidt and Rempe, 2020), it is of great importance to identify the scenarios in which pore coupling must be considered and to describe how it will affect the NMR signals (Grunewald and Knight, 2009). Pore coupling effects arise due to the magnetization exchange between pore environments of different characteristic pore sizes during the NMR measurement time (Ramakrishnan et al., 1998) and can be observed both in T_1 and T_2 distributions as the merging of two peaks (Grunewald and Knight, 2009), as well as in 2D $T_2 - T_2$- or $T_1 - T_2$-maps as off-diagonal peaks (Song, 2012). Figure 1 shows a comparison between uncoupled (Figure 1A) and coupled scenarios (Figure 1B) in relation to the simple modeled behavior of diffusing hydrogen atoms inside pore spaces. In Figure 1B, the diffusing spins of protons (indicated with blue and red) move through both pore environments, resulting in merged relaxation peaks and a reduced amplitude of the fast peak as some protons diffuse out of the micropores without relaxing (Figure 1D). Figures 1C, D show the exemplary relaxation time distributions for uncoupled and coupled scenarios. 2D $T_2 - T_2$-maps (Figures 1E, F) are measurements to detect the chemical shift of protons and monitor the exchange of water among different pore environments during designed experimental mixing periods (Song, 2012). In pore coupled environment (Figure 1F), the magnetization can be exchanged between environments multiple times during the NMR measurement, leading to shifts in the peaks away from the diagonal and asymmetry in cross-peaks (Song, 2012; Zhang and Zhang, 2021).

Pore coupling affects the interpretation of NMR data in respect to the water distribution within the sample and the characterization of the pore network's structure, e.g., leading to an underestimation of total micropore volume and/or the underestimation of the macropore's characteristic sizes (Grunewald and Knight, 2009). The identification of pore coupling-prone materials and the scenarios leading to stronger coupling, as well as the quantification of the pore coupled effects in NMR signals, is of great importance to utilize NMR to its full potential and avoid mistakes in the interpretation of the results.

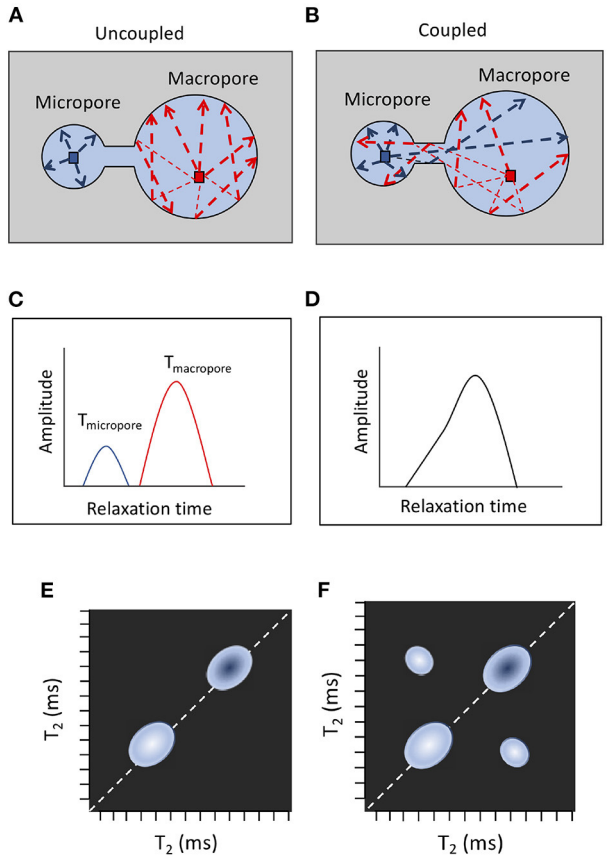


FIGURE 1
Comparison of an uncoupled and a coupled scenario. Panels **(A, B)** illustrate the movement of one exemplary hydrogen atom as it diffuses through the pore spaces. Panels **(C, D)** represent the corresponding relaxation time distribution curves, while panels **(E, F)** represent the corresponding 2D T_2-T_2-maps.

3.1. Approaches to study pore coupling

Pore coupling effects have been studied experimentally and through numerical simulations in fully saturated environments.

A common method used in the laboratory to identify pore coupling effects involves comparing the NMR signals from fully saturated samples with those from centrifuged samples (with drained macropores and saturated micropores) (e.g., Toumelin et al., 2003; Anand and Hirasaki, 2007; Grunewald and Knight, 2009). The measurement obtained from the centrifuged samples provides a baseline for the fast relaxation peak in an uncoupled scenario, which can be used to isolate pore-coupled signals from the fully saturated measurement.

Numerical simulations on digitized or synthetic samples are great tools to study pore coupling as they allow for the targeted exploration of its controlling parameters. The typical numerical modeling workflow is depicted in Figure 2. A popular approach to simulate NMR relaxation using real geological geometries involves the creation of digital porous media from micro-CT images (e.g., Chi and Heidari, 2015). Two main simulation methods are predominant in literature: Random Walk (e.g., Talabi, 2008)

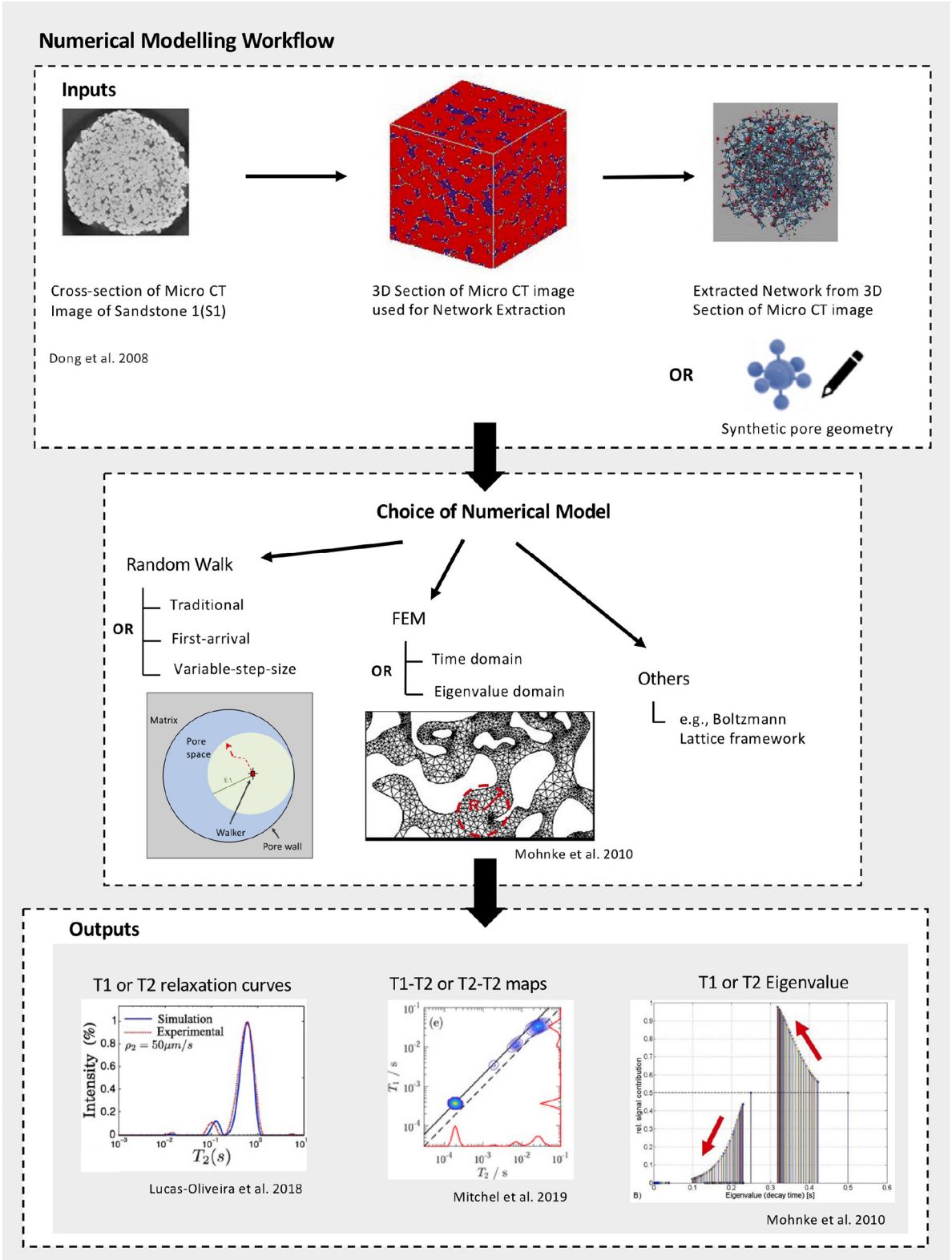


FIGURE 2
Typical workflow for the numerical modeling of pore coupling effects on NMR data. With images from Dong and Blunt (2008); Mohnke and Klitzsch (2010); Lucas-Oliveira et al. (2018), and Mitchell et al. (2019).

and Finite Elements (e.g., Mohnke and Klitzsch, 2010; Mitchell et al., 2019). Furthermore, Mohnke et al. (2014) developed a Boltzmann Lattice framework, allowing for the joint simulation of NMR relaxation and transport properties in unsaturated environments. Several versions of the Random Walk method have been developed deviating from the traditional fix-step-size technique: in the First Arrival or First Passage approach (e.g., Toumelin et al., 2003; Mitchell et al., 2019) the step-size of the walkers depends on its proximity to the pore wall, while, in the Variable-Step-Size approach (e.g., Zhang et al., 2011; Carneiro et al., 2013), each walker moves at changing step-sizes with its own internal clock. Furthermore, Toumelin et al. (2007) developed a Random Walk capable of modeling a CPMG pulse sequence. Mitchell et al. (2019) compares the First Arrival Random Walk to the Finite Elements Method (FEM) and argues that FEM provides greater versatility by allowing for more arbitrary pore geometries, while reducing computational costs and uncertainties by solving the diffusion equations in the eigenvalue domain (thus skipping the inversion step). Simulating NMR relaxation using Finite Elements approaches can be straightforward employing licensed-software such as COMSOL Multiphysics (COMSOL Multiphysics, 1998) or open-software such as OpenFoam (Weller et al., 1998) (e.g., Tandon and Heidari, 2018). The development and publication of open-access simulation codes for open-software would reduce the barriers for using FEM to simulate NMR signals of complicated scenarios.

Moreover, multiple studies have used 2D approaches, such as $T_2 - T_2$-maps (see Figures 1E, F), to detect and quantify pore coupling (e.g., Washburn and Callaghan, 2006; Mitchell et al., 2019; Zhang and Zhang, 2021). Although $T_2 - T_2$-maps are great tools to identify a pore-coupled sample, they require more demanding experimental techniques than 1D approaches (Fleury and Soualem, 2009), with evidence suggesting that no additional pore couple strength information can be obtained from $T_2 - T_2$-maps, in comparison to T_1- or T_2-distributions (Johnson and Schwartz, 2014).

3.2. Factors controlling pore coupling

From literature, two main factors controlling pore coupling are identified in existing studies: surface geochemistry and network connectivity.

The surface geochemistry of the pore's wall, in particular related to the rock's mineralogy, can be quantified through the surface relaxivity value (ρ). Surface relaxivity ρ is an empirical factor describing the wall's ability to enhance relaxation (Bergman et al., 1995). It will increase in the presence of paramagnetic material in the grain such as manganese(II) or iron(III) (Bryar et al., 2000), and is also affected by pore fluid properties (e.g., Anand and Hirasaki, 2007) and temperature (e.g., Godefroy et al., 2002). Higher surface relaxivity values will lead to faster relaxation processes, thus weakening pore coupling effects. The role of surface geochemistry in pore coupling effects has been probed experimentally using iron-coated silica gels by Daughney et al. (2000) and Grunewald and Knight (2009), with changing saturation fluids by Anand and Hirasaki (2007) and Fleury and Soualem (2009), and via numerical simulations (e.g., Mitchell et al., 2019). Furthermore, since both surface relaxivity and the fluid's diffusion coefficient are temperature dependent, a rise in temperature has been found to enhance pore coupling (Toumelin et al., 2003; Anand et al., 2008).

The parameters affecting the pore's network connectivity will change depending on the network's model representing the sample. Three well-studied, pore coupling-prone network geometries can be identified in literature: shales, modeled as systems with macro- and micropores connected through pore throats (e.g., Washburn, 2014), clay-lined sandstones, modeled as systems where NMR relaxation predominantly takes place in the micropores surrounding round-shaped macropores (e.g., Anand and Hirasaki, 2007), and carbonate rocks, often understood through the micrograin consolidation model, with a cubic packing of spherical porous grains, each comprised by non-porous micrograins (e.g., Schwartz et al., 2013). For the simpler geometry of pores connected by throats, the influence of the throat sizes on pore-coupling strength has been studied numerically by Mohnke and Klitzsch (2010) (throat length) and Ghomeshi et al. (2018) (throat diameter). Their results show that better connected geometries with shorter and wider throats have stronger pore coupling.

For clay-lined geometries, Anand and Hirasaki (2007) introduces a parameter, α, to quantify the degree of pore coupling (see Equation 5), with higher α values signaling weaker pore coupling. This parameter recognizes the key fact that the total volume available for a spin to diffuse in (i.e., the macropore and the micropore volume) is decoupled from the available active surface for spin relaxation (i.e., the micropore surface area). Relaxation is assumed to only take place in the micropores because of the much larger surface available in comparison to macropores.

$$\alpha = \beta\eta\mu = \beta\frac{L_2}{L_1}\frac{L_2\rho}{D} = \frac{\beta\rho}{D}\frac{L_2^2}{L_1} \tag{5}$$

Where μ is the Brownstein's parameter, η is the aspect ratio (L_2/L_1), L_2 is defined as the half length to the middle of the macropore, and L_1 as the half-distance between clay flakes. β is a parameter controlled by geometry-factors described in Equation (6), where ϕ is the total porosity of the sample, ϕ_μ the microporosity and ϕ_m the macroporosity.

$$\beta = \frac{\phi_\mu(1-\phi_m)}{\phi} \tag{6}$$

By adjusting the definition of L_2 and L_1, α can also be applied to the micrograin consolidation model (e.g., Souza et al., 2016; Mitchell et al., 2019), as in Equation (7), where R_g is the microporous grain's radius and R_μ the micropore radius, ρ is the surface relaxivity, D is the diffusivity of water, and β is the geometrical parameter described by Equation (6).

$$\alpha_{\text{grain}} = \frac{\rho\beta R_g^2}{DR_\mu} \tag{7}$$

These established equations suggest that increased grain radius leads to the weakening of the pore coupling. This relationship was also tested and verified experimentally and numerically by Anand and Hirasaki (2007) and Mitchell et al. (2019).

4. Discussion—Knowledge gaps

4.1. The vadose zone

The study of pore coupling effects has so far been limited to fully-saturated conditions. The NMR method, nevertheless, has great potential for studying unsaturated conditions and fluid-flow processes in the vadose zone (Costabel and Yaramanci, 2011; Walsh et al., 2014; Flinchum et al., 2019; Tian, 2021); it can provide information regarding water availability, distribution and movement, and has already been used to study infiltration mechanisms (e.g., Walsh et al., 2014), draining processes (e.g., Mohnke et al., 2014), water availability for plants in form of rock moisture (e.g., Rempe and Dietrich, 2018; Schmidt and Rempe, 2020; Zhang and Zhang, 2021), among others.

However, to address some of these research questions, e.g., for the study of physico-chemical processes in the soil (e.g., wetting or swelling), an accurate measurement of water distribution within different pore environments is crucial and only possible if eventual distortions of the NMR signals, related to pore coupling effects, are accounted for. This is also relevant, and must still be examined, e.g., for the estimation of water retention curves from NMR measurements in the saturated material (e.g., Jaeger et al., 2009; Costabel and Hiller, 2021). Furthermore, the complexity and heterogeneity of top-soils may lead to large differences in surface relaxivity values for neighboring pore environments (e.g., Jaeger et al., 2009; Benavides et al., 2017); the effect such heterogeneous ρ distribution will have on pore coupling remains to be investigated. Likewise unexplored, remains whether a potential transition from the slow into the fast diffusion regime, observed in analytical models with a decrease in saturation level (Costabel, 2011), would translate to a reduction on pore coupling effects. Whether pore coupling strength can be used as an estimator of the degree of connectivity between the pores (Carneiro et al., 2014; Tian, 2021) to evaluate hydraulic conductivity and estimate permeability, is, similarly, an interesting open question, with potential applications to determine, e.g., soil contaminants' flow speed and extent.

Due to the inherent characteristics of the vadose zone, the complexity of interpretation and modeling of NMR signals increases. Three further factors controlling NMR relaxation must be considered in relation to pore coupling: level of saturation, water retention function, and pore shape. The pore-filling and draining mechanism, such as the specific order in which pore environments fill or drain, and possible hysteresis are of great interest and can be explored using NMR (e.g., Mohnke and Klitzsch, 2010; Falzone and Keating, 2016; Zhang and Zhang, 2021). Additionally, studies such as Costabel and Yaramanci (2011) and Mohnke et al. (2015) investigated the influence of the pore shape on the hydraulic properties and found that sharp angles are capable of withholding small water quantities as they drain. Understanding how these three factors impact pore coupling effects in unsaturated conditions is essential to expand the use of NMR for practical environmental applications in the vadose zone, which plays important retention and buffering roles and supports crucial ecosystem functions.

4.2. Karst aquifers

Karst aquifers are the source of about a quarter of the world's potable water (Clifford and Williams, 2007) and feature highly heterogeneous networks formed through the chemical weathering of soluble rocks. These creates conduits with free surface flow and high flow rate, as well as confined flow with various porosity and permeability values. Due to these unique features, karst aquifers are sensitive to hydro-meteorological changes and are highly vulnerable to external forcings, such as contamination (Hartmann et al., 2014; Goldscheider et al., 2020). Due to their inter- and intragranular porosity and their low- to mid- surface relaxivity, carbonate systems are highly prone to pore coupling (Ramakrishnan et al., 1998). However, the coupling effects in carbonate system and their effects on the relaxation behavior of mesopores, or of pores in a wider range of sizes, is understudied.

Detailed experimental investigation into pore coupling effects in carbonate rocks is required for full utilization of NMR in karst aquifers. Moreover, the improvement of image processing steps are required to better extract pore networks from carbonate samples via improved segmentation (Lucas-Oliveira et al., 2018), which will allow for advanced simulations of the NMR responses in pore coupled carbonate systems.

5. Closing remarks

Low-field NMR is an emerging technology within the hydrogeophysics community with significant potential for practical applications in near-surface environments and for critical zone research due to its non-invasive, cost-efficient, direct sensitivity to pore water. Much information can be gained from improving and extending the applications of NMR in different environments. In order to achieve its full potential, a better understanding of NMR signals in complex scenarios is required. This includes, though it is not limited to, the understanding of pore coupling effects in fully and partially saturated environments.

In the past years, with improvements in the understanding of NMR signals in partially saturated environments, the application of NMR in the vadose zone has expanded, yet pore coupling effects remain understudied in unsaturated conditions. Technological advances have allowed for more sophisticated numerical simulation methods to study pore coupling in more complex geometries, and the original definition of pore coupling has been expanded to include different pore networks or even diffusion between pores and microfractures (Chi and Heidari, 2015). The development and identification of better techniques to study pore coupling (e.g., FEM over Random Walk or T_2-distributions over $T_2 - T_2$-maps) has brought this emerging field forward.

Developing our understanding of NMR relaxation in pore-coupled unsaturated environments, as well as in complex geometries with a wider range of pore sizes or lower surface relaxivities, would improve the interpretation of NMR data and lead to its broader applications in near-surface environments. Furthermore, studying the relation between coupling degree and network connectivity opens the possibility of using pore coupling as a connectivity estimator in unknown samples or for larger scale

field application (Carneiro et al., 2014) and should be seen as more than a disturbance in the NMR signal.

Author contributions

FS: figures and writing. CZ: concept and writing. All authors contributed to the article and approved the submitted version.

Funding

This work was funded by State Key Laboratory of Geomechanics and Geotechnical Engineering (SKlGGE), Institute of Rock and Soil Mechanics, Chinese Academy of Sciences, provided the funding Z019003.

Conflict of interest

The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Publisher's note

All claims expressed in this article are solely those of the authors and do not necessarily represent those of their affiliated organizations, or those of the publisher, the editors and the reviewers. Any product that may be evaluated in this article, or claim that may be made by its manufacturer, is not guaranteed or endorsed by the publisher.

References

An, R., Zhang, X., Wang, Y., Liu, X., Chen, C., and Gong, J. (2022). Freeze–thaw impact on sandy clay in artificial frozen walls: an investigation of shear strength and pore-size distribution. *Int. J. Geomech.* 22, 04022230. doi: 10.1061/(ASCE)GM.1943-5622.0002489

Anand, V., and Hirasaki, G. (2007). Diffusional coupling between micro and macroporosity for NMR relaxation in sandstones and grainstones. *Petrophysics* 48, 289–307.

Anand, V., Hirasaki, G., and Fleury, M. (2008). NMR diffusional coupling: effects of temperature and clay distribution. *Petrophysics* 49, 362–372.

Behroozmand, A. A., Keating, K., and Auken, E. (2015). A review of the principles and applications of the NMR technique for near-surface characterization. *Surv. Geophys.* 36, 27–85. doi: 10.1007/s10712-014-9304-0

Benavides, F., Leiderman, R., Souza, A., Carneiro, G., and Bagueira, R. (2017). Estimating the surface relaxivity as a function of pore size from NMR t2 distributions and micro-tomographic images. *Comput. Geosci.* 106, 200–208. doi: 10.1016/j.cageo.2017.06.016

Bergman, D. J., Dunn, K.-J., Schwartz, L. M., and Mitra, P. P. (1995). Self-diffusion in a periodic porous medium: a comparison of different approaches. *Phys. Rev. E* 51, 3393–3400. doi: 10.1103/PhysRevE.51.3393

Binley, A., Hubbard, S. S., Huisman, J. A., Revil, A., Robinson, D. A., Singha, K., et al. (2015). The emergence of hydrogeophysics for improved understanding of subsurface processes over multiple scales. *Water Resour. Res.* 51, 3837–3866. doi: 10.1002/2015WR017016

Brownstein, K. R., and Tarr, C. E. (1979). Importance of classical diffusion in NMR studies of water in biological cells. *Phys. Rev. A* 19, 2446–2453.

Bryar, T. R., Daughney, C. J., and Knight, R. J. (2000). Paramagnetic effects of iron(iii) species on nuclear magnetic relaxation of fluid protons in porous media. *J. Magn. Reson.* 142, 74–85. doi: 10.1006/jmre.1999.1917

Carneiro, G., Souza, A., Boyd, A., Schwartz, L., Coutinho, B., Trevizan, W., et al. (2013). "Image analysis and NMR modeling of sedimentary rocks," in *13th International Congress of the Brazilian Geophysical Society and EXPOGEF* (Rio de Janeiro), 938–941.

Carneiro, G., Souza, A., Boyd, A., Schwartz, L., Song, Y.-Q., Trevizan, W., et al. (2014). "Evaluating pore space connectivity by NMR diffusive coupling," in *SPWLA 55th Annual Logging Symposium* (Abu Dhabi).

Chi, L., and Heidari, Z. (2015). Diffusional coupling between microfractures and pore structure and its impact on nuclear magnetic resonance measurements in multiple-porosity systems. *Geophysics* 80, D31–D42. doi: 10.1190/geo2013-0467.1

Clifford, F., and Williams, P. (2007). *Karst Hydrogeology Geomorphology*. Chichester: Wiley

Coates, G., Xiao, L., and Prammer, M. (1999). *NMR Logging Principles and Applications*. Elsevier Science.

COMSOL Multiphysics (1998). "Introduction to comsol multiphysics," in *COMSOL Multiphysics* (Burlington, MA).

Costabel, S. (2011). *Nuclear magnetic resonance on laboratory and field scale for estimating hydraulic parameters in the vadose zone* (Ph.D. thesis).

Costabel, S., and Hiller, T. (2021). Soil hydraulic interpretation of nuclear magnetic resonance measurements based on circular and triangular capillary models. *Vadose Zone J.* 20, e20104. doi: 10.1002/vzj2.20104

Costabel, S., Hiller, T., Dlugosch, R., Kruschwitz, S., and Muller-Petke, M. (2022). Evaluation of single-sided nuclear magnetic resonance technology for usage in geosciences. *Meas. Sci. Technol.* 34, 015112. doi: 10.1088/1361-6501/ac9800

Costabel, S., Weidner, C., Müller-Petke, M., and Houben, G. (2018). Hydraulic characterisation of iron-oxide-coated sand and gravel based on nuclear magnetic resonance relaxation mode analyses. *Hydrol. Earth Syst. Sci.* 22, 1713–1729. doi: 10.5194/hess-22-1713-2018

Costabel, S., and Yaramanci, U. (2011). Relative hydraulic conductivity and effective saturation from Earth's field nuclear magnetic resonance–a method for assessing the Vadose zone. *Near Surface Geophys.* 9, 155–167. doi: 10.3997/1873-0604.2010055

Daughney, C. J., Bryar, T. R., and Knight, R. (2000). Detecting sorbed hydrocarbons in a porous medium using proton nuclear magnetic resonance. *Environ. Sci. Technol.* 34, 332–337. doi: 10.1021/es990637m

Ding, S., Jia, H., Zi, F., Dong, Y., and Yao, Y. (2020). Frost damage in tight sandstone: experimental evaluation and interpretation of damage mechanisms. *Materials* 13, 4617. doi: 10.3390/ma13204617

Dlugosch, R., Gunther, T., Muller-Petke, M., and Yaramanci, U. (2013). Improved prediction of hydraulic conductivity for coarse-grained, unconsolidated material from nuclear magnetic resonance. *Geophysics* 78, EN55–EN64. doi: 10.1190/geo2012-0187.1

Dong, H. M.-C., and Blunt, M. M.-C. (2008). *Imaging and pore network extraction* (Ph.D. thesis) Department of Earth Science and Engineering, Imperial College.

Elsayed, M., Isah, A., Hiba, M., Hassan, A., Al-Garadi, K., El Husseiny, A., et al. (2022). A review on the applications of nuclear magnetic resonance (NMR) in the oil and gas industry: laboratory and field-scale measurements. *J. Petrol. Explorat. Product. Technol.* 12, 2747–2784. doi: 10.1007/s13202-022-01476-3

Falzone, S., and Keating, K. (2016). A laboratory study to determine the effect of pore size, surface relaxivity, and saturation on NMR t2 relaxation measurements. *Near Surface Geophys.* 14, 57–69. doi: 10.3997/1873-0604.2016001

Fleury, M., and Soualem, J. (2009). Quantitative analysis of diffusional pore coupling from T2-store-T2 NMR experiments. *J. Coll. Interface Sci.* 336, 250–259. doi: 10.1016/j.jcis.2009.03.051

Flinchum, B., Holbrook, S., Parsekian, A., and Carr, B. (2019). Characterizing the critical zone using borehole and surface nuclear magnetic resonance. *Vadose Zone J.* 18, 180209. doi: 10.2136/vzj2018.12.0209

Ghomeshi, S., Kryuchkov, S., and Kantzas, A. (2018). An investigation into the effects of pore connectivity on T2 NMR relaxation. *J. Magn. Reson.* 289, 79–91. doi: 10.1016/j.jmr.2018.02.007

Godefroy, S., Fleury, M., Deflandre, F., and Korb, J.-P. (2002). Temperature effect on NMR surface relaxation in rocks for well logging applications. *J. Phys. Chem. B.* 106, 11183–11190. doi: 10.1021/jp0213452

Goldscheider, N., Chen, Z., Auler, A., Bakalowicz, M., Broda, S., Drew, D., et al. (2020). Global distribution of carbonate rocks and karst water resources. *Hydrogeol. J.* 28, 1–17. doi: 10.1007/s10040-020-02139-5

Grunewald, E., and Knight, R. (2009). A laboratory study of NMR relaxation times and pore coupling in heterogeneous media. *Geophysics* 74, E215–E221. doi: 10.1190/1.3223712

Hartmann, A., Goldscheider, N., Wagener, T., Lange, J., and Weiler, M. (2014). Karst water resources in a changing world: review of hydrological modeling approaches. *Rev. Geophys.* 52, 218–242. doi: 10.1002/2013RG000443

Jaeger, F., Bowe, S., As, H., and Schaumann, G. (2009). Evaluation of 1h NMR relaxometry for the assessment of pore size distribution in soil samples. *Eur. J. Soil Sci.* 60, 1052–1064. doi: 10.1111/j.1365-2389.2009.01192.x

Johnson, D. L., and Schwartz, L. M. (2014). Analytic theory of two-dimensional NMR in systems with coupled macro- and micropores. *Phys. Rev. E* 90, 032407. doi: 10.1103/PhysRevE.90.032407

Kass, M. A., Irons, T. P., Minsley, B. J., Pastick, N. J., Brown, D. R. N., and Wylie, B. K. (2017). *In situ* nuclear magnetic resonance response of permafrost and active layer soil in boreal and tundra ecosystems. *Cryosphere* 11, 2943–2955. doi: 10.5194/tc-11-2943-2017

Lucas-Oliveira, E., Araujo-Ferreira, A., Trevizan, W., Fortulan, C., and Bonagamba, T. (2018). Computational approach to integrate 3d x-ray microtomography and NMR data. *J. Magn. Reson.* 292, 16–24. doi: 10.1016/j.jmr.2018.05.001

Mitchell, J., Souza, A., Fordham, E., and Boyd, A. (2019). A finite element approach to forward modeling of nuclear magnetic resonance measurements in coupled pore systems. *J. Chem. Phys.* 150, 154708. doi: 10.1063/1.5092159

Mohnke, O., Jorand, R., Nordlund, C., and Klitzsch, N. (2015). Understanding NMR relaxometry of partially water-saturated rocks. *Hydrol. Earth Syst. Sci.* 19, 2763–2773. doi: 10.5194/hess-19-2763-2015

Mohnke, O., and Klitzsch, N. (2010). Microscale simulations of NMR relaxation in porous media considering internal field gradients. *Vadose Zone J.* 9, 846–857. doi: 10.2136/vzj2009.0161

Mohnke, O., Stiebler, M., and Klitzsch, N. (2014). Joint numerical microscale simulations of multiphase flow and NMR relaxation behavior in porous media using Lattice Boltzmann methods. *Water Resour. Res.* 50, 7378–7393. doi: 10.1002/2013WR014684

Müller-Petke, M., Dlugosch, R., Lehmann-Horn, J., and Ronczka, M. (2015). Nuclear magnetic resonance average pore-size estimations outside the fast-diffusion regimeNMR average pore-size estimations. *Geophysics* 80, D195–D206. doi: 10.1190/geo2014-0167.1

Parsekian, A. D., Grosse, G., Walbrecker, J. O., Muller-Petke, M., Keating, K., et al. (2013). Detecting unfrozen sediments below thermokarst lakes with surface nuclear magnetic resonance. *Geophys. Res. Lett.* 40, 535–540. doi: 10.1002/grl.50137

Ramakrishnan, T. S., Schwartz, L. M., Fordham, E., Kenyon, W. E., and Wilkinson, D. (1998). Forward models for nuclear magnetic resonance in carbonate rocks. *The Log Analyst*, 40.

Rempe, D. M., and Dietrich, W. E. (2018). Direct observations of rock moisture, a hidden component of the hydrologic cycle. *Proc. Natl. Acad. Sci. U.S.A.* 115, 2664–2669. doi: 10.1073/pnas.1800141115

Ryu, S. (2009). *Effect of inhomogeneous surface relaxivity, pore geometry and internal field gradient on NMR logging: exact and perturbative theories and numerical investigations*, in *SPWLA Annual Logging Symposium* (The Woodlands, TX).

Schmidt, L., and Rempe, D. (2020). Quantifying dynamic water storage in unsaturated bedrock with borehole nuclear magnetic resonance. *Geophys. Res. Lett.* 47, e2020GL089600. doi: 10.1029/2020GL089600

Schwartz, L. M., Johnson, D. L., Mitchell, J., Chandrasekera, T. C., and Fordham, E. J. (2013). Modeling two-dimensional magnetic resonance measurements in coupled pore systems. *Phys. Rev. E* 88, 032813. doi: 10.1103/PhysRevE.88.032813

Singer, P. M., Chen, Z., Wang, X., and Hirasaki, G. J. (2020). Diffusive coupling in heptane-saturated kerogen isolates evidenced by NMR t1-t2 and t2-t2 maps. *Fuel* 280, 118626. doi: 10.1016/j.fuel.2020.118626

Song, Y.-Q. (2012). Magnetic resonance of porous media (MRPM): a perspective. *J. Magn. Reson.* 229, 12–24. doi: 10.1016/j.jmr.2012.11.010

Souza, A., Mesquita, P., Carneiro, G., Boyd, A., Ferreira, F., Machado, P., et al. (2016). "Surface relaxivity estimation and NMR-micp matching in diffusionaly coupled rocks," in *30th International Symposium fo the Society of Core Analysts* (Snowmass, CO).

Sullivan, T. (2022). *Geophysical investigation of Arctic and sub-Arctic permafrost* (Ph.D. thesis) Wyoming.

Talabi, O. A. (2008). *Pore-scale simulation of NMR response in porous media* (Ph.D. thesis) London.

Tandon, S., and Heidari, Z. (2018). Effect of internal magnetic-field gradients on nuclear-magnetic-resonance measurements and nuclear-magnetic-resonance-based pore-network characterization. *SPE Reserv. Eval. Eng.* 21, 609–625. doi: 10.2118/181532-PA

Tian, Z. (2021). NMR diffusional coupling of multiple-scale porous rock and its detection. *Chin. J. Geophys.* 64, 1119–1130. doi: 10.6038/cjg2021O0081

Toumelin, E., Torres-Verdon, C., Chen, S., and Fischer, D. (2003). Reconciling NMR measurements and numerical simulations: assessment of temperature and diffusive coupling effects on two-phase carbonate samples. *Petrophysics* 44.

Toumelin, E., Torres-Verdon, C., Sun, B., and Dunn, K.-J. (2007). Random-walk technique for simulating NMR measurements and 2d NMR maps of porous media with relaxing and permeable boundaries. *J. Magn. Reson.* 188, 83–96. doi: 10.1016/j.jmr.2007.05.024

Walsh, D. O., Grunewald, E. D., Turner, P., Hinnell, A., and Ferre, T. P. (2014). Surface NMR instrumentation and methods for detecting and characterizing water in the vadose zone. *Near Surface Geophys.* 12, 271–284. doi: 10.3997/1873-0604.2013066

Washburn, K. (2014). Relaxation mechanisms and shales. *Concepts Magn. Reson. A* 43, 57–78. doi: 10.1002/cmr.a.21302

Washburn, K. E., and Callaghan, P. T. (2006). Tracking pore to pore exchange using relaxation exchange spectroscopy. *Phys. Rev. Lett.* 97, 175502. doi: 10.1103/PhysRevLett.97.175502

Washburn, K. E., and Cheng, Y. (2017). Detection of intermolecular homonuclear dipolar coupling in organic rich shale by transverse relaxation exchange. *J. Magn. Reson.* 278, 18–24. doi: 10.1016/j.jmr.2017.02.022

Watanabe, K., and Wake, T. (2009). Measurement of unfrozen water content and relative permittivity of frozen unsaturated soil using NMR and TDR. *Cold Regions Sci. Technol.* 59, 34–41. doi: 10.1016/j.coldregions.2009.05.011

Weller, H., Tabor, G., Jasak, H., and Fureby, C. (1998). A tensorial approach to computational continuum mechanics using object orientated techniques. *Comput. Phys.* 12, 620–631. doi: 10.1063/1.168744

Zhang, F., and Zhang, C. (2021). Probing water partitioning in unsaturated weathered rock using nuclear magnetic resonance. *Geophysics* 86, WB131–WB147. doi: 10.1190/geo2020-0591.1

Zhang, Z., Johnson, D., and Schwartz, L. (2011). Simulating the time-dependent diffusion coefficient in mixed-pore-size materials. *Phys. Rev. E Stat. Nonlinear Soft Matter Phys.* 84, 031129. doi: 10.1103/PhysRevE.84.031129

01_张弛老师第一作者语言样本.pdf

文献 6/6

Recent research advances in enhanced CO2 mineralization and geologic CO2 storage

Chi Zhang; Yuhang Wang; Zuhao Kou; Liwei Zhang

Advances in Geo-Energy Research (2023); DOI: 10.46690/ager.2023.12.01

证据用途：第一作者直接语言样本；张弛老师角色：第一作者；证据等级：A

Advances in Geo-Energy Research

Vol. 10, No. 3, p. 141-145, 2023

Perspective

Recent research advances in enhanced CO_2 mineralization and geologic CO_2 storage

Chi Zhang[1], Yuhang Wang[2], Zuhao Kou[3], Liwei Zhang[4]*

[1]*Department of Meteorology and Geophysics, University of Vienna, Vienna 1010, Austria*

[2]*School of Environmental Studies, China University of Geosciences, Wuhan 430074, P. R. China*

[3]*Department of Earth & Environmental Engineering, Columbia University, New York 10027, USA*

[4]*State Key Laboratory of Geomechanics and Geotechnical Engineering, Institute of Rock and Soil Mechanics, Chinese Academy of Sciences, Wuhan 430071, P. R. China*

Keywords:
Geologic CO_2 storage
enhanced rock weathering
trapping mechanism
convective mixing
reservoir heterogeneity

Cited as:
Zhang, C., Wang, Y., Kou, Z., Zhang, L. Recent research advances in enhanced CO_2 mineralization and geologic CO_2 storage. Advances in Geo-Energy Research, 2023, 10(3): 141-145.
https://doi.org/10.46690/ager.2023.12.01

Abstract:
Enhanced CO_2 mineralization and geologic CO_2 storage have received increasing attention as two prominent approaches in combating climate change and fostering sustainable development of human society. This paper aims to explore three emerging areas of research within the realm of enhanced CO_2 mineralization and geologic CO_2 storage, including enhanced rock weathering, numerical modeling and validation of CO_2 storage accounting for the interplay of various trapping mechanisms, and the examination of how reservoir heterogeneity influences the migration of CO_2-brine multiphase flow. Discussions highlight the effectiveness of the spectrum induced polarization for monitoring changes in petrophysical and geochemical properties of rocks during enhanced rock weathering. Additionally, the multi-scale heterogeneity of geological formations needs to be carefully characterized, due to the fact that it plays a vital role in CO_2 migration. Further research is required to achieve accurate and reliable simulations of convective mixing for field-scale applications.

1. Introduction

CO_2 is widely recognized as the primary contributor to global warming (Soeder et al., 2021). The implementation of enhanced CO_2 mineralization and geologic CO_2 storage strategies is expected to play an important role in mitigating CO_2 emissions (Zhang et al., 2023). Within this dynamic field, a multitude of new research directions have emerged, including investigation of CO_2 mineralization through enhanced rock weathering, numerical modeling of CO_2 trapping mechanisms in geologic CO_2 storage, and exploration of the effects of reservoir heterogeneity on the migration of CO_2-brine multiphase flow. In subsequent sections, interesting research works related to the aforementioned research directions are discussed in detail.

2. Enhanced rock weathering for permanent CO_2 sequestration

Climate change presents a pervasive global challenge that necessitates the development and implementation of diverse strategies for carbon dioxide removal. One such strategy, known as enhanced rock weathering (ERW), represents a group of methods that accelerates naturally occurring rock weathering, enabling permanent trap of CO_2 (Kelemen et al., 2019; Beerling et al., 2020; Snæbjörnsdóttir et al., 2020). ERW can be implemented through natural and surficial deployments by spreading rock powders on diverse land types, as well as through ex-situ and in-situ engineered mineral sequestration. Silicate and carbonate rock powders on farmland and forests can facilitate the conversion of gaseous CO_2 into aqueous bicarbonate or carbonate ions (HCO_3^-,

Yandy Scientific Press

*Corresponding author.
E-mail address: chi.zhang@univie.ac.at (C. Zhang); wangyuhang17@cug.edu.cn (Y. Wang); zk2283@columbia.edu (Z. Kou); lwzhang@whrsm.ac.cn (L. Zhang).
2207-9963 © The Author(s) 2023.
Received September 28, 2023; revised October 11, 2023; accepted October 18, 2023; available online October 20, 2023.

CO_3^{2-}). These ions can either be mineralized on land or be transported through rivers to the ocean, where they can persist as solutes for an extensive duration (>10,000 years) and finally precipitate as solid carbonate minerals (e.g., calcite, magnesite, dolomite, etc.) in the subsea. The ex-situ process, typically employed on olivine or alkaline industry wastes, is usually performed in high temperature and high pressure reactors. The in-situ process, which involves injecting CO_2 into the subsurface pore spaces of rocks like basalt and peridotite, has been effectively demonstrated in projects like Iceland's Carbfix Project (Snæbjörnsdóttir et al., 2020). Besides removal of CO_2, ERW offers additional benefits such as enhanced agricultural productivity, soil remediation, and ocean alkalinization, thereby enabling sustainable socio-economic advancement (Köhler et al., 2010; Beerling et al., 2020).

During the calcite precipitation, the microstructure of weathering rocks is altered, leading to highly heterogeneous and dynamic fluid-rock interactions, which significantly impacts the enhanced weathering process. Effective and robust monitoring of the ERW processes not only help quantify the amount of removed carbon achieved through enhanced weathering reactions, but it is imperative to mitigate risks such as induced seismicity and groundwater contamination. The mineralization processes within soil, water, and plant biomass are typically monitored through the analysis of collected samples, yielding accurate yet spatially limited information. Geophysical monitoring and imaging techniques have the potential of providing high spatiotemporal resolution information of the evolution of petrophysical properties during calcite precipitation. Within the geophysical methods, complex conductivity or spectral induced polarization (SIP) method has shown great sensitivity to pore fluid and mineral interactions (e.g., Binley and Kemna, 2020; Zhang et al., 2012). During the application of SIP, the polarization of charge carriers like calcite is associated with the polarization of the electrical double layer (EDL) surrounding the mineral particles under an external alternating current supply (Jougnot et al., 2010; Revil et al., 2014). Studies (Wu et al., 2010; Izumoto et al., 2020; Saneiyan et al., 2021) have shown the potential of SIP in characterizing the distribution of the induced calcite precipitation at laboratory scales.

The majority of ERW research centers around silicate rocks. However, enhancing the weathering of carbonate rocks may also have a substantial contribution to carbon dioxide removal, given much higher dissolution rates of carbonates than silicates. Over half of the atmospheric CO_2 absorbed by rock weathering each year is attributed to carbonate rocks (Gaillardet et al., 1999). In a recent carbonate rock ERW study, laboratory columns were utilized to emulate natural carbonate weathering processes, while geophysical responses were correlated with simulated geochemical reactions and solution geochemistry. Such geophysical investigation of ERW processes elucidates and distinguishes different stages of calcite formation from fine grains to larger aggregates. The combination of geophysical monitoring, geochemical analysis, and numerical modeling not only provides deeper insight into calcite precipitation, but also contributes to the broader understanding on ERW.

3. Numerical simulation accounting for the interplay of trapping mechanisms

The understanding of CO_2 trapping mechanisms within geological formations has been well-established, encompassing structural, residual, dissolution, and mineral trapping (Duan and Sun, 2003; Juanes et al., 2006; Szulczewski et al., 2012; Zhang et al., 2019). These trapping mechanisms operate across varying time scales. Structural and dissolution trapping come into effect immediately after CO_2 injection. Dissolution trapping continues to work in the post-injection stage due to the establishment of gravity-induced convective mixing. Mineral trapping is achieved through geochemical reactions, with the rate of these reactions determined by the composition of the host rock. To comprehensively evaluate the interactions among these trapping mechanisms, the construction of efficient and predictive numerical models is imperative.

Incorporation of the aforementioned trapping mechanisms into the numerical model necessitates the integration of hydraulic, thermal, and chemical processes, which pose considerable challenges to large-scale numerical simulations (Gasda et al., 2011; Wang et al., 2022). To improve the computational efficiency, the equations governing different processes can be decoupled and solved in a sequential manner via non-iterative operator splitting method (Postma et al., 2022). Moreover, in a fully coupled system, evaluation of the physical properties and the derivatives regarding the unknowns can be expedited using a parameterized space (Lyu et al., 2023). In addition to the physics-based numerical simulations, the rise of machine learning has sparked enthusiasm for surrogate models. These models can predict the spatiotemporal evolution of the state variables without the need to run numerical simulations (Wen et al., 2021; Yan et al., 2022; Zhao et al., 2023). However, it is worth noting that the developed models have been typically trained based on simplified physical models (e.g., immiscible two-phase system). Also, it is very important to gather data of physical observations that are open to the public for numerical model validation purpose. At the current stage, ensuring the accuracy of numerical models remains challenging due to the limited availability of comprehensive physical observations. An effective approach to address this challenge is to conduct laboratory-scale CO_2 injection benchmark studies with laboratory-designed heterogeneous permeable medium to evaluate the performance of numerical models (Flemisch et al., 2023). Research groups can utilize their respective custom-developed or open-source numerical simulators and validate their numerical simulation results based on the comprehensive characterization of the heterogeneous permeable medium employed in the benchmark studies.

The key take-away messages can be summarized as follows: (i) The efficacy of the classic continuum model, formulated upon Darcy's law with a multiphase extension, is validated in accurately explaining pertinent physical observations. The simulation results effectively capture the characteristics of both structural trapping and residual trapping, which are influenced by the migration of supercritical CO_2. (ii) The dissolution trapping, facilitated by convective mixing driven by gravity, is found to enhance the dissolution process of

CO_2. However, predictions concerning the dynamics of dissolved CO_2 exhibit notable disparities among the participating research groups. This variability can be attributed to several physical and modeling parameters, including CO_2 solubility limits, phase densities, and grid resolutions. Consequently, further investigations are warranted to refine the simulation of convective mixing, particularly when applied to scenarios of a larger scale. (iii) Most benchmark laboratory studies are conducted in a controlled system characterized by well-defined heterogeneity. In contrast, real-world subsurface formations are typically characterized through seismic surveys with significantly coarser resolutions. Given that spatial heterogeneity notably influences plume migration (Jackson and Krevor, 2020), field-scale simulations may experience considerable uncertainties in reservoir properties stemming from the relatively coarse geological characterization.

4. Effects of reservoir heterogeneity on CO_2-brine flow properties

Deep saline aquifers have been regarded as ideal candidates for gigaton-scale CO_2 storage due to their wide geographical spread. Investigation on controlled CO_2 storage in deep saline aquifers is a highly interdisciplinary challenge at the intersection of geology, geochemistry, petrophysics, and engineering. Given the prevalence of reservoir heterogeneity such as the presence of cross-bedding layers, it becomes imperative to examine the influence of reservoir heterogeneity on the multiphase flow properties of CO_2-brine systems, thus paving the way for high-accuracy field-scale simulations.

Given the diverse nature of CO_2 injectivity, flow patterns, trapping mechanisms, migration, and storage capacity across different rock types, it is essential to classify reservoir rocks for accurate geological modeling and reservoir simulation in the context of geological carbon storage (Padhi et al., 2014; Zheng et al., 2021). Rock typing entails classifying specific units of reservoir rocks by considering lithofacies, petrophysical properties of the rock and its pores, interactions between rock and fluid, and/or historical production and injection data (Rebelle et al., 2014). Presently, rock-typing studies often depend on combining two or more clustering methods (Farshi et al., 2019). Unlike oil and gas reservoirs, which have extensive data collected during exploration and development, there is limited data for deep saline aquifers in the context of geological carbon storage. This scarcity underscores the need for reservoir characterization and flow unit classification specifically tailored to saline aquifers.

Following reservoir characterization, studying the impact of rock heterogeneity (i.e., variations in physical and petrophysical properties of rock) on fluid distribution at the sub-core scale enhances our fundamental knowledge of rock relative permeability. It also plays a crucial role in advancing the development of multiphase flow simulation techniques at the field scale. Hence, comprehending how sub-core heterogeneity affects CO_2 flow properties holds paramount importance in the realm of CO_2 geological storage. Currently, the majority of existing studies have concentrated solely on exploring the impact of a singular type of sub-core heterogeneity on CO_2 flow properties, and further research is needed to explore the impact of two or more types of heterogeneity (i.e., the combination of pore distribution and mineral distribution heterogeneities) on CO_2 flow. Also, it is challenging to correlate sub-core heterogeneity with reservoir-scale heterogeneity. Sub-core scale heterogeneities often do not fully represent the entire range of reservoir characteristics, even within a single target layer, due to a large gap between sub-core scale and reservoir-scale (Perrin et al., 2010). Consequently, it is reasonable to anticipate variability in CO_2 sweep efficiency and distribution across different locations within deep saline aquifers during CO_2 injection.

Additionally, when injecting CO_2 into saline aquifers containing reactive minerals, varied interactions between CO_2 and reactive minerals are anticipated within the host rocks. To elaborate, a diverse pore size distribution often indicates widespread heterogeneous dissolution and precipitation within a core sample (Wang et al., 2020). Mineral dissolution tends to happen predominantly in channels or fractures, where fluid flow encounters less resistance (Mohamed et al., 2018). Furthermore, studies have confirmed that mineral dissolution and precipitation within a core sample depend on the specific position during fluid flow-through tests (Huq et al., 2015; Kou et al., 2023). As a result, changes in pore architecture often lead to time-dependent responses in CO_2-brine multiphase flow. After identifying sub-core scale and core-scale properties related to multiphase flow in saline aquifers, these findings are recommended to be integrated into field-scale studies to assess the forthcoming CO_2 injectivity and trapping mechanisms.

5. Summary

As the global energy transition gains momentum and the urgency to effectively remove CO_2 becomes unprecedented, forthcoming research in enhanced CO_2 mineralization and geological CO_2 storage will be paramount in establishing a sufficiently large carbon sink for CO_2, thereby achieving net-zero CO_2 emissions and addressing the challenges of climate change. Future research in this field will focus on improving the efficiency and reliability of CO_2 mineralization, as well as the predictive capability of reservoir simulators for field-scale CO_2 storage operations. Developing better monitoring techniques and ensuring the long-term integrity of storage sites will be key areas of investigation as well. Additionally, efforts will be made to explore new storage options, such as basalts and subsea storage, to increase the total storage capacity.

Currently, research endeavors pertaining to CO_2 mineralization are witnessing a surge, with particular emphasis on swiftly converting CO_2 into stable minerals. Improving the efficiency and reducing the cost of the CO_2 conversion processes will be crucial to make them economically viable, and enhanced rock weathering is a hot research spot in this field. In addition, studies on the numerical modeling that incorporates the interaction between different trapping mechanisms and the impact of multi-scale reservoir heterogeneity are crucial to ensure the efficiency and safety of geologic CO_2 storage.

Acknowledgements

The authors gratefully acknowledge the support of the National Natural Science Foundation of China (Nos. 42172315 and U1967208).

Conflict of interest

The authors declare no competing interest.

Open Access This article is distributed under the terms and conditions of the Creative Commons Attribution (CC BY-NC-ND) license, which permits unrestricted use, distribution, and reproduction in any medium, provided the original work is properly cited.

References

Beerling, D. J., Kantzas, E. P., Lomas, M. R., et al. Potential for large-scale CO_2 removal via enhanced rock weathering with croplands. Nature, 2020, 583(7815): 242-248.

Binley, A., Kemna, A. DC resistivity and induced polarization methods. Hydrogeophysics, 2005, 50: 129-156.

Duan, Z., Sun, R. An improved model calculating CO_2 solubility in pure water and aqueous NaCl solutions from 273 to 533 K and from 0 to 2,000 bar. Chemical Geology, 2003, 193: 257-271.

Farshi, M., Moussavi-Harami, R., Mahboubi, A., et al. Reservoir rock typing using integrating geological and petrophysical properties for the Asmari Formation in the Gachsaran oil field, Zagros basin. Journal of Petroleum Science and Engineering, 2019, 176: 161-171.

Flemisch, B., Nordbotten, J. M., Fernø, M., et al. The FluidFlower validation benchmark study for the storage of CO_2. Transport in Porous Media, 2023, in press, https://doi.org/10.1007/s11242-023-01977-7.

Gaillardet, J., Dupré, B., Louvat, P., et al. Global silicate weathering and CO_2 consumption rates deduced from the chemistry of large rivers. Chemical Geology, 1999, 159(1): 3-30.

Gasda, S. E., Nordbotten, J. M., Celia, M. A. Vertically averaged approaches for CO_2 migration with solubility trapping. Water Resources Research, 2011, 47: W05528.

Huq, F., Haderlein, S. B., Cirpka, O. A. et al. Flow-through Experiments on Water-Rock Interactions in a Sandstone Caused by CO_2 Injection at Pressures and Temperatures Mimicking Reservoir Conditions. Applied Geochemistry, 2015, 58: 136-146.

Izumoto, S., Huisman, J.A., Wu, Y., et al. Effect of solute concentration on the spectral induced polarization response of calcite precipitation. Geophysical Journal International. 2020, 220(2): 1187-1196.

Jackson, S. J., Krevor, S. Small-scale capillary heterogeneity linked to rapid plume migration during CO_2 storage. Geophysical Research Letters, 2020, 47: e2020GL088616.

Jougnot, D., Ghorbani, A., Revil, A., Leroy, P., Cosenza, P. Spectral induced polarization of partially saturated clay-rocks: a mechanistic approach. Geophysical Journal International, 2010, 180(1): 210-224.

Juanes, R., Spiteri, E. J., Orr, F. M., Blunt, M. J. Impact of relative permeability hysteresis on geological CO_2 storage. Water Resources Research, 2006, 42: W12418.

Kelemen, P., Benson, S.M., Pilorgé, H., et al. An overview of the status and challenges of CO_2 storage in minerals and geological formations. Frontiers in Climate, 2019, 1: 9.

Köhler, P., Hartmann, J., Wolf-Gladrow, D. A. Geoengineering potential of artificially enhanced silicate weathering of olivine. Proceedings of the National Academy of Sciences of the United States of America, 2010, 107(47): 20228-20233.

Kou, Z., Wang, H., Alvarado, V., et al. Effects of carbonic acid-rock interactions on CO_2/Brine multiphase flow properties in the upper minnelusa sandstones. SPE Journal, 2023, 28 (2): 754-767.

Lyu, X., Voskov, D. Advanced modeling of enhanced CO_2 dissolution trapping in saline aquifers. International Journal of Greenhouse Gas Control, 2023, 127: 103907.

Mohamed, K., Ali, S., Reza, R. et al. Experimental evaluation of carbonated brine- limestone interactions under reservoir conditions-emphasis on the effect of core scale heterogeneities. International Journal of Greenhouse Gas Control, 2018, 68: 128-145.

Padhi, A., Mallick, S., Behzadi, H., et al. Efficient modeling of seismic signature of patchy saturation for time lapse monitoring of carbon sequestrated deep saline reservoirs. Applied Energy, 2014, 114: 445-455.

Perrin, J.C., Benson, S. An experimental study on the influence of sub-core scale heterogeneities on CO_2 distribution in reservoir rocks. Transport in Porous Media, 2010, 82(1): 93-109.

Postma, T. J. W., Bandilla, K. W., Celia, M. A. Implications of CO_2 mass transport dynamics for large-scale CCS in basalt formations. International Journal of Greenhouse Gas Control, 2022, 121: 103779.

Rebelle, M., Lalanne, B. Rock-typing in carbonates: A critical review of clustering methods. Paper SPE 96812 Presented at Abu Dhabi International Petroleum Exhibition and Conference, Abu Dhabi, UAE, 10–13 November, 2014.

Revil, A., Florsch, N., Camerlynck, C. Spectral induced polarization porosimetry. Geophysical Journal International, 2014, 198(2): 1016-1033.

Saneiyan, S., Ntarlagiannis, D., Colwell, F. Complex conductivity signatures of microbial induced calcite precipitation, field and laboratory scales. Geophysical Journal International, 2021, 224(3): 1811-1824.

Snæbjörnsdóttir, S. O., Sigfússon, B., Marieni, C., et al. Carbon dioxide storage through mineral carbonation. Nature Reviews Earth & Environment, 2020, 1(12): 695.

Szulczewski, M. L., MacMinn, C. W., Herzog, H. J., et al. Lifetime of carbon capture and storage as a climate-change mitigation technology. Proceedings of the National Academy of Sciences of the United States of America, 2012, 109: 5185-5189.

Soeder, D. J. Greenhouse gas sources and mitigation strategies from a geosciences perspective. Advances in Geo-Energy Research, 2021, 5(3): 274-285.

Wang, H., Alvarado, V., Smith, E. R. et al. Link between CO_2-Induced Wettability and Pore Architecture Alteration. Geophysical Research Letter, 2020, 47: e2020GL088490.

Wang, Y., Vuik, C., Hajibeygi, H. Analysis of hydrodynamic trapping interactions during full-cycle injection and migration of CO_2 in deep saline aquifers. Advances in Water Resources, 2022, 159: 104073.

Wen, G., Tang, M., Benson, S. M. Towards a predictor for CO_2 plume migration using deep neural networks. International Journal of Greenhouse Gas Control, 2021, 105: 103223.

Wu, Y., Hubbard, S., Williams, K. H., et al. On the complex conductivity signatures of calcite precipitation. Journal of Geophysical Research: Biogeosciences, 2010, 115(G2): G00G04.

Yan, B., Chen, B., Harp, D. R., Jia, W., Pawar, R. J. A robust deep learning workflow to predict multiphase flow behavior during geological CO_2 sequestration injection and post-injection periods. Journal of Hydrology, 2022, 607: 127542.

Zhang, L., Nowak, W., Oladyshkin, S., et al. Opportunities and challenges in CO_2 geologic utilization and storage. Advances in Geo-Energy Research, 2023, 8(3): 141-145.

Zhang, C., Slater, L., Redden, G. et al. Spectral induced polarization signatures of hydroxide adsorption and mineral precipitation in porous media. Environmental Science & Technology, 2012, 46(8): 4357-4364.

Zhang, L., Wang, Y., Miao, X., et al. Geochemistry in geologic CO_2 utilization and storage: A brief review. Advances in Geo-Energy Research, 2019, 3: 304-313.

Zhao, M., Wang, Y., Gerritsma, M., Hajibeygi, H. Efficient simulation of CO_2 migration dynamics in deep saline aquifers using a multi-task deep learning technique with consistency. Advances in Water Resources, 2023, 178: 104494.

Zheng, T., Guo, B., Shao, H. A hybrid multiscale framework coupling multilayer dynamic reconstruction and full-dimensional models for CO_2 storage in deep saline aquifers. Journal of Hydrology, 2021, 600: 126649.